Has GPT-6-Astra Achieved Embodied Intelligence?

Joint Research Team

Abstract

GPT-6-Astra demonstrates a level of competence in robotic manipulation and navigation that challenges the conventional paradigm of building specialized models for individual tasks and embodiments. In this report, we ask not only how capable GPT-6-Astra is, but what its capabilities imply for the field. **First, we probe the limits of standalone GPT-6-Astra.** We conduct a large-scale evaluation across 12 representative benchmarks spanning tabletop manipulation, humanoids, mobile platforms, and dexterous hands. Under a unified protocol that allows in-context learning (ICL) from training-set demonstrations and substantially increases the number of execution steps, GPT-6-Astra achieves unprecedented performance (Fig. 1). Our analysis suggests that future research should increasingly focus on improving 1) the generality of in-context learning 2) cost and execution time, 3) physical-state verification, 4) recovery from irreversible failures, and 5) long-horizon preservation of progress. **Second, we study the agentic harness for embodied control.** Besides exploring the ideal action representation and chucking parameters, we identify that close-loop coordination between a high-level "System 2" reasoner (*i.e.*, GPT-6-Astra) and a low-level "System 1" controller (*e.g.*, $\pi_{0.5}$) is costly but effective. **Third, we examine GPT-6-Astra on real-world deployment.** Our experiments on 8 robot configurations reveal substantial gaps: 1) robot demonstrations required for in-context learning are expensive to collect, and 2) it is implausible to trade longer execution for higher success rates with physical constraints. **Fourth, we envision the next generation of embodied benchmarks.** We develop a prototype that 1) derives tasks from real-world needs rather than current robot capabilities, 2) provides no in-domain demonstrations, and 3) judges agents on a world clock that keeps running while they deliberate. Taken together, our results suggest that GPT-6-Astra has largely crossed the threshold of embodied competence, but not yet that of practical embodied intelligence. The key remaining challenge is to translate benchmark success into reliable, efficient, and scalable real-world deployment.

Homepage: https://sensenova.ai/has-gpt-6-astra-achieved-embodied-intelligence

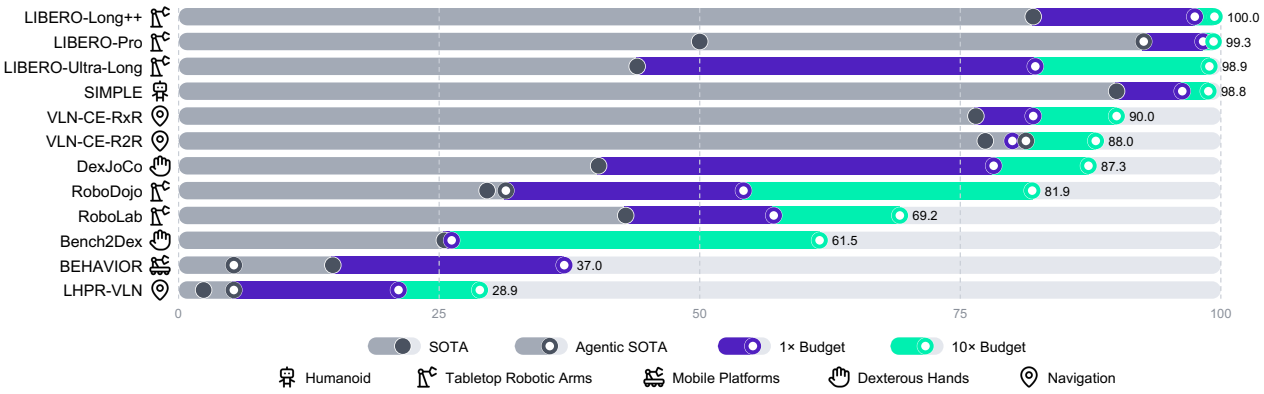


Figure 1 Probing the capability upper bound of GPT-6-Astra across 12 benchmarks under a unified protocol that 1) allows for in-context learning (ICL) on the training set and 2) extends the step limit (budget) up to 10×. RoboLab has no training set and thus no ICL; BEHAVIOR is not extended to 10× due to its large official budget. SOTA cut-off date: October 12, 2026. Details in Tab. 2.

1 Introduction

The advent of GPT-6-Astra marks a major shift in embodied intelligence: a general-purpose model can now tackle robotic manipulation and navigation tasks that have traditionally been dominated by specialized models, such as VLAs and WAMs, while exhibiting capabilities including zero-shot generalization, in-context learning, and failure recovery. In this technical report, we use GPT-6-Astra as a lens to examine a broader question: what does this new level of embodied capability mean for the research community, and where should the field go next? We first conduct a large-scale evaluation across diverse embodied tasks to characterize the extent of its capabilities and identify which problems are already approaching saturation. We then study harness design to understand the performance–cost trade-offs involved in translating these capabilities into effective control. Next, we validate our findings on real robots, asking how far benchmark success carries over to physical deployment and what gaps remain.Finally, we revisit benchmark design and envision what the next generation of embodied benchmarks should look like.

First, we probe the upper bound of GPT-6-Astra's embodied capabilities under a unified protocol. We conduct a large-scale study across 12 representative benchmarks, covering tabletop robotic arms, humanoids, mobile platforms, and dexterous hands. Under this protocol, GPT-6-Astra outperforms existing methods by substantial margins, including recent agentic approaches that also build on GPT-6-Astra. Notably, as shown in Fig. 1, it approaches saturation on the LIBERO series and SIMPLE, while achieving 81.9% on RoboDojo and 87.3% on DexJoCo. The protocol follows two simple and well-motivated principles:

1) We allow in-context learning (ICL) from the training set whenever successful demonstrations are available, which we consider reasonable since such demonstrations are conventionally accessible when training smaller task-specific policies. GPT-6-Astra studies the available training data and constructs a frozen knowledge package containing task guidance, motion and visual references, executable helpers, recovery procedures, and metadata. At test time, it retrieves and adapts these resources to the current scene. This substantially improves performance when the demonstrations closely cover the test distribution, but the benefit weakens as tasks move away from the data used to build the package.

2) We extensively increase the execution budget, allowing up to $10\times$ more interaction steps than the official setting. We find that standard step limits can significantly underestimate GPT-6-Astra's capabilities. Additional interaction yields large gains on tasks that benefit from repeated correction, such as RoboDojo and Bench2Dex, because GPT-6-Astra can detect failed actions, revise its strategy, and recover through further interaction. This emerging failure-recovery capability is the key mechanism that allows a larger execution budget to translate into higher success rates.

Our strong empirical results put us in a position to identify the remaining limitations of GPT-6-Astra more clearly. Specifically, we find that: i) in-context learning transfer poorly beyond the tasks from which the demonstrations are drawn; ii) even simple tasks incur inference cost and execution time far beyond what would be practical for commercial deployment; iii) GPT-6-Astra often struggles to verify physical success and identify which task condition remains unmet; iv) GPT-6-Astra often struggles to identify the cause of task failure; v) it cannot reliably recover from consequential or irreversible failures; and vi) long-horizon execution remains difficult, as the model struggles to preserve and track progress across many subtasks.

Second, we ablate the harness design and find that effectively unleashing GPT-6-Astra's embodied performance depends critically on both the collaboration protocol and the action interface. We first study the interaction between "System 2" (*i.e.*, GPT-6-Astra) and "System 1" (*e.g.*, $\pi_{0.5}$ [12]). Closed-loop collaboration, where GPT-6-Astra reviews and refines VLA-proposed actions before execution, substantially improves performance over standalone or open-loop execution, but frequent System-2 supervision incurs significant token cost. Reviewing longer action sequences improves this trade-off, while combining closed-loop collaboration with ICL yields the strongest performance at the highest cost. We further study action representation and chunking, and identify a representation gap between multimodal reasoning and low-level robot control. Direct joint-angle control underperforms, while more semantically aligned end-effector commands work better; adaptive action chunking further improves performance while reducing unnecessary model calls. Together, these findings suggest that harness design should expose action spaces aligned with the model's reasoning capabilities while delegating precise execution to lower-level controllers, with adaptive interaction to balance performance and inference cost.

Third, we validate GPT-6-Astra on real-world robots spanning single and dual robotic arms, grippers, dexterous hands, a wheeled mobile manipulator, and a humanoid. Our study reveals three key findings. 1) GPT-6-Astra operates effectively across a wide range of embodiments, but its success rate remains highly task-dependent. It performs

particularly well on tasks such as tabletop grasping, while deformable-object manipulation, precise contact, and long-horizon tasks remain challenging. 2) GPT-6-Astra demonstrates surprising active-vision capabilities, intentionally repositioning the camera to obtain observations needed for subsequent actions. 3) GPT-6-Astra shows that a slow "System 2" model can directly collaborate with a high-frequency "System 0" locomotion controller to perform humanoid manipulation that requires high-frequency local control. At the same time, real-world deployment introduces challenges that are largely absent from simulation. GPT-6-Astra can no longer easily trade additional execution steps for higher success rates, due to safety constraints, mechanical limits, and real-time requirements. Moreover, successful trajectory demonstrations, which are central to our ICL setup in simulation, are often expensive or unavailable in real-world applications.

Fourth, we showcase a prototype of what we envision as the next generation of benchmarks for embodied intelligence. We propose three principles. 1) Tasks should be derived from human needs and expressed as people would naturally state them, with evaluation based on the utility of the outcome rather than a fixed goal state. 2) No in-domain training episodes or scene-specific demonstrations should be provided; instead, agents should acquire experience during deployment. 3) The world should continue to evolve while the agent deliberates, making reasoning time and its physical consequences part of the evaluation. The prototype makes several remaining limitations more apparent in a deployment-oriented setting: i) GPT-6-Astra focuses on the current step but can lose track of other changes it has caused in the environment, such as leaving a refrigerator open; ii) without demonstrations, it struggles to figure out how unfamiliar objects and mechanisms work, or how to position its own body to interact with them; and iii) because the world continues to change while it reasons, its observations can become outdated before the resulting action is executed. Building such benchmarks also raises practical challenges in reliable judging, simulation fidelity, real-time evaluation, and evaluation cost.

In summary, by pushing GPT-6-Astra toward its capability limits, dissecting the harness that enables this performance, validating its behavior on real robots, and rethinking how embodied intelligence should be evaluated, we aim to clarify both what has changed and what remains fundamentally unsolved.

2 Large-Scale Study under A Unified Protocol

We study GPT-6-Astra across twelve embodied settings, spanning manipulation, humanoid loco-manipulation, household activities, and navigation, under a single protocol in which its model weights remain fixed. Two conditions shape the study: in-context learning (ICL), which gives GPT-6-Astra access to the same training demonstrations from which conventional policies are learned, and an extended interaction budget, which gives it more opportunity to correct mistakes within an episode.

2.1 Protocol

Our study covers twelve evaluation settings, grouped into five categories: tabletop manipulation (RoboDojo [26], RoboLab [161], LIBERO-Long++ and LIBERO-Ultra-Long [162], and LIBERO-PRO [183]), dexterous manipulation (Bench2Dex [163] and DexJoCo [144]), humanoid loco-manipulation (SIMPLE [152]), long-horizon household activities (BEHAVIOR [8]), and vision-language navigation (VLN-CE R2R and RxR [87], and LHPR-VLN [134]).

We evaluate GPT-6-Astra under a demonstration-informed protocol with two interaction budgets ($1\times$ and $10\times$), as illustrated in Fig. 2. The protocol separates preparation using training demonstrations from test-time interaction. We report performance under the standard budget and an extended budget that allows more interaction steps for execution and recovery.

Motivation. Learned robotic policies are commonly trained or fine-tuned on demonstrations provided by the benchmark. Evaluating GPT-6-Astra without access to these demonstrations would therefore measure its performance under a different information regime. We use in-context learning (ICL) to investigate how effectively GPT-6-Astra can exploit the available training demonstrations without updating its model weights. Our objective is to characterize its capabilities when the task-relevant information used to train conventional policies is also available to it through contextual preparation.

We additionally investigate the effect of the interaction budget. In our evaluations, GPT-6-Astra often identifies unsuccessful actions, revises its strategy, and recovers through further interaction. A strict step limit can terminate this

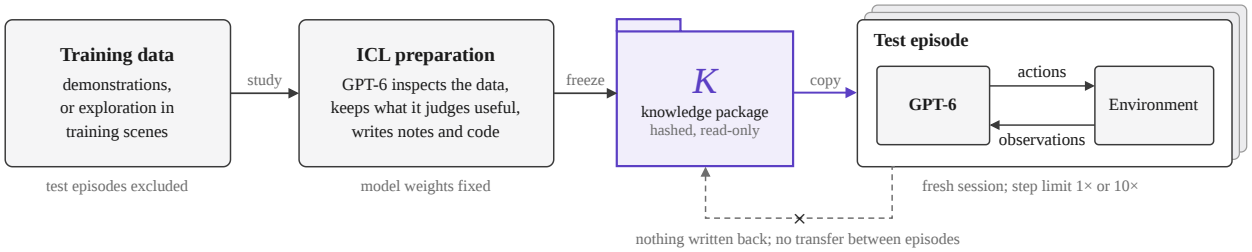


Figure 2 In-context learning protocol. For each task, GPT-6-Astra studies the training data and writes a knowledge package K, which is frozen before testing. Each test episode starts a fresh session with K and runs under the standard ($1\times$) or extended ($10\times$) budget. Model weights remain fixed throughout, and K remains unchanged during testing, and no information passes between episodes.

process before recovery succeeds. We therefore evaluate under two budgets. The standard budget ($1\times$) is the interaction budget prescribed by each benchmark and measures performance under its intended conditions; the extended budget ($10\times$) allows up to ten times as many action steps and measures how task completion improves when the policy has more opportunities to correct mistakes.

In-context learning preparation. Before testing, GPT-6-Astra undergoes a separate ICL preparation phase for each task, using the available training demonstrations represented through benchmark-appropriate observations and demonstrated actions. During preparation, GPT-6-Astra may analyze the demonstrations and develop task-specific notes and reusable code. These materials form a knowledge package, which is frozen before evaluation and provided to every test episode of that task. GPT-6-Astra's model weights remain unchanged. Preparation excludes held-out test episodes and evaluator-private information.

Episode evaluation. Each test episode begins in a fresh, isolated session with the task instruction, the frozen knowledge package, and the permitted observation and action interfaces. At each decision point, GPT-6-Astra examines the current observations and specifies the next action or action sequence. The environment executes these actions and returns new observations, which GPT-6-Astra uses to assess the outcome and determine its next action. The interaction history is retained within the episode, allowing GPT-6-Astra to diagnose mistakes and attempt recovery. Neither the knowledge package nor the model weights are updated during testing, and information from one test episode is not transferred to another. Observation access, robot-control action semantics, control frequency, and success criteria remain fixed across budget conditions.

Extended interaction budget. The extended budget allows ten times as many action steps as the standard budget. We report outcomes at $1\times$ and at cumulative cutoffs of $2\times, \ldots, 10\times$ the standard budget, using the same initialization, policy configuration, and success criterion. An extension continues the same episode, preserving its interaction history and accumulated environmental consequences; additional attempts occur within that episode rather than through resets. Episodes terminate upon verified success, an applicable terminal failure, or exhaustion of the allotted budget. Reporting success across these cutoffs quantifies the contribution of additional interaction while retaining the standard-budget result as a separate comparison point.

2.2 In-context Learning

With in-context learning, GPT-6-Astra achieves strong performance across embodied benchmarks without any update to its model weights. Figure 1 compares GPT-6-Astra with the best published result (SOTA) and the best published agentic method (Agentic SOTA) on all 12 settings, both reported under the standard budget. Already at the standard ($1\times$) budget, GPT-6-Astra matches or exceeds the strongest published result on most settings, with the largest margins on LIBERO-Ultra-Long and DexJoCo, where it nearly doubles the best published success rate. To isolate the contribution of ICL, we compare zero-shot and ICL conditions on SIMPLE: task-level ICL raises standard-budget success on the four common tasks from 70.8% to 90.8% and shortens successful episodes (Table 12). The benefit of ICL, however, depends on how closely the training data covers the test tasks. Where test tasks closely follow the training distribution, as in the LIBERO family, SIMPLE and VLN-CE, the package can retrieve and adapt demonstrated motions or routes almost directly, and success approaches saturation. DexJoCo's contact-rich dexterous tasks are demanding, but phase-indexed

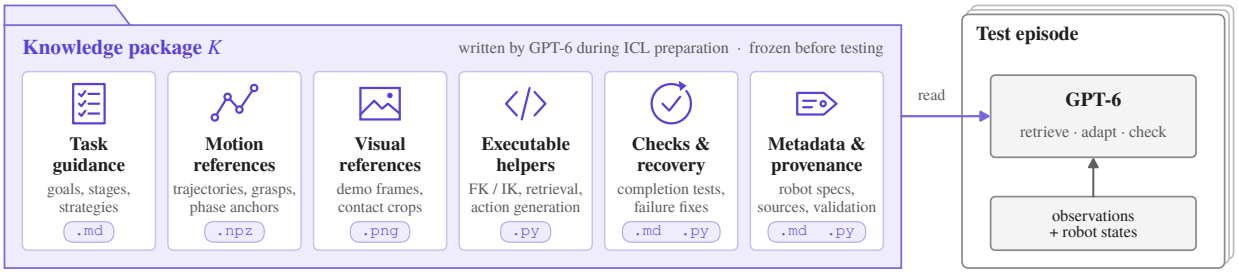


Figure 3 Contents of an ICL knowledge package. GPT-6-Astra decides what to retain, yet packages share six recurring components; file types are typical and content varies by benchmark. At test time, K is read alongside current observations to retrieve, adapt and check actions.

motion references and controller helpers cover them well. RoboLab, in contrast, provides no training set from which a task-level package can be built, and its success remains moderate; Sec. 2.4 discusses this limitation further.

Package contents. Each knowledge package is organized around six recurring components (Fig. 3): task guidance, motion references, visual references, executable helpers, checks and recovery, and metadata and provenance. This structure is not prescribed. During ICL preparation, GPT-6-Astra independently chooses what to inspect and retain within each benchmark's permitted training data and tool interfaces, and we do not specify which demonstrations, frames, or state/action records it should examine or save; the six components nonetheless recur across benchmarks.

- **Task guidance.** Written instructions describe task goals, execution stages, and strategies. They specify subtask ordering, object roles, and constraints that guide the selection and use of stored references.
- **Motion references.** Demonstration-derived trajectories, grasp configurations, and phase anchors provide numerical examples of how actions unfold. The agent retrieves relevant segments or poses and adapts them to the current scene and robot state.
- **Visual references.** Selected demonstration frames and contact crops illustrate object appearance, grasp regions, and intermediate task states. These examples support interpretation of the stored motions and comparison with current observations.
- **Executable helpers.** Reusable Python programs support forward and inverse kinematics, reference retrieval, and action generation. They convert stored references and current geometric estimates into candidate actions.
- **Checks & recovery.** Completion tests describe observable evidence that a grasp, subtask, or overall task has succeeded. Recovery procedures associate failure symptoms with corrective actions, such as adjusting an approach or selecting an alternative grasp.
- **Metadata & provenance.** Robot specifications and interface conventions define how stored states and actions should be interpreted. Source records and validation notes document the origin of the retained material and the scope of its offline checks.

Within this shared structure, the retained information reflects each benchmark's demands. Bench2Dex [163] and DexJoCo [144] emphasize hand configurations, phase-indexed motion examples, and helpers for dexterous control. LIBERO-Long++ and LIBERO-Ultra-Long [162] organize selected demonstrations around composable atomic skills, while LIBERO-PRO [183] primarily retains textual strategies and recovery knowledge from exploration. SIMPLE [152] preserves complete numerical training trajectories and helpers for extracting and modifying candidate action chunks under current feedback. LHPR-VLN [134] and VLN-CE [87] retain navigation strategies, landmark references, stopping guidance, and recovery notes from training-scene practice. BEHAVIOR [8] emphasizes object-specific grasping affordances, bimanual and tool-contact geometry, and conditional verification and recovery procedures.

Package usage. Retrospective usage analyses reveal a common workflow on every benchmark we audited: GPT-6-Astra reads the task guidance, retrieves a candidate motion through a prepared helper, and checks its execution against the visual references. Numerical demonstrations are frequently loaded internally by these helpers, linking stored examples

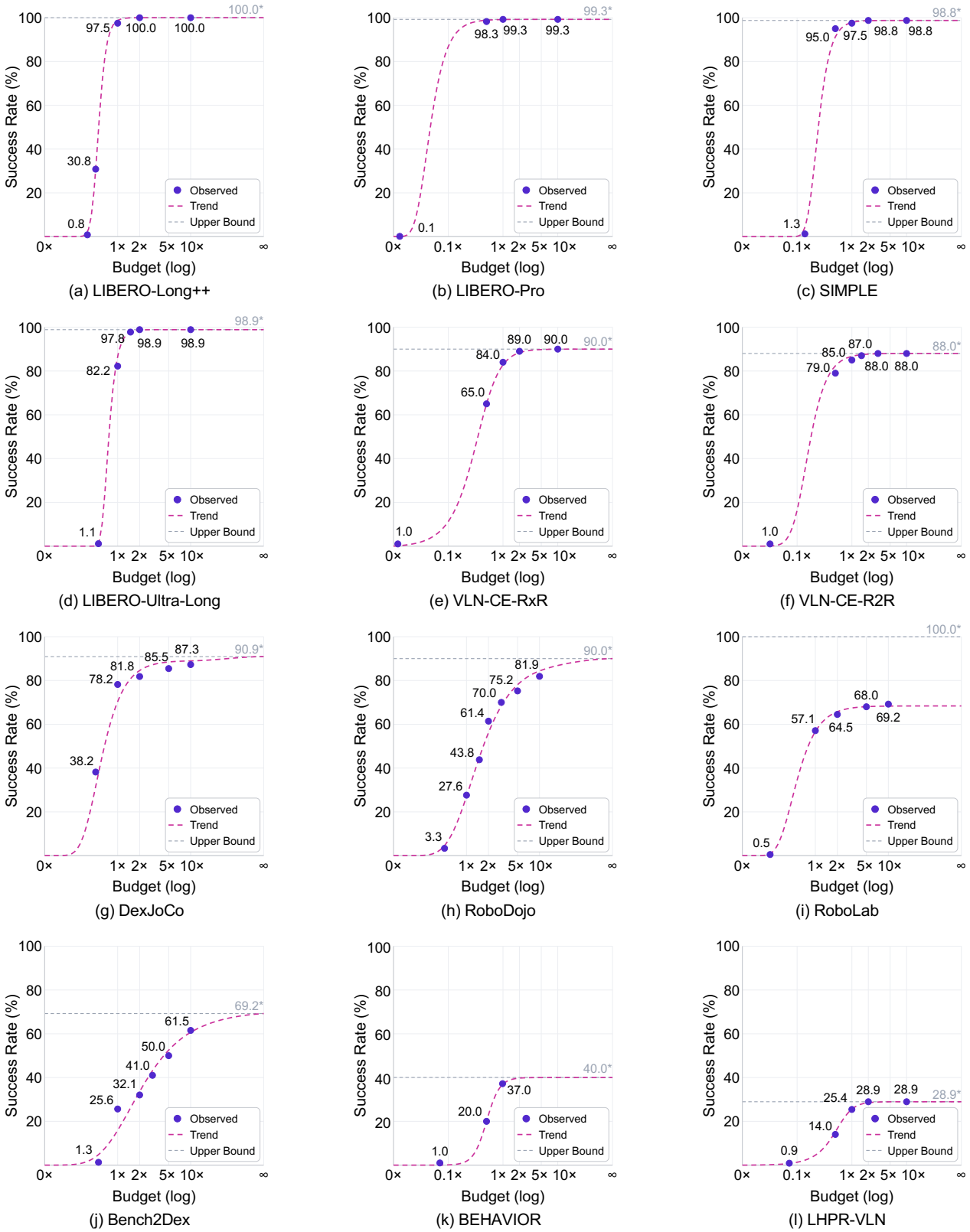
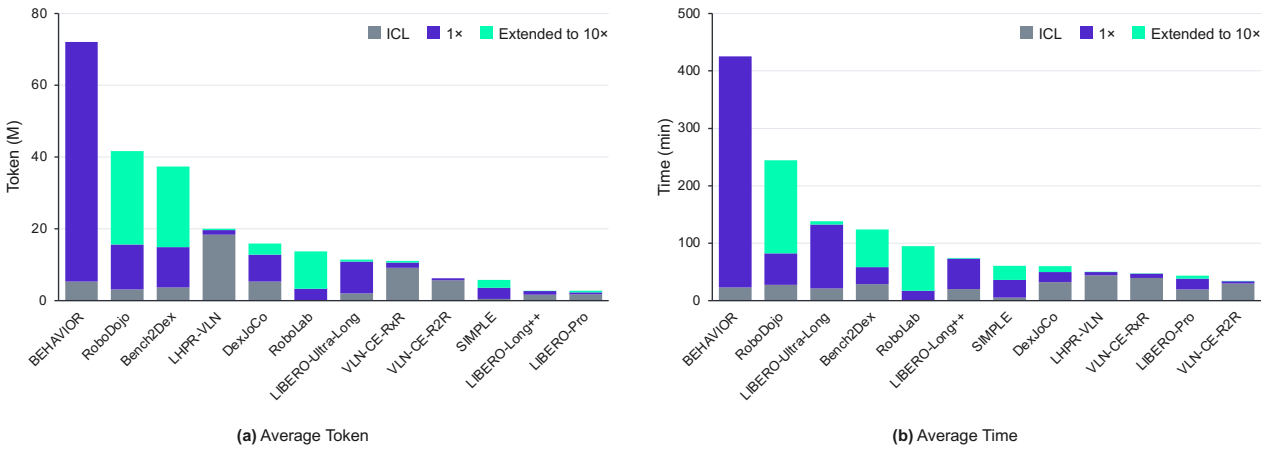


Figure 4 Success rate versus step budget. The budget is expressed as a multiple of each benchmark's official step limit; dots show observed success rates and pink curves show fitted trends. We evaluate budgets up to 10× the official limit (1× for BEHAVIOR). The gray line marks the upper bound, which assumes that every episode that exhausts the evaluated budget without succeeding has the potential to succeed with an unlimited budget.



(a) Average Token **(b)** Average Time

Figure 5 (a) Average per-task token usage and **(b) average per-task time** for each benchmark, stacked from bottom to top as ICL preparation, the $1\times$ run, and the additional cost of extending to $10\times$. Here, $1\times$ and $10\times$ denote maximum action-step budgets, not token or time budgets. The relatively small additional token and time costs primarily reflect early termination of extended runs upon success or terminal failure, before the full $10\times$ budget is exhausted. We observe that the vast majority of tokens consumed are cached input tokens (approximately 98% in our runs). Accordingly, we use a simple rule of thumb for cost estimation: **1M tokens $\approx$ 1 USD**.

directly to action generation; on DexJoCo, controllers and visual references are used repeatedly during both initial execution and continuation. Rather than replaying demonstrations, GPT-6-Astra adapts them to the current scene. In Bench2Dex Ball Box Loading, it remaps reference joints and compensates for a 4–5 cm difference in workspace height while preserving the demonstrated bimanual support geometry; in DexJoCo Photograph, it selects an alternative reference pickup after a slip. The packages thus serve as reusable resources for proposing, adapting, and revising actions under feedback, although these analyses do not isolate the causal contribution of individual components.

2.3 Extending Interaction Budget

Beyond the coverage of training data (Sec. 2.2), the demands of the tasks themselves limit performance at $1\times$, leaving four benchmarks far from solved even with ICL: RoboDojo, Bench2Dex, BEHAVIOR, and LHPR-VLN. RoboDojo and Bench2Dex require precise bimanual or dexterous manipulation, often involving repeated corrections. BEHAVIOR and LHPR-VLN require long sequences of household manipulation or navigation, with subgoals tracked across many stages, and have the lowest success rates of all settings.

To assess how much the standard step limit contributes to these shortfalls, we extend the budget to $10\times$ while keeping model weights and ICL packages fixed (Figs. 1 and 4). Additional interaction yields large gains on RoboDojo and Bench2Dex. On RoboDojo, success roughly triples, and about three quarters of the episodes unfinished at $1\times$ are completed, with the Precision group gaining most. Success on Bench2Dex more than doubles. These gains indicate that the standard budget can interrupt productive execution and recovery, while additional interaction allows GPT-6-Astra to complete further corrective attempts.

On LHPR-VLN, however, the extended budget yields only a modest gain, and success remains the lowest of all settings. This suggests that the demands of long-horizon navigation, including tracking subgoals across many stages, are not resolved by additional interaction alone (Sec. 2.4). We do not extend the budget on BEHAVIOR, whose standard budget is already long.

Benchmarks that already perform well at $1\times$ also benefit from additional interaction. DexJoCo and VLN-CE reach success rates of around 90%, while the LIBERO family and SIMPLE exceed 95% and approach saturation (Fig. 1).

2.4 Limitations and Failure Modes

Despite the substantial gains from ICL and the extended budget, several limitations and recurring failure modes remain. We first discuss two limitations of the approach, transfer and cost, and then three behavioral failure modes identified from episode videos and agent logs (Fig. 6). Infrastructure interruptions remain scored as failures but are considered

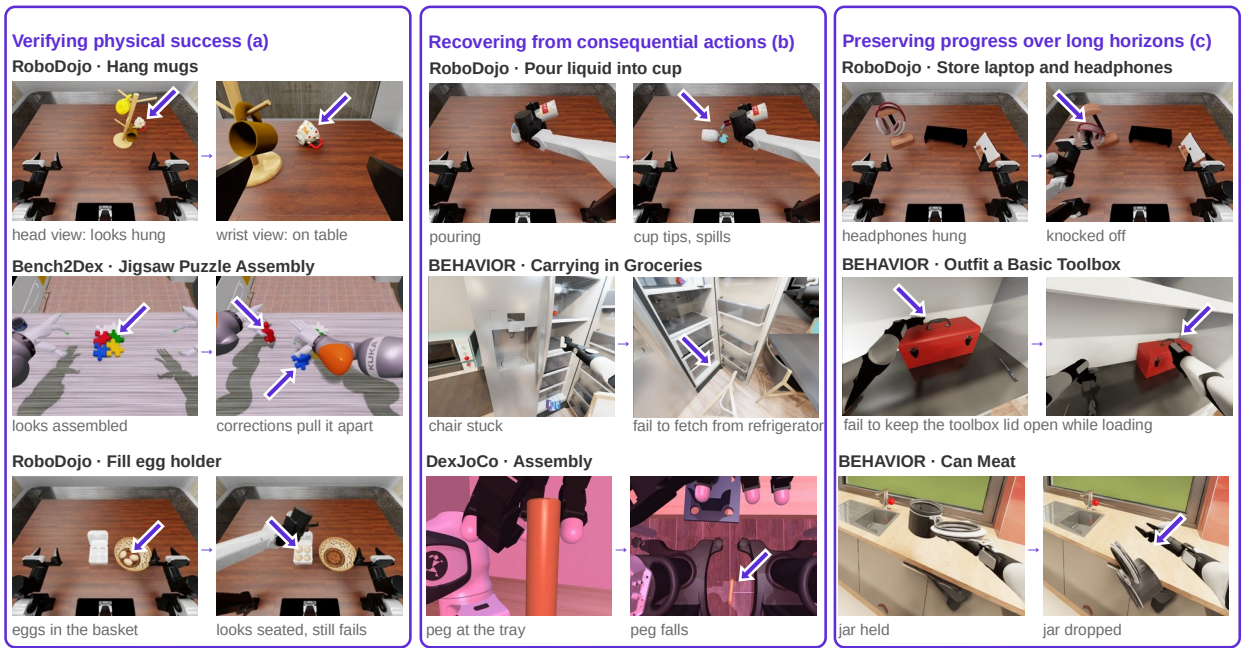


Figure 6 Residual failures under ICL with the extended budget. Each pair shows the state before a key event and its outcome; purple arrows mark the relevant objects. *(a) Verifying physical success:* apparent or self-reported completion diverges from the actual task state. *(b) Recovering from consequential actions:* an early mishap, such as a spill or a dropped part, cannot be undone and derails later subtasks. *(c) Preserving progress over long horizons:* progress that later steps depend on, such as an open lid or a placed object, is lost and blocks the task.

separately from the behavioral analysis.

ICL packages are largely task-specific, which limits transfer. Although ICL packages substantially improve performance across benchmarks, they are built from each benchmark's training data and mostly encode task-level guidance, motion references, and helpers, so their benefit weakens when test tasks move away from the data they were built from. In the RoboDojo Open group, which has no task-specific demonstrations and relies on a package studied from the other four groups, GPT-6-Astra + ICL still benefits from additional interaction but attains the lowest success among all groups (Appendix C.3, Table 5). Similarly, RoboLab provides no training set from which a package can be built; GPT-6-Astra leads the listed baselines there, yet its absolute success remains moderate (Table 6). Building packages that generalize beyond their source tasks remains an open direction.

Time and cost remain high. Strong performance comes at a substantial cost in tokens and wall-clock time, even on benchmarks that GPT-6-Astra nearly solves (Fig. 5). On LIBERO-Long++, where success at the standard budget already exceeds 97%, each episode still consumes about a million tokens and nearly an hour of inference and tool use; on LIBERO-Ultra-Long, these costs rise to roughly nine million tokens and nearly two hours (Table 4). Extending the budget increases the cost further, and much of the additional cost is concentrated in difficult episodes that often still fail: on RoboDojo, a single episode under the extended budget takes more than three hours on average. Since the $10\times$ setting only caps the number of action steps and episodes may end earlier upon success or terminal failure, these costs reflect the steps actually executed. Because inference latency is not counted in the action-step budget, this cost also limits deployment in settings that require real-time control.

Verifying physical success. GPT-6-Astra can detect that a task is not yet complete without identifying which physical condition remains unmet; consequently, its correction process fails to converge (Fig. 6(a)). In Bench2Dex Jigsaw Puzzle Assembly, it reports that all four pieces form the required rectangle while noting that the episode has not ended, and continues to adjust the residual offsets, but its corrections pull the pieces apart again instead of closing the remaining gaps. In RoboDojo Hang mugs, GPT-6-Astra takes an overlap between a mug's handle and the peg in its head camera as evidence that the mug is hung, without confirming that the peg passes through the handle; the mug ends up on the table,

and in its later regrasps the closing of its fingers is likewise taken as contact that has not occurred. In RoboDojo Fill egg holder, GPT-6-Astra moves all four eggs from the basket into the holder and reports that they remain seated after it withdraws, yet the episode ends as a failure with only partial credit. Visual plausibility and self-reported progress can therefore diverge from the task's actual completion conditions, sustaining repeated correction or premature confidence. Reliable execution requires checks that resolve the task's actual contact, support, and alignment constraints.

Recovering from consequential actions. Each attempt changes the state from which recovery must proceed, and some changes cannot be undone (Fig. 6(b)). In RoboDojo Pour liquid into cup, the cup tips over early in the episode and the liquid spills onto the table; GPT-6-Astra rights the cup and resumes pouring, but the spill cannot be reversed, and the episode ends with most of its budget unused. In DexJoCo Assembly, a peg slips from the grasp and falls to the floor; GPT-6-Astra judges it unreachable and parks the arms with most of the budget remaining. Recovery is also costly when the failed step is a prerequisite for later ones, such as opening a container or securing an object: until it succeeds, downstream subtasks cannot proceed, and repeated retries consume the budget without advancing the task. Such dependencies are especially prevalent in long-horizon benchmarks like BEHAVIOR and LHPR-VLN, where a single unresolved subtask can stall progress toward the overall goal. In LHPR-VLN, for example, an agent may become stuck trying to reach an intermediate destination, lose track of its navigation plan, and resort to aimless exploration without making meaningful progress. Additional budget therefore helps only when a recovery path still exists, and successful recovery usually involves a meaningful change of strategy rather than repetition. In DexJoCo Fold Glasses, for example, GPT-6-Astra switches to a side approach, verifies the grasp with a test lift, and completes the task after exceeding the standard budget.

Preserving progress over long horizons. Long-horizon tasks require earlier progress to remain intact while later actions are executed, and small deviations compound across stages (Fig. 6(c)). In RoboDojo Store laptop and headphones, GPT-6-Astra has already stored the laptop and hung the headphones on their stand, then returns to adjust one earcup; the correction pulls the headband off the stand, and the headphones are not re-hung before the budget runs out. In BEHAVIOR can_meat, a jar that is already held beside the counter tilts during the next motion and falls to the floor, so progress has to be rebuilt before the task can continue. Beyond such direct disruptions, small errors in grasping, placement, or navigation accumulate over successive stages and alter the conditions for later actions, forcing GPT-6-Astra to re-localize and correct earlier mistakes. Reliable long-horizon execution therefore requires tracking and rechecking conditions established earlier, rather than treating completed subgoals as permanently resolved.

These observations motivate task-relevant verification, recovery strategies that account for irreversible consequences and prerequisite dependencies, and persistent tracking of earlier progress. They suggest that further progress depends on how effectively the agent uses additional interaction to revise its strategy and verify the resulting state.

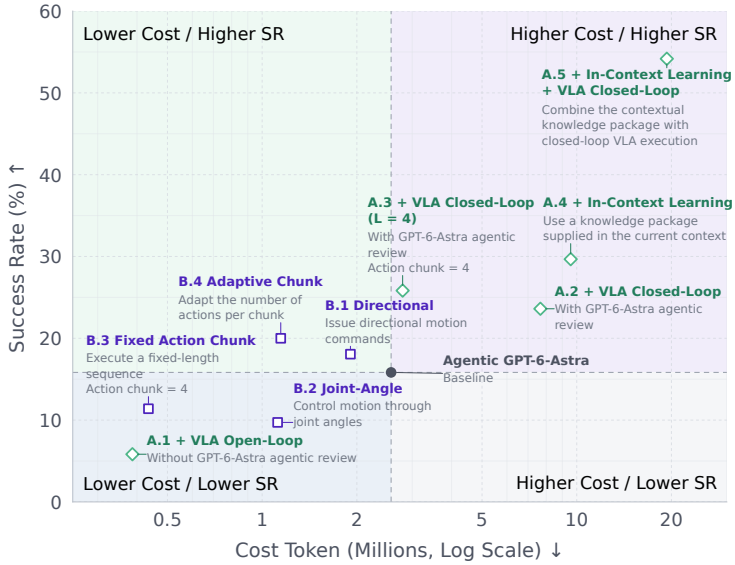
3 Harness Ablation

Section 2.2 shows that additional interaction can substantially improve task completion, but its benefits depend on how effectively the agent uses that interaction. Longer execution incurs substantial inference costs and does not reliably resolve failures in physical verification, recovery, or preservation of subtask progress. These findings motivate examining the harness that connects GPT-6-Astra's reasoning to observations, actions, and feedback.

We therefore conduct controlled ablations to investigate which harness designs improve task performance and at what inference cost. We study two aspects: System 1–System 2 collaboration and action representations and chunking.

3.1 System 1–System 2 Collaboration

We use GPT-6-Astra as System 2 and $\pi_{0.5}$ [12] fine-tuned on RoboDojo as System 1 (VLA). We compare three interaction settings: (i) standalone VLA, which executes actions without GPT supervision; (ii) open-loop collaboration, where GPT specifies subgoals and the VLA directly executes joint-action sequences; and (iii) closed-loop collaboration, where the VLA proposes actions that are converted into candidate EEF motions through forward kinematics, allowing GPT to accept, modify, or reject them before execution. For closed-loop collaboration, we vary the review horizon $L \in \{1, 4\}$, where L denotes the number of VLA-proposed EEF actions reviewed per GPT call. Finally, we evaluate in-context learning (ICL) following the setup described in Section 2.2, both for the direct-control agent and in combination with closed-loop VLA collaboration ($L = 2$).



Setting	SR (%) ↑	Tokens (M) ↓
A. System 1–System 2 Collaboration		
Baseline	15.56	2.560
VLA ($\pi_{0.5}$)	6.91	–
A.1 + VLA Open-loop	5.83	0.387
A.2 + VLA Closed-loop	23.61	7.664
A.3 + VLA Closed-loop ($L = 4$)	25.83	2.796
A.4 + ICL	29.67	9.567
A.5 + ICL & VLA Closed-loop	**54.17**	19.335
B. Action Representation and Chunking		
Baseline	15.56	2.560
B.1 Directional motion commands	18.06	1.908
B.2 Joint-angle	9.72	1.120
B.3 fixed action chunk ($L = 4$)	11.39	0.435
B.4 adaptive action chunk	20.00	1.146

Figure 7 Token cost versus performance on RoboDojo for different System 1–System 2 collaboration strategies and action interfaces. Each point represents an experimental configuration, with token cost measured in millions and performance measured by the unweighted mean of category success rate. Dashed lines intersect at the Agentic GPT-6-Astra baseline reference point, defining four shaded quadrants.

As shown in Figure 7, the standalone VLA achieves an average success rate of 6.91%, indicating limited performance on this benchmark. Open-loop collaboration similarly achieves 5.83% while using 0.387 million GPT-6-Astra tokens. In contrast, closed-loop collaboration, where System 2 reviews and refines VLA-proposed EEF actions, improves the success rate by up to 20.00 percentage points over open-loop execution. However, per-action review requires 7.664 million tokens. Increasing the review horizon to $L = 4$ reduces token consumption by 63.5% to 2.796 million tokens, while improving the success rate from 23.61% to 25.83%. Combining ICL with closed-loop VLA achieves the highest success rate of 54.17%, exceeding the best non-ICL closed-loop configuration by 28.34 percentage points, but consumes 19.335 million tokens, approximately 6.9 times as many as that configuration.

These findings suggest that delegating execution to System 1 alone is insufficient for reliable task completion, while frequent System 2 supervision incurs substantial token consumption. Reviewing longer action sequences improves both performance and token efficiency relative to per-action review, highlighting the importance of the collaboration protocol. The configuration combining ICL with closed-loop VLA achieves substantially higher performance, although at a much larger token cost. **A key direction is therefore to make verification and recovery more selective and to incorporate task-relevant experience more efficiently, balancing reliable execution with token consumption.**

3.2 Action Representation and Chunking

We study action representations and execution horizons under the same direct-control framework. Our baseline follows the RoboRSI [119] agent framework, where GPT-6-Astra observes RGB images and robot states and generates one Cartesian end-effector (EEF) action per decision. For action representations, we compare absolute Cartesian EEF targets (baseline), directional motion commands that specify relative movement directions and distances, and direct joint-angle control. For execution horizons, we compare single-action execution ($L = 1$), fixed action chunks ($L = 4$), and adaptive chunking, where GPT-6-Astra selects $L \in \{1, 2, 3, 4\}$. Here, L denotes the number of EEF actions generated per GPT decision, and token consumption includes both input and output tokens.

As shown in Figure 7, directional motion commands improve the average success rate by 2.50 percentage points, whereas

	Embodiment	#	Main question	Tasks
	Single arm, gripper Franka 3, Franka Panda, PiPER + Pika gripper	**3**	Do simulation results transfer, from rigid to deformable and precise tasks?	Stacking and container nesting; cloth and towel folding; plug insertion
	Dual arm, grippers AgileX Piper	**1**	Can GPT-6-Astra coordinate two arms on harder zero-shot tasks?	Ordered folding; flower insertion; bimanual holding; sorting
	Dual arm, dexterous hands Tianji M6S	**1**	Can GPT-6-Astra control high-DoF hands?	Toy and bottle grasping
	Wheeled mobile manipulator Galaxea R1 Pro, AgileX Tracer + xArm7	**2**	Can GPT-6-Astra act over room-scale distances and long horizons?	Find washer and press button; flower collection; tray arrangement; navigation and search
	Humanoid Unitree G1	**1**	Can GPT-6-Astra act as System 2 over low-level locomotion and arm control?	Tabletop pick-and-drop; turn, approach and stop

Table 1 Real-world platforms, ordered from simple to complex embodiments. Each class probes a different question; settings and per-task results are in App. E.

direct joint-angle control leads to a 5.84-percentage-point drop. This suggests a potential representation gap between the embodied intelligence of multimodal models and low-level robot control spaces. Action chunking further reveals a trade-off between success rate and token consumption. Fixed action chunking ($L = 4$) uses 0.435 million tokens but decreases the average success rate by 4.17 percentage points relative to the baseline. In contrast, adaptive action chunking improves the average success rate by 4.44 percentage points while using 1.146 million tokens. Thus, adaptive chunking achieves a higher success rate than fixed chunking at a higher token cost. These results indicate that both action representation and execution horizon influence the success rate and token consumption of embodied control.

These findings suggest that effectively translating embodied intelligence into physical actions requires interfaces aligned with the model's reasoning capabilities [28]. **A promising direction is to expose semantically meaningful action commands while delegating precise motion execution to robot controllers, with adaptive execution horizons to further balance task performance and token consumption.**

4 Real-World Validation

Since the simulation protocol of Sec. 2.1 rests on two assumptions that do not hold on physical robots, namely that successful demonstrations of the test task are available and that additional execution steps are free, we further study a real-world variant of the protocol that revises both. First, the context changes: since new tasks often come without demonstrations in real-world deployment, GPT-6-Astra operates in a zero-shot setting and receives only structured information provided by a human as context. Second, the budget changes: the step limit is replaced by wall-clock time, because on real hardware the time a task takes determines not only token cost but also safety, hardware health and human labor.

4.1 Setting: The Real-World Variant

We deploy GPT-6-Astra on eight platforms spanning five embodiment classes, ordered from simple to complex in Table 1. The real-world variant departs from the simulation protocol in three respects; per-platform settings, context and results are summarized in Table 14 and detailed in App. E.

(1) Zero-shot. GPT-6-Astra receives no demonstration and no knowledge package distilled from data, and performs as a generalist, without task- or robot-specific training.

(2) Structured human priors as context. In place of the knowledge package, GPT-6-Astra receives context written by a human. This context may encode information that is routinely available when a robot is deployed, such as tool schemas, joint limits, camera placement, safety constraints and open/close presets for dexterous hands, and, on some

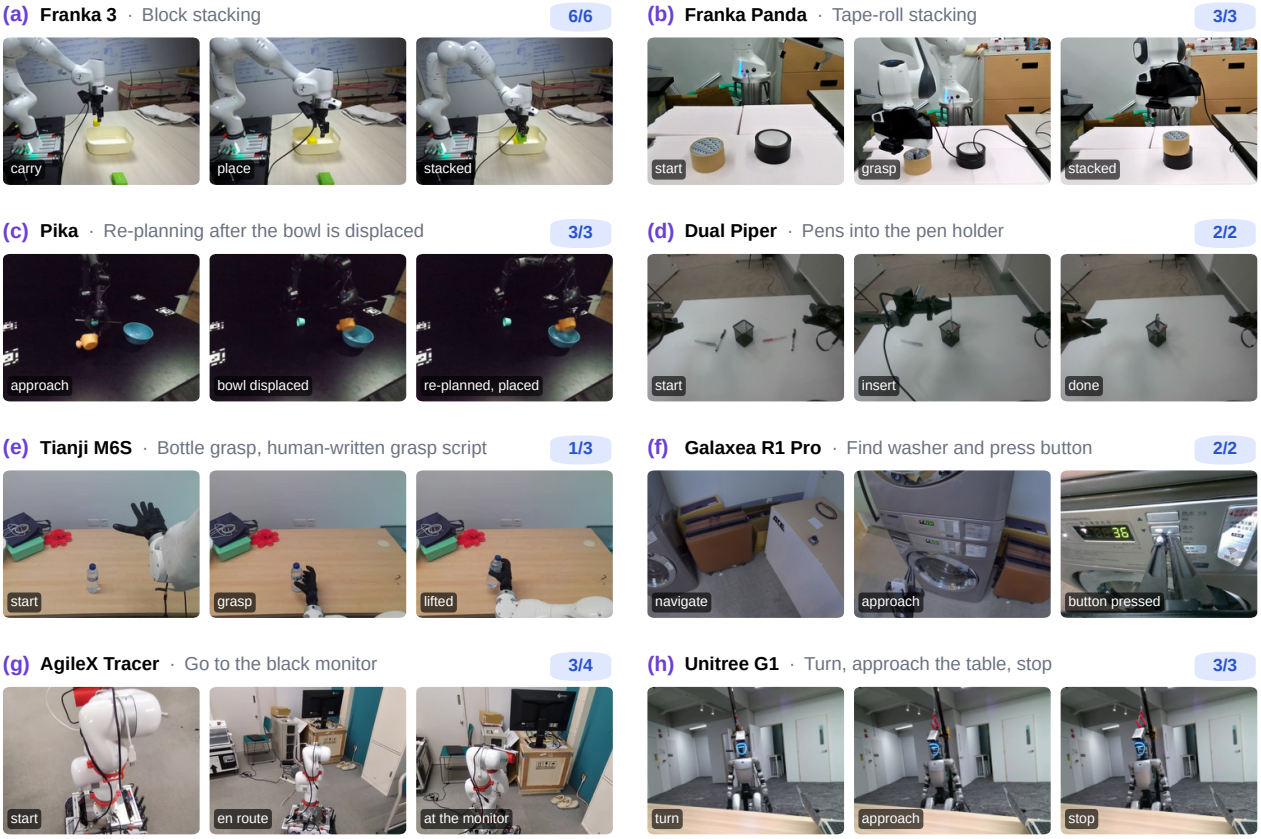


Figure 8 Zero-shot real-world control, from simple to complex embodiments. Frames from released videos in temporal order. Chips give successes over all counted trials. (c) GPT-6-Astra re-plans after contact displaces the receiving bowl. (e) Dexterous grasping with the Wuji hands of the Tianji M6S, given human-written grasp scripts (the successful bottle run; the toy grasp succeeds in 3/3). The full gallery is in Fig. 13, and all tasks per platform in App. E.

platforms, task-level guidance such as admissible grasp points. No platform exposes high-level semantic skills such as pick, place or insert.

(3) Wall-clock budget and success criteria. Trials are budgeted in wall-clock time rather than in steps, normally 60 minutes; per-platform exceptions are listed in App. E. An operator holds an emergency stop throughout, and a person judges success from the physical outcome rather than from the model's own completion statement. For deformable tasks such as cloth folding, a single success criterion is hard to define across platforms, so their success rates indicate that the task was carried through rather than a common level of quality.

4.2 Findings

Finding 1: one model drives every embodiment; success depends on the task. Without platform-specific training, the same model controls all eight platforms, from single arms to a humanoid (Fig. 8). What limits success is the nature of the task: rigid pick-and-place is reliable (17/18 trials) and matches simulation, whereas deformable objects (e.g., cloth folding: 2/4 on the Franka 3), precise contact (e.g., plug insertion: 0/11 across three platforms) and long-horizon mobile tasks (e.g., navigation and search: 9/23 on the AgileX Tracer) remain markedly harder. A second arm extends the range of zero-shot tasks GPT-6-Astra completes (Fig. 13), and GPT-6-Astra re-plans from execution feedback when the scene changes (Fig. 8c). Per-platform results are in App. E.

Finding 2: what GPT-6-Astra can do depends on what it can see. Beyond the prompt and the available primitives, real-world success depends critically on the view configuration, that is, on whether the cameras available to the model can observe the region in which the task is decided. The flower insertion task illustrates this dependence. The gripper

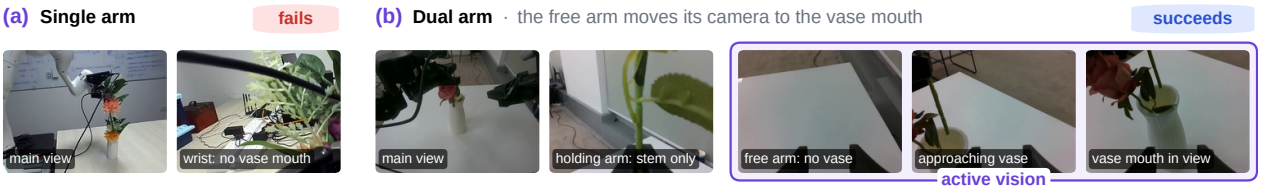


Figure 9 What GPT-6-Astra can do depends on what it can see. Flower insertion, main view and wrist views. (a) Single arm (Franka 3): the gripper holds the stem vertically, so its wrist camera cannot see the vase mouth, and the insertion fails. (b) Dual-arm Piper: the wrist view of the holding arm shows only the stem; GPT-6-Astra moves the free arm until its wrist camera sees the vase mouth (boxed frames: no vase, approaching the vase, vase mouth in view), and the insertion succeeds.

holds the stem vertically from above, so the wrist camera, which moves with the gripper, cannot see the mouth of the vase below the flower. On a single arm (Franka 3), the wrist camera is the only close-up view: GPT-6-Astra never sees the vase mouth during the insertion, and the attempt fails, with the flower dropped beside the vase (Fig. 9a). On the dual-arm Piper, the wrist view of the arm holding the flower is limited in the same way, but GPT-6-Astra moves the free arm until its wrist camera looks at the vase mouth, and then completes the insertion (Fig. 9b). The difference is active vision: the viewpoint becomes part of the action space. This behavior is largely absent from teleoperated demonstrations: a human operator typically watches the scene from their own vantage point rather than through the robot's cameras, so demonstrations rarely move a camera to resolve an occlusion, and policies learned from them have little to imitate. A zero-shot agent that perceives only through the robot's cameras has to discover it.

Finding 3: GPT-6-Astra as System 2 over existing System 0 motor control. On a humanoid, GPT-6-Astra deliberates far more slowly than local motor control runs, so we test whether it can act directly through existing System 0 controllers, without a task-trained System 1 policy. In SIMPLE, GPT-6-Astra issues joint, body and locomotion targets that decoupled WBC or AMO [91] controllers execute. On a real Unitree G1 with Wuji hands, it commands arm-joint targets and binary hand open/close states for manipulation (2/3), and turns and walks through the vendor velocity interface (3/3); the two tasks are evaluated separately (Fig. 8h and Fig. 13, B3). This extends the System 2/System 1 coordination of Sec. 3 down to existing motor-control interfaces, but it establishes feasibility under slow, feedback-driven supervision rather than reliable real-time visuomotor control.

4.3 What Changes in the Real World

The two ingredients behind the simulation results in Sec. 2 are each constrained on real-world setting.

Physical constraints limit extended execution. Extending the step limit by up to 10× works in simulation because additional steps and failed attempts carry no consequence; the physical world imposes three constraints that simulation does not model. (i) Safety. Contacts beyond the controller's limits trigger protective stops that usually end the trial, for instance when the gripper touches the table while folding thin cloth on the Franka 3, or when a plug-insertion attempt on the PiPER arm ends in a joint collision fault (2 of its 3 trials), and GPT-6-Astra cannot recover from such hardware-level faults. Some outcomes, such as a dropped object, are irreversible. (ii) Mechanical limits. GPT-6-Astra has no direct feedback on the load its commands impose: on the Tianji M6S, finger joints of the Wuji hand stalled under load during repeated grasp attempts, and its only guard was a temperature threshold it had set for itself, which the hand reached (62 °C), interrupting the attempt. Actuators overheating can terminate prolonged trials: a torso over-temperature fault ended both flower-collection runs on the Galaxea R1 Pro. (iii) Real time. In simulation the environment pauses while the model deliberates. On a physical robot, elapsed time is the quantity through which every other cost accumulates: tokens, exposure to safety risks, hardware stress and human supervision. Episodes are long: on the G1, GPT-6-Astra spends about 90% of each run deliberating, and a single manipulation trial takes over an hour. The trade-off between longer execution and higher success that GPT-6-Astra exploits in simulation is therefore not freely available on real hardware (App. E).

Missing demonstrations reshape in-context learning. The ICL protocol of Sec. 2.1 presumes that successful demonstrations of the test task, on the test robot, already exist. In real-world deployment new tasks often come without them. Human-written context suffices for a meaningful range of tasks, but it is still prompt engineering by an expert and its quality directly affects the outcome: on the Tianji hands, without predefined grasp primitives GPT-6-Astra did not

complete a single grasp (0/3), whereas with human-written grasp primitives it succeeded in 3/3. This context also need not come from a person: on the Galaxea R1 Pro, GPT-6-Astra wrote its own robot profile in a motion-free familiarization session, from vendor documentation and live probes of sensors and controllers, and every task session starts from it. Human-written context represents only one point in a broader design space. Between a complete set of in-domain demonstrations and none at all, a range of alternative context sources may be available at deployment time, which we have not yet evaluated: cross-embodiment demonstrations, self-generated demonstrations in simulation, human demonstrations, the robot's own deployment history, and knowledge packages shared across robots. The principled construction of context for embodied tasks from such heterogeneous sources remains an open problem, and constitutes a substantial research opportunity for in-context learning in real-world deployment.

5 Towards a Next-Generation Embodied Benchmark

The preceding sections show that GPT-6-Astra's results on established embodied benchmarks depend heavily on conditions that the benchmarks grant. Its success rises with in-context access to the benchmark's demonstrations and with an extended step budget in which failures cost nothing (Sec. 2), and depends on how a general agent is adapted to an action interface built for a learned policy (Sec. 3). These conditions follow from how most embodied benchmarks were designed, for policies trained by imitation: demonstrations of a task are collected, a policy is fitted to them, and it is evaluated on held-out episodes of the same task [8, 26, 99, 112]. On real robots, by contrast, neither demonstrations of a new task nor extra time comes for free (Sec. 4). For a general-purpose agent, such benchmarks therefore measure in part how well it exploits what they provide, and raising their difficulty, the usual response once benchmarks saturate [20, 80], would not change this. We instead propose three principles that change how embodied benchmarks are designed and judged, and illustrate them with a household proof of concept.

5.1 Principles for Next-Generation Benchmarks

Following the protocol now common for language agents, in which an agent acts in an environment by any means and is scored on the outcome [74, 110, 130, 155], we target a setting in which the agent perceives the world only through its robot's sensors, acts through the commands its hardware exposes, and is evaluated on a single deployment in a world that keeps evolving; everything it builds above that interface is part of the system under test. Current embodied benchmarks judge success as a planning domain language would: BDDL [135], inspired by PDDL [50], states a task as a goal condition over the scene's objects that must hold in a single state (Fig. 10, left). Such goals inherit the assumptions of classical planning [51, 126]: a fully specified goal with binary success, a world that changes only through the agent's actions, time counted in steps, and a given world model. Extensions of PDDL relax some of these [45, 49, 59], but embodied benchmarks have rarely adopted them, and each principle below breaks at least one.

Tasks derived from human needs. Benchmark tasks are usually ones on which a robot can be shown to succeed, because demonstrations must be collected for them; even generalist policies that operate zero-shot in unseen homes [12] are evaluated on tasks chosen from the robot's perspective. A deployed robot must instead do what the people it serves need done, whether or not any robot has done it before. Following BEHAVIOR-1K [90], we propose that tasks be derived from such needs and stated as the person would state them. Such tasks also test whether the agent knows the limits of its embodiment: a humanoid may fetch a step stool to reach a ceiling bulb, whereas a wheeled base should ask for help. Real needs also carry obligations that no instruction mentions, such as an unattended stove, a running tap or a cabinet the agent itself left open. These have been studied in narrower forms, as tidying without instructions [76, 127], mitigating risks the agent itself creates [106] and rescuing objects from a spreading hazard [182]; in deployment they also arise while the robot is busy with something else, and the agent must notice them unprompted and decide whether to interrupt its work. We therefore place these obligations (Duty) in the same deployment as the explicit request (Intent), with deadlines on the world clock.

A request stated as a need cannot be judged as a goal. "Make breakfast" resembles a high-level action in hierarchical planning [40, 126]: it admits many refinements, from a slice of toast to a cooked meal, of which a goal formula keeps only the one its author chose (Fig. 10). Hierarchical languages accept any decomposition of the task into an executable plan [59], yet the refinements serve the person to different degrees, which calls for a utility over outcomes [77] rather than the fraction of satisfied goal predicates reported as partial credit [8, 135]. A fixed check can even contradict its own instruction: Bench2Dex's condiment-box-loading task [163] asks the agent to place three condiments into a box, but

```
(define (problem cook_bacon-0)
  (:domain behavior-1k)
  (:objects
    bacon.n.01_1 ... bacon.n.01_6 - bacon.n.01
    tray.n.01_1 - tray.n.01
    electric_refrigerator.n.01_1 - ...
    stove.n.01_1 - stove.n.01
    frying_pan.n.01_1 - frying_pan.n.01 ...)
  (:init
    (ontop bacon.n.01_1 tray.n.01_1) ...
    (not (cooked bacon.n.01_1)) ...
    (inside tray.n.01_1
            electric_refrigerator.n.01_1)
    (ontop frying_pan.n.01_1 stove.n.01_1) ...)
  (:goal
    (and
      (forall
        (?bacon.n.01 - bacon.n.01)
        (cooked ?bacon.n.01))
      (not
        (open ?electric_refrigerator.n.01_1)))))
```

```
(define (request make_breakfast)
  (:say "Could you make me breakfast?")
  (:context (resident-leaves-at 15 min))
  (:clock world)  ; deliberation advances it
  (:observe until (+ done 5 min))
  (:acceptable              ; any one, graded
    (toast)                         0.5
    (egg toast)                     0.8
    (egg toast coffee)              1.0)
  (:value  1 if served by 15 min,
           0 after 30 min, linear between)
  (:hidden                  ; never shown to agent
    (cooked x) = (core-temp x >= 70 C)
    (burnt x)  = (surface-temp x >= 200 C))
  (:never  (on_fire) (touches robot person))
  (:whenever (open fridge)
     -> (closed fridge) within 1 min)
  (:whenever (on burner) (empty burner)
     -> (off burner) within 2 min)
  (:whenever (water floor)
     -> (dry floor) within 5 min))
```

Figure 10 Two specifications of a breakfast task. Left: the BEHAVIOR-1K activity `cook_bacon` [8], with its complete goal. The goal is tested against the current state at every step, and the episode succeeds at the first step at which it holds. Nothing requires the stove to be turned off, and since `cooked` holds once an object's maximum temperature has passed a threshold, it remains true for bacon that has burned. Right: what a judge for the request "make breakfast" needs to express. Any acceptable refinement earns credit, graded by what is served and by when, with a value that decays once the resident has left [56]; items count as served only if they never burned. Predicates are defined on physical quantities known only to the judge. The `:never` and `:whenever` clauses are the absence [38] and bounded-response [84] specification patterns, expressible in metric [85] and signal [108] temporal logic and checked over the whole trajectory on the world clock, which keeps running after the agent declares that it is done. All constants are illustrative.

its success check also requires the soy-sauce and vinegar bottles to stand upright, which the instruction never states. Because a capable agent may also satisfy a request in ways a fixed check did not anticipate, or exploit a check weaker than the request [86], the judge should itself be validated against how people rank alternative outcomes [32, 179].

Experience acquired at test time. Demonstrations are often unavailable for new tasks in deployment, and they also bound what an agent learns: recorded from the operator's viewpoint, they rarely move a camera to resolve an occlusion, whereas an agent that understands its embodiment can make its viewpoint part of the action (Sec. 4). We therefore propose that the benchmark provide no in-domain training episodes or scene-specific demonstrations, so that the agent must ground its general priors in an unseen home and embodiment through its own exploration, mapping the layout, testing affordances and figuring out idiosyncratic appliances, and may find strategies that no demonstrator used. Planning languages presume the opposite, a given world model [126], and benchmark goals retain a trace of it: predicates such as `cooked` or `toggled_on` tell the agent which variables matter, whereas a real air fryer has only a winding timer and a toaster's dial yields a continuum of results. The judge should therefore define its predicates on physical quantities kept private from the agent (Fig. 10, right), and charge exploration the same world time and side effects as any other action. Since experience carries over between requests within a deployment and is reset between deployments, the unit of assessment is the deployment, and how quickly the agent becomes competent in it is a quantity of interest [7, 31].

A world that keeps running. An extended step budget raises GPT-6-Astra's success at no cost in simulation, because the simulator pauses while it deliberates: on RoboDojo, the standard budgets correspond to 8–76 s of commanded motion against an average of 55 minutes of computation (Sec. 2.3). On a physical robot that time passes in the world, and on the G1 GPT-6-Astra spends about 90% of each run deliberating (Sec. 4.3). Following recent evaluations in which the environment keeps evolving while the agent reasons [153, 180], we propose that the world not wait for the agent: the time between an observation and the next command, whether spent on inference, tool use or communication, is simulated under the previous command, as on a physical robot [14], while the host's simulation and rendering time is not charged. The judge must then observe the whole trajectory on the world clock, including some time after the agent declares that it is done, since a final-state goal cannot see an egg that was ready and then burned. Obligations become constraints on the trajectory, conditions to be avoided throughout or restored within a deadline once triggered [38, 84]; PDDL3 states such constraints for plans [49], and PARTNR and IS-Bench check ordering and triggered safety conditions

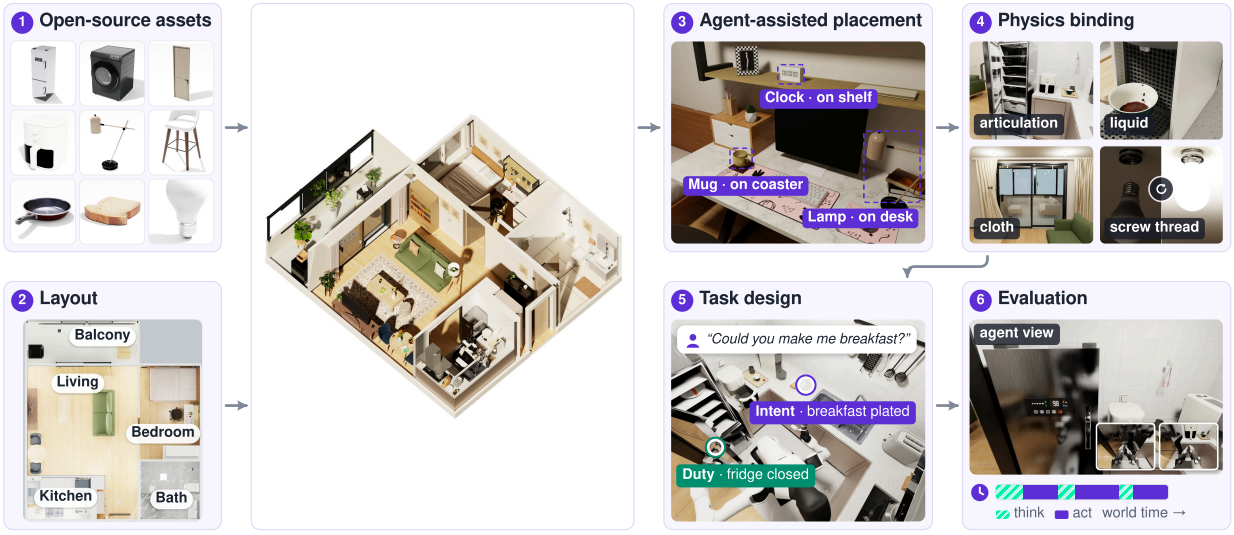


Figure 11 Building and using the household environment. Publicly available assets (1) and a floor plan (2) are composed into an interactive home. Coding agents place everyday items at real-world scale on named supports under human review (3), and every part is bound to physics: articulation, liquids, cloth and contact (4). A task states only a need (5): asked to make breakfast, the agent must work out the fridge, air fryer and toaster by itself, and is to be judged both on the request (Intent) and on obligations no one stated, such as closing the fridge (Duty). The agent sees only through the robot's onboard cameras, and its deliberation advances world time (6).

in embodied tasks [21, 106], but without deadlines. Time also sets the value of an outcome: a resident leaving in fifteen minutes is better served by eggs and toast in ten than by a full breakfast in an hour. The judge should therefore assign a utility that depends jointly on what was achieved and when [56] (Fig. 10, right), which makes the agent's thinking time part of its performance.

5.2 A Household Proof of Concept

We ground these principles in the household. Requests there are open-ended, chores chain many steps across rooms and appliances, much of the work is contact-rich, and the home keeps changing while the robot works, which are the conditions under which GPT-6-Astra still falls short (Secs. 2 and 4). We built a fully interactive one-bedroom apartment in NVIDIA Isaac Sim [114] through a reproducible pipeline (Fig. 11): publicly available assets [75, 90, 130, 167] and a floor plan form the home, coding agents furnish it at real-world scale under human review, and every part is bound to physics, including articulation, contact, position-based fluids and cloth [107, 111], and the heating and cooking processes of everyday appliances. Because judged outcomes are only as meaningful as the physics behind them, the environment is built for causal fidelity: whatever an agent's decisions depend on, from an object's mass and friction to heat flowing from burner to pan to food, should be simulated with the right cause and effect and grounded in real-world references.

Tasks in the apartment state only a need, such as making breakfast before the resident leaves (Fig. 11, 5–6). We ran GPT-6-Astra on such tasks in the zero-shot setting of Sec. 4: it controls an AgiBot G2 [167] through the robot's own interface, observes only the onboard cameras and proprioception, and receives no demonstrations, only a description of the embodiment. The question is whether a model that comes close to saturating the benchmarks of Sec. 2 meets the demands of Sec. 5.1 once the benchmark supplies nothing. Fig. 12 shows five representative failures.

Finding 1: GPT-6-Astra attends to the object in hand and loses track of the rest of the home. Tasks derived from human needs ask the agent to serve the whole request, including obligations that no one states; GPT-6-Astra serves only the step in front of it. Tidying a desk, it moves the book in its gripper while its arm knocks other items to the floor and pushes the monitor out of place (Fig. 12a). Putting groceries away, it opens the refrigerator to store the milk, then turns to carrying fruit to the table and leaves the door open from 300 s to 765 s of world time (Fig. 12b). In both episodes, the rest of the home, including the robot's own arm and the states it has changed, seems to enter the model's decisions only

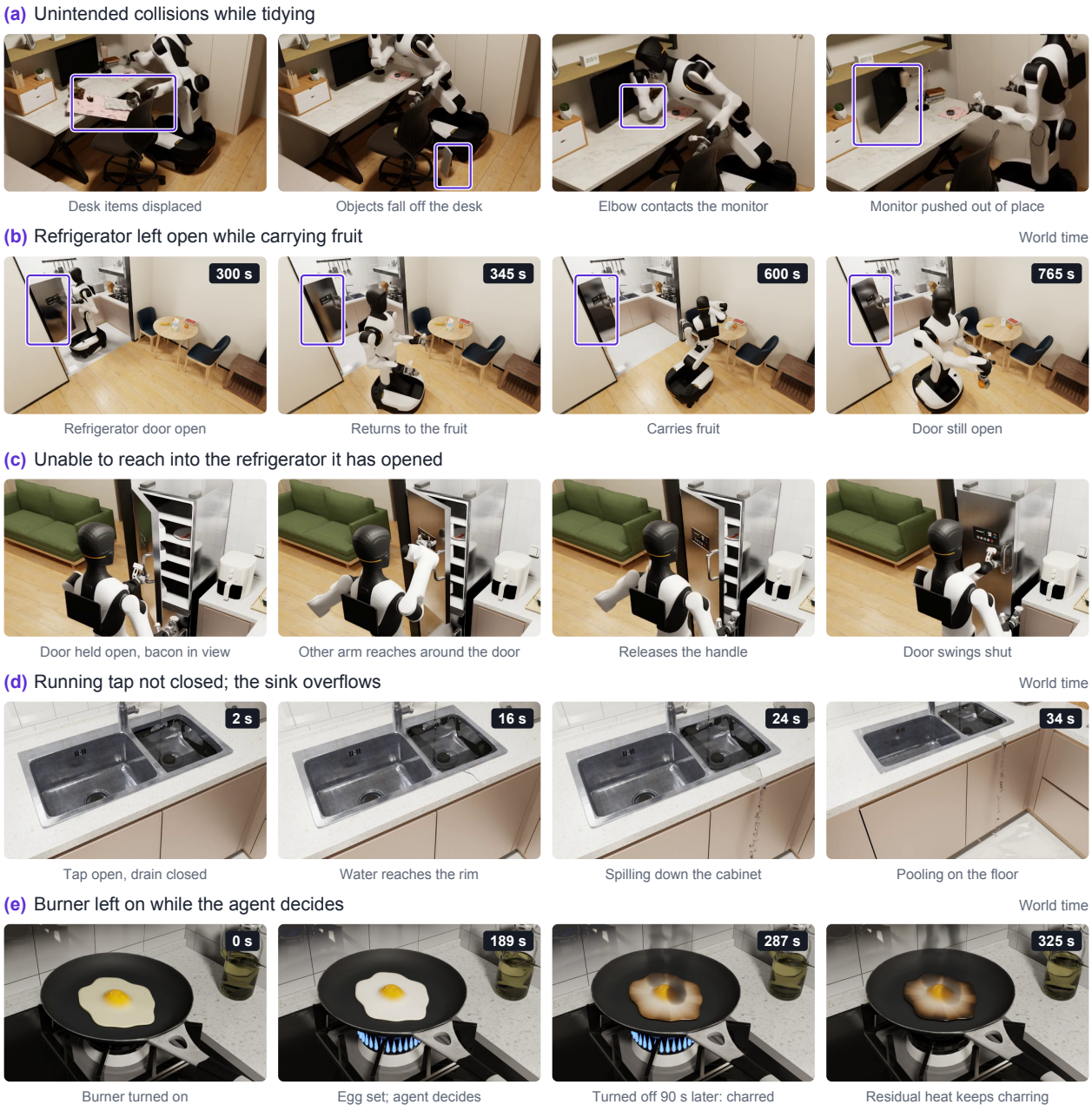


Figure 12 GPT-6-Astra in the household proof of concept. All runs are zero-shot, with contact-based grasping and no demonstrations. (a) While moving a book during tidying, the robot displaces desk items, knocks objects onto the floor and pushes the monitor out of place with its elbow. (b) Having opened the refrigerator to store the milk, the robot turns to carrying fruit to the table, and the door stays open from 300 s to 765 s of world time. (c) Making breakfast, the robot holds the refrigerator door open with one arm, but from where it stands the door blocks its other arm from the bacon inside; when it releases the handle to reach in, the door swings shut. (d) The running tap is not closed, and with the drain closed the sink fills until water spills down the cabinet onto the floor. (e) GPT-6-Astra watches an egg through the robot's head camera and decides to turn the burner off once it has set; deciding takes 8 s of world time with the burner still on, and by the time the burner is turned off, about 90 s after the egg has set, the egg has charred. Purple outlines highlight relevant objects; timestamps denote elapsed world time. External camera views are used for visualization; the agent sees only through the robot's onboard cameras.

when it blocks the next step, much as on the real robots of Sec. 4.3. A check on the final state would miss both: the refrigerator is closed by the end of the episode, and where the book ends up says nothing about the rest of the desk. The judge in Fig. 10 therefore tracks obligations over the whole trajectory rather than only at the end.

Finding 2: Without demonstrations, GPT-6-Astra does not work out how unfamiliar fixtures move or where its own body has to be. With no demonstration to follow, the agent has to discover both through its own interaction. Making breakfast, GPT-6-Astra has to take bacon from the refrigerator (Fig. 12c). It pulls the door open and holds it with one arm, but from where its base stands in the narrow aisle, the open door lies between its free arm and the shelf, and the other arm can only reach around it. Getting to the bacon would require placing the base so that the swinging door and the robot's own body both fit between the refrigerator and the counter behind it. GPT-6-Astra plans for the gripper and the handle but not for the rest of its body: it lets go of the handle to reach in, the door swings shut, and it is back where it started. Faced with an unfamiliar tap, it likewise keeps pushing the handle instead of turning it while the sink fills and overflows (Fig. 12d). In both cases it repeats and adjusts the same motion. On the benchmarks of Sec. 2, the demonstrations it distills in context show both, so this weakness does not appear there.

Finding 3: GPT-6-Astra bases decisions on observations that become outdated while it thinks. In a world that keeps running, an action must suit the state of the world when it takes effect. Watching an egg through the head camera, GPT-6-Astra correctly decides to turn the burner off once the egg has set, but each decision takes seconds of world time with the burner still on, and by the time the burner is off the egg has charred (Fig. 12e). The model reasons as though the world waits for it, which in the simulators of Sec. 2 it does.

These episodes are few and qualitative, but they point the same way. On the benchmarks of Sec. 2, GPT-6-Astra's ability to decide what to do is enough. In a home that states only needs, offers no demonstrations and does not pause, it must also act on that decision, and this is where it currently falls short.

5.3 Open Challenges

Realizing these principles in a full benchmark raises several open problems, which building our prototype made concrete. The first challenge is evidence for the judge. To score what happened rather than check a list of predicates, the judge needs physical facts, such as whether an egg was flipped or burned, which simulators still model with heuristic state changes [90]. Learned judges do not yet fill this gap: vision-language success detectors reach at most 60% balanced accuracy on contact-rich assembly [113], and vision-language judges of agent trajectories can be misled by the agent's own reasoning [78]. The second is simulation fidelity. Because the judge reads the simulator's state, a modeling error looks the same as agent behavior: a wrongly modeled joint can make an agent appear to succeed, and a contact artifact can make it appear to fail, and simulation has so far been shown to predict real-robot results for short tabletop tasks, including a few with deformable objects [70, 95, 173], but not for long-horizon household activities. The third is the cost of a world that does not pause. If the world keeps running while the agent thinks, every physical process must be simulated during inference, and scores then depend on latency, hardware and service load; existing benchmarks avoid this by pausing the environment [171], which removes inference speed from the score, or by advancing time per generated token [153], which is reproducible but does not reflect deployed latency. Finally, a deployment lasting an hour or more is expensive to repeat, while even short robot trials are typically run too few times to separate policies reliably [133].

6 Conclusion

In conclusion, GPT-6-Astra represents a clear leap in embodied intelligence, but not its endpoint. Our study shows that strong harness design with in-context learning and sufficient execution budget can push many isolated benchmarks close to saturation, while the remaining challenges shift toward transfer across tasks, long-horizon reasoning, failure recovery, and real-world constraints on time, cost, and safety. These findings also suggest that benchmark success should be interpreted with increasing care: the next frontier is not simply to solve more isolated tasks, but to build embodied systems that can operate reliably and efficiently under realistic deployment conditions.

References

[1] AgentCanvas. AgentCanvas: Visual Agent Design Platform for Embodied AI. Software framework, 2026. URL https://github.com/jianzhou0420/AgentCanvas.

[2] Peter Anderson, Angel Chang, Devendra Singh Chaplot, Alexey Dosovitskiy, Saurabh Gupta, Vladlen Koltun, Jana Kosecka, Jitendra Malik, Roozbeh Mottaghi, Manolis Savva, and Amir R. Zamir. On Evaluation of Embodied Navigation Agents. arXiv preprint arXiv:1807.06757, 2018. URL https://arxiv.org/abs/1807.06757.

[3] Peter Anderson, Qi Wu, Damien Teney, Jake Bruce, Mark Johnson, Niko Sünderhauf, Ian Reid, Stephen Gould, and Anton van den Hengel. Vision-and-Language Navigation: Interpreting Visually-Grounded Navigation Instructions in Real Environments. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 3674–3683, 2018. URL https://openaccess.thecvf.com/content_cvpr_2018/html/Anderson_Vision-and-Language_Navigation_Interpreting_CVPR_2018_paper.html.

[4] ARC Prize Foundation. ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. arXiv preprint arXiv:2603.24621, 2026. URL https://arxiv.org/abs/2603.24621.

[5] Artificial Analysis. Welcome to the Era of Embodied AI: Introducing AA-Robotics v0.1. Online research preview, 2026. URL https://artificialanalysis.ai/articles/aa-robotics.

[6] Awomo-WAM Team. Awomo-0.5: World action model pre-trained at scale. GitHub repository, 2026. URL https://github.com/Awomo-WestlakeDI/Awomo-0.5.

[7] Jakob Bauer, Kate Baumli, Feryal Behbahani, Avishkar Bhoopchand, Nathalie Bradley-Schmieg, Michael Chang, Natalie Clay, Adrian Collister, Vibhavari Dasagi, Lucy Gonzalez, Karol Gregor, Edward Hughes, Sheleem Kashem, Maria Loks-Thompson, Hannah Openshaw, Jack Parker-Holder, Shreya Pathak, Nicolas Perez-Nieves, Nemanja Rakicevic, Tim Rocktäschel, Yannick Schroecker, Satinder Singh, Jakub Sygnowski, Karl Tuyls, Sarah York, Alexander Zacherl, and Lei M Zhang. Human-Timescale Adaptation in an Open-Ended Task Space. In Proceedings of the 40th International Conference on Machine Learning, volume 202 of Proceedings of Machine Learning Research, pages 1887–1935. PMLR, 2023. URL https://proceedings.mlr.press/v202/bauer23a.html.

[8] BEHAVIOR Team. BEHAVIOR-1K 2026 Challenge Leaderboard. Hugging Face Space, 2026. URL https://huggingface.co/spaces/behavior-1k/2026-challenge-leaderboard.

[9] Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschannen, Emanuele Bugliarello, Thomas Unterthiner, Daniel Keysers, Skanda Koppula, Fangyu Liu, Adam Grycner, Alexey Gritsenko, Neil Houlsby, Manoj Kumar, Keran Rong, Julian Eisenschlos, Rishabh Kabra, Matthias Bauer, Matko Bošnjak, Xi Chen, Matthias Minderer, Paul Voigtlaender, Ioana Bica, Ivana Balazevic, Joan Puigcerver, Pinelopi Papalampidi, Olivier Henaff, Xi Xiong, Radu Soricut, Jeremiah Harmsen, and Xiaohua Zhai. PaliGemma: A versatile 3B VLM for transfer. arXiv preprint arXiv:2407.07726, 2024. URL https://arxiv.org/abs/2407.07726.

[10] Hongzhe Bi, Hengkai Tan, Shenghao Xie, Zeyuan Wang, Shuhe Huang, Haitian Liu, Ruowen Zhao, Yao Feng, Chendong Xiang, Yinze Rong, Hongyan Zhao, Hanyu Liu, Zhizhong Su, Lei Ma, Hang Su, and Jun Zhu. Motus: A Unified Latent Action World Model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 35101–35113, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Bi_Motus_A_Unified_Latent_Action_World_Model_CVPR_2026_paper.html.

[11] Hongzhe Bi, Lingxuan Wu, Tianwei Lin, Hengkai Tan, Zhizhong Su, Hang Su, and Jun Zhu. H-RDT: Human Manipulation Enhanced Bimanual Robotic Manipulation. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 40, pages 18135–18143, 2026. doi: 10.1609/aaai.v40i22.38875. URL https://ojs.aaai.org/index.php/AAAI/article/view/38875.

[12] Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, brian ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, and Ury Zhilinsky. $\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization. In Proceedings of the 9th Conference on Robot Learning, volume 305 of Proceedings of Machine Learning Research, pages 17–40. PMLR, 2025. URL https://proceedings.mlr.press/v305/black25a.html.

[13] Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, Laura Smith, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury

Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.010. URL https://www.roboticsproceedings.org/rss21/p010.html.

[14] Kevin Black, Manuel Galliker, and Sergey Levine. Real-Time Execution of Action Chunking Flow Policies. In Advances in Neural Information Processing Systems, volume 38, pages 33383–33407, 2025. doi: 10.52202/085713-1122. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/300ccb2187dedd4edcc07f7e76d8e553-Abstract-Conference.html.

[15] Black Forest Labs. FLUX 3 Action: A 7b world action model for robot control. Technical blog, 2026. URL https://bfl.ai/models/flux-3-action.

[16] Abdelaziz Bounhar, Abhijeet Somani, Aditi Kabra, Adrian Valente, Adrien Petralia, et al. Robostral Navigate. arXiv preprint arXiv:2607.20785, 2026.

[17] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S Ryoo, Grecia Salazar, Pannag R Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong Tran, Vincent Vanhoucke, Steve Vega, Quan H Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: Robotics Transformer for Real-World Control at Scale. In Proceedings of Robotics: Science and Systems, 2023. doi: 10.15607/RSS.2023.XIX.025. URL https://www.roboticsproceedings.org/rss19/p025.html.

[18] Rui Cai, Jun Guo, Xinze He, Piaopiao Jin, Jie Li, Bingxuan Lin, Futeng Liu, Wei Liu, Fei Ma, Kun Ma, Feng Qiu, Heng Qu, Yifei Su, Qiao Sun, Dong Wang, Donghao Wang, Yunhong Wang, Rujie Wu, Diyun Xiang, Yu Yang, Hangjun Ye, Yuan Zhang, and Quanyun Zhou. Xiaomi-Robotics-0: An Open-Sourced Vision-Language-Action Model with Real-Time Execution. arXiv preprint arXiv:2602.12684, 2026. URL https://arxiv.org/abs/2602.12684.

[19] Zhongang Cai, Yubo Wang, Qingping Sun, Ruisi Wang, Chenyang Gu, Wanqi Yin, Zhiqian Lin, Zhitao Yang, Chen Wei, Oscar Qian, Hui En Pang, Xuanke Shi, Kewang Deng, Xiaoyang Han, Zukai Chen, Jiaqi Li, Xiangyu Fan, Hanming Deng, Lewei Lu, Bo Li, Ziwei Liu, Quan Wang, Dahua Lin, and Lei Yang. Holistic Evaluation of Multimodal LLMs on Spatial Intelligence. arXiv preprint arXiv:2508.13142, 2025. URL https://arxiv.org/abs/2508.13142.

[20] Center for AI Safety, Scale AI, and HLE Contributors Consortium. A benchmark of expert-level academic questions to assess AI capabilities. Nature, 649:1139–1146, 2026. doi: 10.1038/s41586-025-09962-4. URL https://www.nature.com/articles/s41586-025-09962-4.

[21] Matthew Chang, Gunjan Chhablani, Alexander Clegg, Mikael Dallaire Cote, Ruta Desai, Michal Hlavac, Vladimir Karashchuk, Jacob Krantz, Roozbeh Mottaghi, Priyam Parashar, Siddharth Patki, Ishita Prasad, Xavier Puig, Akshara Rai, Ram Ramrakhya, Daniel Tran, Joanne Truong, John M. Turner, Eric Undersander, and Tsung-Yen Yang. PARTNR: A Benchmark for Planning and Reasoning in Embodied Multi-agent Tasks. In International Conference on Learning Representations, 2025. URL https://arxiv.org/abs/2411.00081.

[22] Chi-Lam Cheang, Guangzeng Chen, Ya Jing, Tao Kong, Hang Li, Yifeng Li, Yuxiao Liu, Hongtao Wu, Jiafeng Xu, Yichu Yang, Hanbo Zhang, and Minzhao Zhu. GR-2: A Generative Video-Language-Action Model with Web-Scale Knowledge for Robot Manipulation. arXiv preprint arXiv:2410.06158, 2024. URL https://arxiv.org/abs/2410.06158.

[23] Bingao Chen, Haoquan Fang, and C. Karen Liu. Can Astra Solve RoboMME without Breaking the Bank? Online research report, 2026. URL https://github.com/bingaochen/Astra-on-RoboMME.

[24] Jiaqi Chen, Bingqian Lin, Ran Xu, Zhenhua Chai, Xiaodan Liang, and Kwan-Yee Wong. MapGPT: Map-Guided Prompting with Adaptive Path Planning for Vision-and-Language Navigation. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 9796–9810. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.529. URL https://aclanthology.org/2024.acl-long.529/.

[25] Jintao Chen, Junjun Hu, Haochen Bai, Minghua Luo, Xinda Xue, Botao Ren, Chengyu Bai, Shichao Xie, Ziyi Chen, Fei Liu, Zedong Chu, Xiaolong Wu, Mu Xu, and Shanghang Zhang. AstraNav-World: World Model for Foresight Control and Consistency. arXiv preprint arXiv:2512.21714, 2025. doi: 10.48550/arXiv.2512.21714. URL https://arxiv.org/abs/2512.21714v2.

[26] Tianxing Chen, Yue Chen, Zixuan Li, Junyuan Tang, Kailun Su, Haoran Lu, Weijie Wan, Baijun Chen, Songling Liu, Haowen Yan, Honghao Su, Zhiyang Dou, Kaixuan Wang, Dandan Zhang, Yunze Liu, Yan Qin, Qiwei Liang, Qiwei Wu, Zijian Lin, Wenwei Lin, Yuran Wang, Minghua He, Tianshu Wu, Ruihai Wu, Jingquan Zhou, Kai-Chong Lei, Haibao Yu, Yuanfeng Ji,

Weiyang Jin, Guanyu Lin, Xiaofan Li, Qi Xiong, Renjing Xu, Zhongyu Li, Wenhao Chai, Enze Xie, Ziwei Wang, Yao Mu, Hao Dong, Wojciech Matusik, Mingyu Ding, Wenbo Ding, Ping Luo, and Masayoshi Tomizuka. RoboDojo: A Unified Sim-and-Real Benchmark for Comprehensive Evaluation of Generalist Robot Manipulation Policies. arXiv preprint arXiv:2607.04434, 2026. URL https://arxiv.org/abs/2607.04434.

[27] Xinyi Chen, Yilun Chen, Yanwei Fu, Ning Gao, Jiaya Jia, Weiyang Jin, Hao Li, Yao Mu, Jiangmiao Pang, Yu Qiao, Yang Tian, Bin Wang, Bolun Wang, Fangjing Wang, Hanqing Wang, Tai Wang, Ziqin Wang, Xueyuan Wei, Chao Wu, Shuai Yang, Jinhui Ye, Junqiu Yu, Jia Zeng, Jingjing Zhang, Jinyu Zhang, Shi Zhang, Feng Zheng, Bowen Zhou, and Yangkun Zhu. InternVLA-M1: A Spatially Guided Vision-Language-Action Framework for Generalist Robot Policy. arXiv preprint arXiv:2510.13778, 2025. doi: 10.48550/arXiv.2510.13778. URL https://arxiv.org/abs/2510.13778.

[28] Yanzhe Chen, Zechen Bai, Zhijun Cao, Wenzheng Zeng, Kevin Qinghong Lin, Yiqi Lin, Guoqiang Liang, Kevin Yuchen Ma, Qiming Huang, and Mike Zheng Shou. Show-Harness: Just a VLM Agent Can Play Robots. arXiv preprint arXiv:2609.10522, 2026. URL https://arxiv.org/abs/2609.10522.

[29] Dongzhou Cheng, Taoran Yi, Ye Fang, Xingwu Zhang, Fan Feng, Yixuan Li, Gengxiong Zhuang, Rongze Wang, Shuai Yang, Wei Song, Weizhi Xue, Minyan Wu, Jie Gui, Jiaqi Wang, and Tong Wu. In-Context Robot Learning with VLM Agents. arXiv preprint arXiv:2609.19138, 2026. URL https://arxiv.org/abs/2609.19138.

[30] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin CM Burchfiel, and Shuran Song. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In Proceedings of Robotics: Science and Systems, 2023. doi: 10.15607/RSS.2023.XIX.026. URL https://www.roboticsproceedings.org/rss19/p026.html.

[31] François Chollet. On the Measure of Intelligence. arXiv preprint arXiv:1911.01547, 2019. URL https://arxiv.org/abs/1911.01547.

[32] Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep Reinforcement Learning from Human Preferences. In Advances in Neural Information Processing Systems, volume 30, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/d5e2c0adad503c91f91df240d0cd4e49-Abstract.html.

[33] Guangzhao Dai, Qianru Sun, Qi Wu, and Bin Zhu. GPT-6-Astra Lights Up Embodied Navigation: Evaluation in Zero-Shot Vision-and-Language Navigation in Continuous Environments. arXiv preprint arXiv:2609.29861, 2026. URL https://arxiv.org/abs/2609.29861.

[34] Guangzhao Dai, Qi Wu, and Bin Zhu. How Far Can GPT-6-Astra Go? Evaluating Capabilities in Zero-Shot Vision-and-Language Navigation. arXiv preprint arXiv:2609.20116, 2026. URL https://arxiv.org/abs/2609.20116.

[35] Zijie Diao, Yitong Chen, Sicheng Xie, Tianyi Lu, Wujian Peng, Guojin Zhong, Houze Xu, Ziyi Ye, Zuxuan Wu, and Yu-Gang Jiang. LIBERO-Agent: Evaluating General-Purpose Agents for Direct Embodied Manipulation. arXiv preprint arXiv:2609.39507, 2026. URL https://arxiv.org/abs/2609.39507.

[36] Xin Ding, Liang Mi, Mingzhe Huang, Zixuan Wang, Chao Zhang, Zixu Hao, Fu Chen, Xiangyu Li, Yikai Zheng, Yaoyu Guo, Weijun Wang, Kun Li, Hao Wu, Yunxin Liu, and Ting Cao. Zetta (ζ): An Efficient Closed-Loop Embodied Harness for Self-Evolving Physical Intelligence. arXiv preprint arXiv:2608.16590, 2026. URL https://arxiv.org/abs/2608.16590.

[37] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Josh Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning Universal Policies via Text-Guided Video Generation. In Advances in Neural Information Processing Systems, volume 36, pages 9156–9172, 2023. doi: 10.52202/075280-0403. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract-Conference.html.

[38] Matthew B. Dwyer, George S. Avrunin, and James C. Corbett. Patterns in property specifications for finite-state verification. In Proceedings of the IEEE/ACM International Conference on Software Engineering, pages 411–420, 1999. doi: 10.1145/302405.302672. URL https://doi.org/10.1145/302405.302672.

[39] Karim Elmaaroufi, Justin Svegliato, Sarunas Kalade, Graham Schelle, Sanjit A. Seshia, and Matei Zaharia. RHO: Your Coding Agent is Secretly a Roboticist. arXiv preprint arXiv:2606.16458, 2026. URL https://arxiv.org/abs/2606.16458.

[40] Kutluhan Erol, James Hendler, and Dana S. Nau. HTN Planning: Complexity and Expressivity. In Proceedings of the Twelfth National Conference on Artificial Intelligence, pages 1123–1128, 1994. URL https://aaai.org/papers/01123-aaai94-173-htn-planning-complexity-and-expressivity/.

[41] Haoquan Fang, Jiafei Duan, Donovan Clay, Sam Wang, Shuo Liu, Weikai Huang, Xiang Fan, Wei-Chuan Tsai, Shirui Chen, Yi Ru Wang, Shanli Xing, Jaemin Cho, Jae Sung Park, Ainaz Eftekhar, Peter Sushko, Karen Farley, Angad Wadhwa, Cole Harrison, Winson Han, Ying-Chun Lee, Eli VanderBilt, Rose Hendrix, Suveen Ellawela, Lucas Ngoo, Joyce Chai, Zhongzheng

Ren, Ali Farhadi, Dieter Fox, and Ranjay Krishna. MolmoAct2: Action Reasoning Models for Real-world Deployment. In Conference on Robot Learning, 2026. URL https://hq-fang.github.io/. Accepted; to appear.

[42] Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. LIBERO-Plus: A Progressive Robustness Benchmark for Visual-Language-Action Models. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 38574–38583, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Fei_LIBERO-Plus_A_Progressive_Robustness_Benchmark_for_Visual-Language-Action_Models_CVPR_2026_paper.html.

[43] Hongyan Feng, Sunlai Chen, Xuanyu Liu, Miao Pan, Yangfan Xie, Yuxiang Cui, Zhongxiang Zhou, Rong Xiong, Wenqi Zhang, Jianwei Yin, Yueting Zhuang, and Xuhong Zhang. Embodied-Navigator: Point, Think, Memorize, and Align for Efficient Navigation. arXiv preprint arXiv:2608.17512, 2026. doi: 10.48550/arXiv.2608.17512. URL https://arxiv.org/abs/2608.17512v2.

[44] Pete Florence, Corey Lynch, Andy Zeng, Oscar A Ramirez, Ayzaan Wahid, Laura Downs, Adrian Wong, Johnny Lee, Igor Mordatch, and Jonathan Tompson. Implicit Behavioral Cloning. In Proceedings of the 5th Conference on Robot Learning, volume 164 of Proceedings of Machine Learning Research, pages 158–168. PMLR, 2022. URL https://proceedings.mlr.press/v164/florence22a.html.

[45] Maria Fox and Derek Long. Modelling Mixed Discrete-Continuous Domains for Planning. Journal of Artificial Intelligence Research, 27:235–297, 2006. doi: 10.1613/jair.2044. URL https://kclpure.kcl.ac.uk/portal/en/publications/modelling-mixed-discrete-continuous-domains-for-planning/.

[46] Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, Huang Huang, Wenli Xiao, Fei-Fei Li, Guanya Shi, Jiajun Wu, Shankar Sastry, Yuke Zhu, Ken Goldberg, and Linxi Fan. CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation. In Proceedings of the 43rd International Conference on Machine Learning, volume 306 of Proceedings of Machine Learning Research, pages 32155–32212. PMLR, 2026. URL https://proceedings.mlr.press/v306/fu26u.html.

[47] Galbot Team, Xuchuan Chen, Xiaoqian Cheng, Yu Deng, Lihe Ding, Shaocong Dong, Xiangjun Gao, Haozhe Jia, Zekai Li, Zhoujian Li, Yunrui Lian, Sikai Liang, Chenghuai Lin, Dairu Liu, Jiahang Liu, Qingtao Liu, Yuxuan Ma, Zekun Qi, Jiayi Su, He Wang, Ruochen Xu, Tianyu Xu, Xudong Xu, Zhe Xu, Mi Yan, Siming Yan, Li Yi, Ruixi Yu, Jinlu Zhang, Yintianrun Zhang, Zhikai Zhang, Zhizheng Zhang, Yixin Zheng, and Weiyi Zhu. Systematically Exploring the Capabilities of GPT-6 Astra as Embodied Policies. arXiv preprint arXiv:2609.38537, 2026. URL https://arxiv.org/abs/2609.38537.

[48] Gemini Robotics Team, Saminda Abeyruwan, Joshua Ainslie, Jean-Baptiste Alayrac, Montserrat Gonzalez Arenas, Travis Armstrong, Ashwin Balakrishna, Robert Baruch, Maria Bauza, Michiel Blokzijl, Steven Bohez, Konstantinos Bousmalis, Anthony Brohan, Thomas Buschmann, Arunkumar Byravan, Serkan Cabi, Ken Caluwaerts, Federico Casarini, Oscar Chang, Jose Enrique Chen, Xi Chen, Hao-Tien Lewis Chiang, Krzysztof Choromanski, David D'Ambrosio, Sudeep Dasari, Todor Davchev, Coline Devin, Norman Di Palo, Tianli Ding, Adil Dostmohamed, Danny Driess, Yilun Du, Debidatta Dwibedi, Michael Elabd, Claudio Fantacci, Cody Fong, Erik Frey, Chuyuan Fu, Marissa Giustina, Keerthana Gopalakrishnan, Laura Graesser, Leonard Hasenclever, Nicolas Heess, Brandon Hernaez, Alexander Herzog, R. Alex Hofer, Jan Humplik, Atil Iscen, Mithun George Jacob, Deepali Jain, Ryan Julian, Dmitry Kalashnikov, M. Emre Karagozler, Stefani Karp, Chase Kew, Jerad Kirkland, Sean Kirmani, Yuheng Kuang, Thomas Lampe, Antoine Laurens, Isabel Leal, Alex X. Lee, Tsang-Wei Edward Lee, Jacky Liang, Yixin Lin, Sharath Maddineni, Anirudha Majumdar, Assaf Hurwitz Michaely, Robert Moreno, Michael Neunert, Francesco Nori, Carolina Parada, Emilio Parisotto, Peter Pastor, Acorn Pooley, Kanishka Rao, Krista Reymann, Dorsa Sadigh, Stefano Saliceti, Pannag Sanketi, Pierre Sermanet, Dhruv Shah, Mohit Sharma, Kathryn Shea, Charles Shu, Vikas Sindhwani, Sumeet Singh, Radu Soricut, Jost Tobias Springenberg, Rachel Sterneck, Razvan Surdulescu, Jie Tan, Jonathan Tompson, Vincent Vanhoucke, Jake Varley, Grace Vesom, Giulia Vezzani, Oriol Vinyals, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Fei Xia, Ted Xiao, Annie Xie, Jinyu Xie, Peng Xu, Sichun Xu, Ying Xu, Zhuo Xu, Yuxiang Yang, Rui Yao, Sergey Yaroshenko, Wenhao Yu, Wentao Yuan, Jingwei Zhang, Tingnan Zhang, Allan Zhou, and Yuxiang Zhou. Gemini Robotics: Bringing AI into the Physical World. arXiv preprint arXiv:2503.20020, 2025. URL https://arxiv.org/abs/2503.20020.

[49] Alfonso E. Gerevini, Patrik Haslum, Derek Long, Alessandro Saetti, and Yannis Dimopoulos. Deterministic Planning in the Fifth International Planning Competition: PDDL3 and Experimental Evaluation of the Planners. Artificial Intelligence, 173 (5–6):619–668, 2009. URL https://dblp.org/rec/journals/ai/GereviniHLSD09.

[50] Malik Ghallab, Adele Howe, Craig Knoblock, Drew McDermott, Ashwin Ram, Manuela Veloso, Daniel Weld, and David Wilkins. PDDL – The Planning Domain Definition Language, Version 1.2. Technical Report CVC TR-98-003/DCS TR-1165, Yale Center for Computational Vision and Control, oct 1998. URL https://www.cs.cmu.edu/~mmv/planning/readings/98aips-PDDL.pdf.

[51] Malik Ghallab, Dana Nau, and Paolo Traverso. Automated Planning: Theory and Practice. Morgan Kaufmann, 2004. ISBN 9781558608566. URL https://projects.laas.fr/planning/aptp/.

[52] Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An Open-Source Generalist Robot Policy. In Proceedings of Robotics: Science and Systems, 2024. doi: 10.15607/RSS.2024.XX.090. URL https://www.roboticsproceedings.org/rss20/p090.html.

[53] Ruiyan Gong, Yingnan Guo, Junjun Hu, Jintao Kong, Xiaoxu Leng, Tianlun Li, Weize Li, Fei Liu, Zhicheng Liu, Jia Lu, Minghua Luo, Chenlin Ming, Yanfen Shen, Jiyue Tao, Zhengbo Wang, Mingyang Yin, Minqi Gu, Zihao Guan, Wei Guo, Guoqing Liu, Huachong Pang, Menglin Yang, Zeqian Ye, Xiaoxiao Geng, Zhining Gu, Honglin Han, Di Jing, Hongyu Pan, Mingchao Sun, Kuan Yang, Jianfang Zhang, Yanghong Chen, Ye He, Wei Mei, Jiahao Shi, Xiangpo Yang, Yanqing Zhu, Yang Cai, Jingjing Ma, Shihui Su, Zixiao Tang, Linbo Zheng, Zedong Chu, Xiaolong Wu, Wenbin Tang, and Mu Xu. ABot-N1: Toward a General Visual Language Navigation Foundation Model. arXiv preprint arXiv:2607.10383, 2026. doi: 10.48550/arXiv.2607.10383. URL https://arxiv.org/abs/2607.10383.

[54] Meng-Hao Guo, Zhe-Han Mo, Jia-Jun Wang, Yi Zhang, Kejin Wang, Yi-Xuan Deng, Jia-Peng Zhang, Yongming Rao, and Shi-Min Hu. Transferring the Intelligence of VLMs to Robotic Control. arXiv preprint arXiv:2609.22966, 2026. URL https://arxiv.org/abs/2609.22966.

[55] Wenxuan Guo, Xiuwei Xu, Yichen Liu, Xiangyu Li, Hang Yin, Huangxing Chen, Wenzhao Zheng, Jianjiang Feng, Jie Zhou, and Jiwen Lu. AwareVLN: Reasoning with Self-awareness for Vision-Language Navigation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 4065–4075, 2026. URL https://openaccess.thecvf.com/content/CVPR2026/html/Guo_AwareVLN_Reasoning_with_Self-awareness_for_Vision-Language_Navigation_CVPR_2026_paper.html.

[56] Peter Haddawy and Steve Hanks. Utility Models for Goal-Directed, Decision-Theoretic Planners. Computational Intelligence, 14(3):392–429, 1998. URL https://onlinelibrary.wiley.com/toc/14678640/1998/14/3.

[57] Sida He, Lingxi Xie, Yunning Cao, Pengfei Chen, Kaiwen Duan, Jiannan Ge, Xinyue Huo, Jiacheng Shao, and Qi Tian. Robot Manipulation with GPT-6-Astra: Body Knowledge, Experience Reuse, Emergent Skills, and Sim2Real Transfer. arXiv preprint arXiv:2609.31770, 2026. URL https://arxiv.org/abs/2609.31770.

[58] HKU MMLab and KAI. Physical RSI 1.0: Recursive Self-Harness for Scaling Embodied Skills. Project website, 2026. URL https://mmlab.hk/research/PhysicalRSI. Accessed: 2026-10-15.

[59] Daniel Höller, Gregor Behnke, Pascal Bercher, Susanne Biundo, Humbert Fiorino, Damien Pellier, and Ron Alford. HDDL: An Extension to PDDL for Expressing Hierarchical Planning Problems. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 34, pages 9883–9891, 2020. doi: 10.1609/aaai.v34i06.6542. URL https://ojs.aaai.org/index.php/AAAI/article/view/6542.

[60] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video Prediction Policy: A Generalist Robot Policy with Predictive Visual Representations. In Proceedings of the 42nd International Conference on Machine Learning, volume 267 of Proceedings of Machine Learning Research, pages 24328–24346. PMLR, 2025. URL https://proceedings.mlr.press/v267/hu25g.html.

[61] Yucheng Hu, Jianke Zhang, Yuanfei Luo, Yanjiang Guo, Xiaoyu Chen, Sun Xinshu, Kun Feng, Qingzhou Lu, Sheng Chen, Yangang Zhang, Wei Li, and Jianyu Chen. BagelVLA: Enhancing Long-Horizon Manipulation via Interleaved Vision-Language-Action Generation. In Proceedings of Robotics: Science and Systems, 2026. doi: 10.15607/RSS.2026.XXII.083. URL https://roboticsproceedings.org/rss22/p083.html.

[62] Haojian Huang, Pukun Zhao, Zexi Li, Yehang Zhang, Yangkai Wei, Wenqian Li, Han Yang, Kaiwen Zhou, Ying-Cong Chen, and Yinchuan Li. Are Frontier VLM Agents Ready to Be Robot Generalists? An Empirical Study with the Embodied Agent Arena. arXiv preprint arXiv:2610.00854, 2026. URL https://arxiv.org/abs/2610.00854.

[63] Jingzhi Huang, Junkai Huang, Wenxuan Song, Yuhang Peng, Hailong Huang, Haoyang Yang, Lujia Wang, and Yi Wang. SEDualVLN: A Spatially-Enhanced Dual-System Framework for Vision-Language Navigation. arXiv preprint arXiv:2605.17249, 2026. doi: 10.48550/arXiv.2605.17249. URL https://arxiv.org/abs/2605.17249v3.

[64] Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. In Proceedings of the 7th Conference on Robot Learning, volume 229 of Proceedings of Machine Learning Research, pages 540–562. PMLR, 2023. URL https://proceedings.mlr.press/v229/huang23b.html.

[65] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and brian ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models. In Proceedings of the 6th Conference on Robot Learning, volume 205 of Proceedings of Machine Learning Research, pages 1769–1782. PMLR, 2023. URL https://proceedings.mlr.press/v205/huang23c.html.

[66] Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. ReKep: Spatio-Temporal Reasoning of Relational Keypoint Constraints for Robotic Manipulation. In Proceedings of the 8th Conference on Robot Learning, volume 270 of Proceedings of Machine Learning Research, pages 4573–4602. PMLR, 2025. URL https://proceedings.mlr.press/v270/huang25g.html.

[67] HumanCLAW Team. HumanCLAW: Can Vision-Language Models Act Through a Body? Project website, 2026. URL https://human-claw.github.io/. GPT-6 Astra: Results and Analysis; accessed 2026-10-15.

[68] brian ichter, Anthony Brohan, Yevgen Chebotar, Chelsea Finn, Karol Hausman, Alexander Herzog, Daniel Ho, Julian Ibarz, Alex Irpan, Eric Jang, Ryan Julian, Dmitry Kalashnikov, Sergey Levine, Yao Lu, Carolina Parada, Kanishka Rao, Pierre Sermanet, Alexander T Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Mengyuan Yan, Noah Brown, Michael Ahn, Omar Cortes, Nicolas Sievers, Clayton Tan, Sichun Xu, Diego Reyes, Jarek Rettinghouse, Jornell Quiambao, Peter Pastor, Linda Luu, Kuang-Huei Lee, Yuheng Kuang, Sally Jesmonth, Nikhil J. Joshi, Kyle Jeffrey, Rosario Jauregui Ruano, Jasmine Hsu, Keerthana Gopalakrishnan, Byron David, Andy Zeng, and Chuyuan Kelly Fu. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. In Proceedings of the 6th Conference on Robot Learning, volume 205 of Proceedings of Machine Learning Research, pages 287–318. PMLR, 2023. URL https://proceedings.mlr.press/v205/ichter23a.html.

[69] Gabriel Ilharco, Vihan Jain, Alexander Ku, Eugene Ie, and Jason Baldridge. General Evaluation for Instruction Conditioned Navigation using Dynamic Time Warping. In NeurIPS Visually Grounded Interaction and Language Workshop, 2019. URL https://arxiv.org/abs/1907.05446.

[70] Arhan Jain, Mingtong Zhang, Kanav Arora, William Chen, Marcel Torne, Muhammad Zubair Irshad, Sergey Zakharov, Yue Wang, Sergey Levine, Chelsea Finn, Wei-Chiu Ma, Dhruv Shah, Abhishek Gupta, and Karl Pertsch. PolaRiS: Scalable Real-to-Sim Evaluations for Generalist Robot Policies. In Proceedings of Robotics: Science and Systems, 2026. doi: 10.15607/RSS.2026.XXII.062. URL https://www.roboticsproceedings.org/rss22/p062.html.

[71] Eric Jang, Alex Irpan, Mohi Khansari, Daniel Kappler, Frederik Ebert, Corey Lynch, Sergey Levine, and Chelsea Finn. BC-Z: Zero-Shot Task Generalization with Robotic Imitation Learning. In Proceedings of the 5th Conference on Robot Learning, volume 164 of Proceedings of Machine Learning Research, pages 991–1002. PMLR, 2022. URL https://proceedings.mlr.press/v164/jang22a.html.

[72] Jack B. Jedlicki, Amish Sethi, and Heng Yang. Can a Frontier VLM Serve as a Robot Policy for Memory-Dependent Manipulation? arXiv preprint arXiv:2610.17350, 2026. URL https://arxiv.org/abs/2610.17350.

[73] Mengzhao Jia, Yang Lin, Xixin Zhang, Zhihan Zhang, Xiaobai Liu, and Meng Jiang. Agent as Policy for Robotic Manipulation. arXiv preprint arXiv:2609.12541, 2026. URL https://arxiv.org/abs/2609.12541.

[74] Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can Language Models Resolve Real-World GitHub Issues? In International Conference on Learning Representations, 2024. URL https://arxiv.org/abs/2310.06770.

[75] Zhao Jin, Zhengping Che, Tao Li, Zhen Zhao, Kun Wu, Yuheng Zhang, Yinuo Zhao, Zehui Liu, Qiang Zhang, Xiaozhu Ju, Jing Tian, Yousong Xue, and Jian Tang. ArtVIP: Articulated Digital Assets of Visual Realism, Modular Interaction, and Physical Fidelity for Robot Learning. In International Conference on Learning Representations, pages 75272–75301, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/7a03c2bf486e56d8a25e8d5bb72ff1a2-Abstract-Conference.html.

[76] Yash Kant, Arun Ramachandran, Sriram Yenamandra, Igor Gilitschenski, Dhruv Batra, Andrew Szot, and Harsh Agrawal. Housekeep: Tidying Virtual Households using Commonsense Reasoning. In European Conference on Computer Vision, pages 355–373, 2022. URL https://arxiv.org/abs/2205.10712.

[77] Ralph L. Keeney and Howard Raiffa. Decisions with Multiple Objectives: Preferences and Value Trade-offs. Cambridge University Press, 1993. doi: 10.1017/CBO9781139174084. URL https://www.cambridge.org/core/books/abs/decisions-with-multiple-objectives/index/68594E19FFA7CFA694829FE502DA80BE.

[78] Muhammad Khalifa, Lajanugen Logeswaran, Jaekyeom Kim, Sungryull Sohn, Yunxiang Zhang, Moontae Lee, Hao Peng, Lu Wang, and Honglak Lee. Gaming the Judge: Unfaithful Chain-of-Thought Can Undermine Agent Evaluation. arXiv preprint arXiv:2601.14691, 2026. URL https://arxiv.org/abs/2601.14691.

[79] Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth Karamcheti, Soroush Nasiriany, Mohan Kumar Srirama, Lawrence Yunliang Chen, Kirsty Ellis, Peter David Fagan, Joey Hejna, Masha Itkina, Marion Lepert, Yecheng Jason Ma, Patrick Tree Miller, Jimmy Wu, Suneel Belkhale, Shivin Dass, Huy Ha, Arhan Jain, Abraham Lee, Youngwoon Lee, Marius Memmel, Sungjae Park, Ilija Radosavovic, Kaiyuan Wang, Albert Zhan, Kevin Black, Cheng Chi, Kyle Beltran Hatch, Shan Lin, Jingpei Lu, Jean Mercat, Abdul Rehman, Pannag R Sanketi, Archit Sharma, Cody Simpson, Quan Vuong, Homer Rich Walke, Blake Wulfe, Ted Xiao, Jonathan Heewon Yang, Arefeh Yavary, Tony Z. Zhao, Christopher Agia, Rohan Baijal, Mateo Guaman Castro, Daphne Chen, Qiuyu Chen, Trinity Chung, Jaimyn Drake, Ethan Paul Foster, Jensen Gao, David Antonio Herrera, Minho Heo, Kyle Hsu, Jiaheng Hu, Donovon Jackson, Charlotte Le, Yunshuang Li, Roy Lin, Zehan Ma, Abhiram Maddukuri, Suvir Mirchandani, Daniel Morton, Tony Nguyen, Abigail O'Neill, Rosario Scalise, Derick Seale, Victor Son, Stephen Tian, Emi Tran, Andrew E. Wang, Yilin Wu, Annie Xie, Jingyun Yang, Patrick Yin, Yunchu Zhang, Osbert Bastani, Glen Berseth, Jeannette Bohg, Ken Goldberg, Abhinav Gupta, Abhishek Gupta, Dinesh Jayaraman, Joseph J Lim, Jitendra Malik, Roberto Martín-Martín, Subramanian Ramamoorthy, Dorsa Sadigh, Shuran Song, Jiajun Wu, Michael C. Yip, Yuke Zhu, Thomas Kollar, Sergey Levine, and Chelsea Finn. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset. In Proceedings of Robotics: Science and Systems, volume 20, 2024. doi: 10.15607/RSS.2024.XX.120. URL https://roboticsproceedings.org/rss20/p120.html.

[80] Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Waseem, Pontus Stenetorp, Robin Jia, Mohit Bansal, Christopher Potts, and Adina Williams. Dynabench: Rethinking Benchmarking in NLP. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 4110–4124. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.naacl-main.324. URL https://aclanthology.org/2021.naacl-main.324/.

[81] Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.017. URL https://www.roboticsproceedings.org/rss21/p017.html.

[82] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P Foster, Pannag R Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An Open-Source Vision-Language-Action Model. In Proceedings of the 8th Conference on Robot Learning, volume 270 of Proceedings of Machine Learning Research, pages 2679–2713. PMLR, 2025. URL https://proceedings.mlr.press/v270/kim25c.html.

[83] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-Tuning Video Models for Visuomotor Control and Planning. In International Conference on Learning Representations, pages 71531–71552, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/748becc400a57c0e31cfe6a2e7951467-Abstract-Conference.html.

[84] Sascha Konrad and Betty H. C. Cheng. Real-Time Specification Patterns. In Proceedings of the IEEE/ACM International Conference on Software Engineering, pages 372–381, 2005. doi: 10.1145/1062455.1062526. URL https://doi.org/10.1145/1062455.1062526.

[85] Ron Koymans. Specifying Real-Time Properties with Metric Temporal Logic. Real-Time Systems, 2(4):255–299, 1990. doi: 10.1007/BF01995674. URL https://doi.org/10.1007/BF01995674.

[86] Victoria Krakovna, Jonathan Uesato, Vladimir Mikulik, Matthew Rahtz, Tom Everitt, Ramana Kumar, Zac Kenton, Jan Leike, and Shane Legg. Specification Gaming: The Flip Side of AI Ingenuity. DeepMind Blog, 2020. URL https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/. Published 21 April 2020.

[87] Jacob Krantz, Erik Wijmans, Arjun Majumdar, Dhruv Batra, and Stefan Lee. Beyond the Nav-Graph: Vision-and-Language Navigation in Continuous Environments. In Computer Vision – ECCV 2020, volume 12373 of Lecture Notes in Computer Science, pages 104–120. Springer, 2020. doi: 10.1007/978-3-030-58604-1_7. URL https://link.springer.com/chapter/10.1007/978-3-030-58604-1_7.

[88] Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie, and Jason Baldridge. Room-Across-Room: Multilingual Vision-and-Language Navigation with Dense Spatiotemporal Grounding. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, pages 4392–4412. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.356. URL https://aclanthology.org/2020.emnlp-main.356/.

[89] Teyun Kwon, Norman Di Palo, and Edward Johns. Language Models as Zero-Shot Trajectory Generators. IEEE Robotics and Automation Letters, 9(7):6728–6735, 2024. doi: 10.1109/LRA.2024.3410155. URL https://doi.org/10.1109/LRA.2024.3410155.

[90] Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, Wensi Ai, Benjamin Martinez, Hang Yin, Michael Lingelbach, Minjune Hwang, Ayano Hiranaka, Sujay Garlanka, Arman Aydin, Sharon Lee, Jiankai Sun, Mona Anvari, Manasi Sharma, Dhruva Bansal, Samuel Hunter, Kyu-Young Kim, Alan Lou, Caleb R Matthews, Ivan Villa-Renteria, Jerry Huayang Tang, Claire Tang, Fei Xia, Yunzhu Li, Silvio Savarese, Hyowon Gweon, C. Karen Liu, Jiajun Wu, and Li Fei-Fei. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. arXiv preprint arXiv:2403.09227, 2024. URL https://arxiv.org/abs/2403.09227.

[91] Jialong Li, Xuxin Cheng, Tianshu Huang, Shiqi Yang, Ri-Zhao Qiu, and Xiaolong Wang. AMO: Adaptive Motion Optimization for Hyper-Dexterous Humanoid Whole-Body Control. In Proceedings of Robotics: Science and Systems, Los Angeles, CA, USA, June 2025. doi: 10.15607/RSS.2025.XXI.061. URL https://roboticsproceedings.org/rss21/p061.html.

[92] Kailing Li, Yu Han, Tianwen Qian, Yuqian Fu, Jingyu Gong, Jiangming Shi, and Xiaoling Wang. GroundingVLN: Reasoning and Acting with Grounding for Vision-Language Navigation. arXiv preprint arXiv:2609.18581, 2026. doi: 10.48550/arXiv.2609.18581. URL https://arxiv.org/abs/2609.18581.

[93] Lin Li, Qihang Zhang, Yiming Luo, Shuai Yang, Ruilin Wang, Luyao Zhang, Mingrui Yu, Zelin Gao, Nan Xue, Boyu Zhou, Xing Zhu, Mingyu Ding, Yujun Shen, and Yinghao Xu. Causal World Modeling for Robot Control. In Proceedings of Robotics: Science and Systems, 2026. doi: 10.15607/RSS.2026.XXII.016. URL https://www.roboticsproceedings.org/rss22/p016.html.

[94] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified Video Action Model. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.074. URL https://www.roboticsproceedings.org/rss21/p074.html.

[95] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Oier Mees, Karl Pertsch, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. Evaluating Real-World Robot Manipulation Policies in Simulation. In Proceedings of the 8th Conference on Robot Learning, volume 270 of Proceedings of Machine Learning Research, pages 3705–3728. PMLR, 2025. URL https://proceedings.mlr.press/v270/li25c.html.

[96] Zechu Li, Yufeng Jin, Xiaoyang Liu, Puze Liu, Vignesh Prasad, Carlo D'Eramo, and Georgia Chalvatzaki. HARBOR: A Harness Framework for Agentic Robot Reinforcement Learning. arXiv preprint arXiv:2606.08610, 2026. URL https://arxiv.org/abs/2606.08610.

[97] Zexi Li, Yehang Zhang, Wenqian Li, Haojian Huang, Chenxu Wang, Shiyuan Deng, Yangkai Wei, Tianyi Zhang, Binghui Xie, Bohan Zhou, Yifan Chang, Kaiwen Zhou, Ying-Cong Chen, James Cheng, and Yinchuan Li. Robo-Harness K1: Harnessing Robot-Use Agents via Perception Augmentation. arXiv preprint arXiv:2609.29389, 2026. URL https://arxiv.org/abs/2609.29389.

[98] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as Policies: Language Model Programs for Embodied Control. In Proceedings of the IEEE International Conference on Robotics and Automation, pages 9493–9500, 2023. doi: 10.1109/ICRA48891.2023.10160591. URL https://code-as-policies.github.io/.

[99] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In Advances in Neural Information Processing Systems, Datasets and Benchmarks Track, volume 36, pages 44776–44791, 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.

[100] Fangcheng Liu, Yeqing Shen, Anda Cheng, Weishi Mi, Chao Tang, Chenyuan Liu, Yushun Xiang, Tingguang Li, Yong-Lu Li, and Yehui Tang. RoboICL: Embodied in-context learning with GPT-6 Astra. arXiv preprint arXiv:2609.34261, 2026. URL https://arxiv.org/abs/2609.34261.

[101] Haowen Liu, Xirui Li, Shaoxiong Yao, Peng Shi, Tianyi Zhou, Jia-Bin Huang, Furong Huang, and Jiayuan Mao. Guava: Distilling Frontier VLMs into a Compact Agent through a Robotic Manipulation Harness. arXiv preprint arXiv:2606.18363, 2026. URL https://arxiv.org/abs/2606.18363.

[102] Renhang Liu, Navonil Majumder, Tej Deep Pala, and Soujanya Poria. RoboQuest: Generalist Physical Agents that Search, Inspect and Test. arXiv preprint arXiv:2610.10388, 2026. URL https://arxiv.org/abs/2610.10388.

[103] Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. RDT-1B: a Diffusion Foundation Model for Bimanual Manipulation. In International Conference on Learning

Representations, pages 29982–30009, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/49f80e4d2471ad4f2edf4f5f1ab62339-Abstract-Conference.html.

[104] Lin Long and Jaemin Cho. Astra in Robot Control. Blog post, 2026. URL https://llong-cs.github.io/llms-for-robot-control/.

[105] Runyu Lu, Yubo Wu, Ethan Kou, Letian Fu, Wenli Xiao, Ajay Mandlekar, Yinzhen Xu, Guanya Shi, Ken Goldberg, Ang Chen, Mosharaf Chowdhury, Yuke Zhu, Linxi "Jim" Fan, and Guanzhi Wang. Aspire: Agentic /skills discovery for robotics, 2026. URL https://arxiv.org/abs/2607.00272.

[106] Xiaoya Lu, Zeren Chen, Xuhao Hu, Yijin Zhou, Weichen Zhang, Dongrui Liu, Lu Sheng, and Jing Shao. IS-Bench: Evaluating Interactive Safety of VLM-Driven Embodied Agents in Daily Household Tasks. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 40, pages 35680–35688, 2026. doi: 10.1609/aaai.v40i42.40880. URL https://ojs.aaai.org/index.php/AAAI/article/view/40880.

[107] Miles Macklin and Matthias Müller. Position Based Fluids. ACM Transactions on Graphics, 32(4):104:1–104:12, 2013. doi: 10.1145/2461912.2461984. URL https://doi.org/10.1145/2461912.2461984.

[108] Oded Maler and Dejan Nickovic. Monitoring Temporal Properties of Continuous Signals. In Formal Techniques, Modelling and Analysis of Timed and Fault-Tolerant Systems (FORMATS/FTRTFT), volume 3253 of Lecture Notes in Computer Science, pages 152–166. Springer, 2004. doi: 10.1007/978-3-540-30206-3_12. URL https://doi.org/10.1007/978-3-540-30206-3_12.

[109] Achu Menon, Sravanthi Machcha, Sabrina Zou, Tzu Kit Chan, and Jay Chooi. GPT-6 Astra on Robotic Manipulation. Robocurve online research report, September 2026. URL https://openai.robocurve.org/gpt-6-astra/. Published September 4, 2026.

[110] Mike Merrill, Alexander Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Shin, Thomas Walshe, E. Kelly Buchanan, Junhong Shen, Guanghao Ye, Haowei Lin, Jason Poulos, Maoyu Wang, Marianna Nezhurina, Di Lu, Orfeas Menis Mastromichalakis, Zhiwei Xu, Zizhao Chen, Yue Liu, Robert Zhang, Leon Liangyu Chen, Anurag Kashyap, Jan-Lucas Uslu, Jeffrey Li, Jianbo Wu, Minghao Yan, Song Bian, Vedang Sharma, Ke Sun, Steven Dillmann, Akshay Anand, Andrew Lanpouthakoun, Bardia Koopah, Changran Hu, Etash Guha, Gabriel Dreiman, Jiacheng Zhu, Karl Krauth, Li Zhong, Niklas Muennighoff, Robert Amanfu, Shangyin Tan, Shreyas Pimpalgaonkar, Tushar Aggarwal, Xiangning Lin, Xin Lan, Xuandong Zhao, Yiqing Liang, Yuanli Wang, Zilong (Ryan) Wang, Changzhi Zhou, David Heineman, Hange Liu, Harsh Trivedi, John Yang, Junhong Lin, Manish Shetty, Michael Yang, Nabil Omi, Negin Raoof, Shanda Li, Terry Yue Zhuo, Wuwei Lin, Yiwei Dai, Yuxin Wang, Wenhao Chai, Shang Zhou, Dariush Wahdany, Ziyu She, Jiaming Hu, Zhikang Dong, Yuxuan Zhu, Sasha Cui, Ahson Saiyed, Arinbjörn Kolbeinsson, Christopher Rytting, Ryan Marten, Yixin Wang, Jenia Jitsev, Alex Dimakis, Andy Konwinski, and Ludwig Schmidt. Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces. In International Conference on Learning Representations, pages 40903–40986, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/444a3737adaee10d86ad2ef5f74468e6-Abstract-Conference.html.

[111] Matthias Müller, Bruno Heidelberger, Marcus Hennix, and John Ratcliff. Position based dynamics. Journal of Visual Communication and Image Representation, 18(2):109–118, 2007. doi: 10.1016/j.jvcir.2007.01.005. URL https://www.sciencedirect.com/science/article/pii/S1047320307000065.

[112] Soroush Nasiriany, Sepehr Nasiriany, Abhiram Maddukuri, and Yuke Zhu. RoboCasa365: A Large-Scale Simulation Framework for Training and Benchmarking Generalist Robots. In International Conference on Learning Representations, 2026. URL https://arxiv.org/abs/2603.04356.

[113] Zaruhi Navasardyan, Tatul Danielyan, and Hrant Davtyan. FailBench: How Reliable are VLMs at Judging Robot Task Success? arXiv preprint arXiv:2609.03611, 2026. URL https://arxiv.org/abs/2609.03611.

[114] NVIDIA. Isaac Sim. https://github.com/isaac-sim/IsaacSim, 2025. URL https://github.com/isaac-sim/IsaacSim. Software repository. Accessed: 2026-10-15.

[115] NVIDIA. Cosmos 3: Omnimodal World Models for Physical AI. arXiv preprint arXiv:2606.02800, 2026. URL https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf.

[116] NVIDIA, :, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi "Jim" Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, Joel Jang, Zhenyu Jiang, Jan Kautz, Kaushil Kundalia, Lawrence Lao, Zhiqi Li, Zongyu Lin, Kevin Lin, Guilin Liu, Edith Llontop, Loic Magne, Ajay Mandlekar, Avnish Narayan, Soroush Nasiriany, Scott Reed, You Liang Tan, Guanzhi Wang, Zu Wang, Jing Wang, Qi Wang, Jiannan Xiang, Yuqi Xie, Yinzhen Xu, Zhenjia Xu, Seonghyeon Ye, Zhiding Yu, Ao Zhang, Hao Zhang, Yizhou Zhao, Ruijie Zheng, and Yuke Zhu. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. arXiv preprint arXiv:2503.14734, 2025. URL https://arxiv.org/abs/2503.14734.

[117] Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, Albert Tung, Alex Bewley, Alex Herzog, Alex Irpan, Alexander Khazatsky, Anant Rai, Anchit Gupta, Andrew Wang, Anikait Singh, Animesh Garg, Aniruddha Kembhavi, Annie Xie, Anthony Brohan, Antonin Raffin, Archit Sharma, Arefeh Yavary, Arhan Jain, Ashwin Balakrishna, Ayzaan Wahid, Ben Burgess-Limerick, Beomjoon Kim, Bernhard Schölkopf, Blake Wulfe, Brian Ichter, Cewu Lu, Charles Xu, Charlotte Le, Chelsea Finn, Chen Wang, Chenfeng Xu, Cheng Chi, Chenguang Huang, Christine Chan, Christopher Agia, Chuer Pan, Chuyuan Fu, Coline Devin, Danfei Xu, Daniel Morton, Danny Driess, Daphne Chen, Deepak Pathak, Dhruv Shah, Dieter Büchler, Dinesh Jayaraman, Dmitry Kalashnikov, Dorsa Sadigh, Edward Johns, Ethan Foster, Fangchen Liu, Federico Ceola, Fei Xia, Feiyu Zhao, Freek Stulp, Gaoyue Zhou, Gaurav S. Sukhatme, Gautam Salhotra, Ge Yan, Gilbert Feng, Giulio Schiavi, Glen Berseth, Gregory Kahn, Guanzhi Wang, Hao Su, Hao-Shu Fang, Haochen Shi, Henghui Bao, Heni Ben Amor, Henrik I Christensen, Hiroki Furuta, Homer Walke, Hongjie Fang, Huy Ha, Igor Mordatch, Ilija Radosavovic, Isabel Leal, Jacky Liang, Jad Abou-Chakra, Jaehyung Kim, Jaimyn Drake, Jan Peters, Jan Schneider, Jasmine Hsu, Jeannette Bohg, Jeffrey Bingham, Jeffrey Wu, Jensen Gao, Jiaheng Hu, Jiajun Wu, Jialin Wu, Jiankai Sun, Jianlan Luo, Jiayuan Gu, Jie Tan, Jihoon Oh, Jimmy Wu, Jingpei Lu, Jingyun Yang, Jitendra Malik, João Silvério, Joey Hejna, Jonathan Booher, Jonathan Tompson, Jonathan Yang, Jordi Salvador, Joseph J. Lim, Junhyek Han, Kaiyuan Wang, Kanishka Rao, Karl Pertsch, Karol Hausman, Keegan Go, Keerthana Gopalakrishnan, Ken Goldberg, Kendra Byrne, Kenneth Oslund, Kento Kawaharazuka, Kevin Black, Kevin Lin, Kevin Zhang, Kiana Ehsani, Kiran Lekkala, Kirsty Ellis, Krishan Rana, Krishnan Srinivasan, Kuan Fang, Kunal Pratap Singh, Kuo-Hao Zeng, Kyle Hatch, Kyle Hsu, Laurent Itti, Lawrence Yunliang Chen, Lerrel Pinto, Li Fei-Fei, Liam Tan, Linxi Jim Fan, Lionel Ott, Lisa Lee, Luca Weihs, Magnum Chen, Marion Lepert, Marius Memmel, Masayoshi Tomizuka, Masha Itkina, Mateo Guaman Castro, Max Spero, Maximilian Du, Michael Ahn, Michael C. Yip, Mingtong Zhang, Mingyu Ding, Minho Heo, Mohan Kumar Srirama, Mohit Sharma, Moo Jin Kim, Naoaki Kanazawa, Nicklas Hansen, Nicolas Heess, Nikhil J Joshi, Niko Suenderhauf, Ning Liu, Norman Di Palo, Nur Muhammad Mahi Shafiullah, Oier Mees, Oliver Kroemer, Osbert Bastani, Pannag R Sanketi, Patrick Tree Miller, Patrick Yin, Paul Wohlhart, Peng Xu, Peter David Fagan, Peter Mitrano, Pierre Sermanet, Pieter Abbeel, Priya Sundaresan, Qiuyu Chen, Quan Vuong, Rafael Rafailov, Ran Tian, Ria Doshi, Roberto Martín-Martín, Rohan Baijal, Rosario Scalise, Rose Hendrix, Roy Lin, Runjia Qian, Ruohan Zhang, Russell Mendonca, Rutav Shah, Ryan Hoque, Ryan Julian, Samuel Bustamante, Sean Kirmani, Sergey Levine, Shan Lin, Sherry Moore, Shikhar Bahl, Shivin Dass, Shubham Sonawani, Shuran Song, Sichun Xu, Siddhant Haldar, Siddharth Karamcheti, Simeon Adebola, Simon Guist, Soroush Nasiriany, Stefan Schaal, Stefan Welker, Stephen Tian, Subramanian Ramamoorthy, Sudeep Dasari, Suneel Belkhale, Sungjae Park, Suraj Nair, Suvir Mirchandani, Takayuki Osa, Tanmay Gupta, Tatsuya Harada, Tatsuya Matsushima, Ted Xiao, Thomas Kollar, Tianhe Yu, Tianli Ding, Todor Davchev, Tony Z. Zhao, Travis Armstrong, Trevor Darrell, Trinity Chung, Vidhi Jain, Vincent Vanhoucke, Wei Zhan, Wenxuan Zhou, Wolfram Burgard, Xi Chen, Xiaolong Wang, Xinghao Zhu, Xinyang Geng, Xiyuan Liu, Xu Liangwei, Xuanlin Li, Yao Lu, Yecheng Jason Ma, Yejin Kim, Yevgen Chebotar, Yifan Zhou, Yifeng Zhu, Yilin Wu, Ying Xu, Yixuan Wang, Yonatan Bisk, Yoonyoung Cho, Youngwoon Lee, Yuchen Cui, Yue Cao, Yueh-Hua Wu, Yujin Tang, Yuke Zhu, Yunchu Zhang, Yunfan Jiang, Yunshuang Li, Yunzhu Li, Yusuke Iwasawa, Yutaka Matsuo, Zehan Ma, Zhuo Xu, Zichen Jeff Cui, Zichen Zhang, and Zipeng Lin. Open X-Embodiment: Robotic Learning Datasets and RT-X Models : Open X-Embodiment Collaboration. In Proceedings of the IEEE International Conference on Robotics and Automation, pages 6892–6903, 2024. doi: 10.1109/ICRA57147.2024.10611477. URL https://doi.org/10.1109/icra57147.2024.10611477.

[118] OpenAI. Public Summary of Training Content for GPT-6 Astra. Published pursuant to Article 53(1)(d) of the EU AI Act, 2026. URL https://cdn.openai.com/pdf/gpt-6-astra-eu-ai-act-public-summary-of-training-content.pdf. Version v1, last updated 2 September 2026.

[119] OpenSenseNova Team. SenseNova-RoboRSI. https://github.com/OpenSenseNova/SenseNova-RoboRSI, 2026. URL https://github.com/OpenSenseNova/SenseNova-RoboRSI. Software repository. Accessed: 2026-10-15.

[120] Simon Ott, Adriano Barbosa-Silva, Kathrin Blagec, Jan Brauner, and Matthias Samwald. Mapping global dynamics of benchmark creation and saturation in artificial intelligence. Nature Communications, 13:6793, 2022. doi: 10.1038/s41467-022-34591-0. URL https://doi.org/10.1038/s41467-022-34591-0.

[121] Pantograph. Benchmarking Frontier LLMs on Robots. Online research report, 2026. URL https://pantograph.com/journal/vlm-harness.

[122] Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. FAST: Efficient Action Tokenization for Vision-Language-Action Models. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.012. URL https://www.roboticsproceedings.org/rss21/p012.html.

[123] Physical Intelligence, Bo Ai, Ali Amin, Raichelle Aniceto, Ashwin Balakrishna, Greg Balke, Kevin Black, George Bokinsky, Shihao Cao, Thomas Charbonnier, Vedant Choudhary, Foster Collins, Ken Conley, Grace Connors, James Darpinian, Karan Dhabalia, Maitrayee Dhaka, Jared DiCarlo, Danny Driess, Michael Equi, Adnan Esmail, Yunhao Fang, Chelsea Finn, Catherine Glossop, Thomas Godden, Ivan Goryachev, Lachlan Groom, Haroun Habeeb, Hunter Hancock, Karol Hausman, Gashon Hussein, Victor Hwang, Brian Ichter, Connor Jacobsen, Szymon Jakubczak, Rowan Jen, Tim Jones, Gregg Kammerer, Ben

Katz, Liyiming Ke, Mairbek Khadikov, Chandra Kuchi, Marinda Lamb, Devin LeBlanc, Brendon LeCount, Sergey Levine, Xinyu Li, Adrian Li-Bell, Vladislav Lialin, Zhonglin Liang, Wallace Lim, Yao Lu, Enyu Luo, Vishnu Mano, Nandan Marwaha, Aikys Mongush, Liam Murphy, Suraj Nair, Tyler Patterson, Karl Pertsch, Allen Z. Ren, Gavin Schelske, Charvi Sharma, Baifeng Shi, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, Will Stoeckle, Jiaming Tang, Jimmy Tanner, Shalom Tekeste, Marcel Torne, Kyle Vedder, Quan Vuong, Anna Walling, Haohuan Wang, Jason Wang, XuDong Wang, Chris Whalen, Samuel Whitmore, Blake Williams, Charles Xu, Sukwon Yoo, Lili Yu, Wuming Zhang, Zhuoyang Zhang, and Ury Zhilinsky. $\pi_{0.7}$: a Steerable Generalist Robotic Foundation Model with Emergent Capabilities. arXiv preprint arXiv:2604.15483, 2026. URL https://arxiv.org/abs/2604.15483.

[124] Yanyuan Qiao, Wenqi Lyu, Hui Wang, Zixu Wang, Zerui Li, Yuan Zhang, Mingkui Tan, and Qi Wu. Open-Nav: Exploring Zero-Shot Vision-and-Language Navigation in Continuous Environment with Open-Source LLMs. In Proceedings of the IEEE International Conference on Robotics and Automation, pages 6710–6717, 2025. doi: 10.1109/ICRA55743.2025.11127584. URL https://github.com/YanyuanQiao/Open-Nav.

[125] Hanoona Rasheed, Mohammed Irfan Kurpath, Bin Ren, Hisham Cholakkal, Fahad Shahbaz Khan, and Salman Khan. Hard Vision, Easy Vision: What GPT-6 Astra Reveals Across Computer Vision. arXiv preprint arXiv:2609.35718, 2026. URL https://arxiv.org/abs/2609.35718.

[126] Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach. Pearson, 4 edition, 2021. ISBN 9780137505135. URL https://www.pearson.com/en-us/subject-catalog/p/artificial-intelligence-a-modern-approach/P200000003500/9780134610993.

[127] Gabriel Sarch, Zhaoyuan Fang, Adam W. Harley, Paul Schydlo, Michael J. Tarr, Saurabh Gupta, and Katerina Fragkiadaki. TIDEE: Tidying Up Novel Rooms using Visuo-Semantic Commonsense Priors. In European Conference on Computer Vision, pages 480–496, 2022. doi: 10.1007/978-3-031-19842-7_28. URL https://arxiv.org/abs/2207.10761.

[128] Nur Muhammad Shafiullah, Zichen Cui, Ariuntuya (Arty) Altanzaya, and Lerrel Pinto. Behavior Transformers: Cloning k modes with one stone. In Advances in Neural Information Processing Systems, volume 35, pages 22955–22968, 2022. doi: 10.52202/068431-1668. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/90d17e882adbdda42349db6f50123817-Abstract.html.

[129] Qiuhong Shen, Shihua Zhang, Yue Liao, Qi Li, Zhenxiong Tan, Shizun Wang, Shuicheng Yan, and Xinchao Wang. World Action Models: A Survey. arXiv preprint arXiv:2606.20781, 2026. URL https://arxiv.org/abs/2606.20781.

[130] Zeyu Shen, Haoxiang You, Yilang Liu, Zhicheng Zheng, Lihan Zha, Kashu Yamazaki, Mingtong Zhang, Suning Huang, Jiankai Sun, Qianzhong Chen, Lucy He, Kaiyuan Liu, Haoran Chang, Katerina Fragkiadaki, Dhruv Shah, Mac Schwager, Peter Henderson, Ian Abraham, and Canwen Xu. EmbodiedSWE: Coding Agents for Long Horizon Dexterous Robotics. arXiv preprint arXiv:2609.27308, 2026. URL https://arxiv.org/abs/2609.27308.

[131] Xiangyu Shi, Zerui Li, Wenqi Lyu, Jiatong Xia, Feras Dayoub, Yanyuan Qiao, and Qi Wu. SmartWay: Enhanced Waypoint Prediction and Backtracking for Zero-Shot Vision-and-Language Navigation. In IEEE/RSJ International Conference on Intelligent Robots and Systems, 2025. URL https://github.com/sxyxs/SmartWay-Code.

[132] Ruiyang Si, Jianxin Bi, Shunyu Yang, Rui Ni, Wenbo Huang, Qiang Wang, Shulong Jiang, Yintianrun Zhang, Feixiang Ruan, Kewei Zhang, Duomin Wang, Xiuyu Li, Haiwen Feng, Zhen Dong, and Daquan Zhou. Fewer Tokens, Better Action: GPT-6 Astra Robot Agents with 14% Higher Success Rate but 65% Fewer Tokens. arXiv preprint arXiv:2610.01939, 2026. URL https://arxiv.org/abs/2610.01939.

[133] David Snyder, Asher James Hancock, Apurva Badithela, Emma Dixon, Patrick Miller, Rares Andrei Ambrus, Anirudha Majumdar, Masha Itkina, and Haruki Nishimura. Is Your Imitation Learning Policy Better than Mine? Policy Comparison with Near-Optimal Stopping. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.077. URL https://www.roboticsproceedings.org/rss21/p077.html.

[134] Xinshuai Song, Weixing Chen, Yang Liu, Weikai Chen, Guanbin Li, and Liang Lin. Towards Long-Horizon Vision-Language Navigation: Platform, Benchmark and Method. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 12078–12088, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Song_Towards_Long-Horizon_Vision-Language_Navigation_Platform_Benchmark_and_Method_CVPR_2025_paper.html.

[135] Sanjana Srivastava, Chengshu Li, Michael Lingelbach, Roberto Martín-Martín, Fei Xia, Kent Elliott Vainio, Zheng Lian, Cem Gokmen, Shyamal Buch, Karen Liu, Silvio Savarese, Hyowon Gweon, Jiajun Wu, and Li Fei-Fei. BEHAVIOR: Benchmark for Everyday Household Activities in Virtual, Interactive, and Ecological Environments. In Proceedings of the 5th Conference on Robot Learning, volume 164 of Proceedings of Machine Learning Research, pages 477–490. PMLR, 2022. URL https://proceedings.mlr.press/v164/srivastava22a.html.

[136] Edward Sun, Sravanthi Machcha, Sabrina Zou, Tzu Kit Chan, and Jay Chooi. RoboHarm: Do Frontier Robot Policies Refuse Unsafe Instructions? Robocurve, https://robocurve.org/roboharm/, 2026. URL https://robocurve.org/roboharm/. Published 18 September 2026.

[137] Jingwen Sun, Wenyao Zhang, Zekun Qi, Shaojie Ren, Zezhi Liu, Hanxin Zhu, Guangzhong Sun, Xin Jin, and Zhibo Chen. VLA-JEPA: Enhancing Vision-Language-Action Model with Latent World Model. In Computer Vision – ECCV 2026, volume 17064 of Lecture Notes in Computer Science, pages 478–497. Springer, 2026. doi: 10.1007/978-3-032-37196-6_26. URL https://link.springer.com/chapter/10.1007/978-3-032-37196-6_26.

[138] Yiyou Sun, Xinyang Han, Weichen Zhang, Xiang Liu, Yuanbo Pang, Tianyu Wang, Yuhan Cao, Yixiao Huang, Zhaowei Zhang, Chris Duroiu, Haoyun Zhang, Jeffrey Lin, Weishu Zhang, Tyler Zeng, Ying Yan, Bo Liu, Hanson Wen, Mingyang Xu, Xiaoyuan Liu, Zimeng Chen, Weiyan Shi, Amanda Dsouza, Vincent Sunn Chen, Jingxu Xie, Sheldon Huang, Dewen Lei, Hanxuan Zhang, Zonglin Di, Patrick Bryant, Carl Boettiger, Yamini Rangan, Bradley Rothenberg, Kyle Steinfeld, Arvind Rao, Tapio Schneider, Georgios Yannakakis, Laure Zanna, Kaan Ozbay, Ida Sim, Tarek Zohdi, George Em Karniadakis, Jack Gallant, Teresa Head-Gordon, Chris Yuhao Liu, Rongwang Hu, Yafei Cheng, Aras Bacho, Zengyi Qin, Alan Yuille, Yixiong Chen, Hao Liu, Lin Zeng, Bo Chen, Zhendong Li, Litian Gong, Shashank Muralidhar Bharadwaj, Thorben Jansen, Haixiang Tang, Jianhong Tu, Jixian Liu, Maojia Song, Ruheng Wang, Shengcao Cao, Yanxin Zhang, Yingxuan Yang, Zongzheng Zhang, Honglin Bao, Hyunjun Kim, Peiran Xu, Shuo Lu, Zheng Zhang, Zhonghua Wang, Aditya Prakash, Bohan Lyu, Chang Ma, Haoxuan Chen, Hengduan Fan, Molei Tao, Roman Sitelew, Russell Alan Poldrack, Yang Liu, Yang Liu, Yichen Wu, Yuanze Lin, Zhenglu Li, Zibu Wei, Zijiao Chen, Abderrahim Bendahi, Alexander Schubert, Chase Broderick, Chengzhi Liu, Haoran Sun, Haoyang Li, Haoyue Hua, Jiayuan Ding, Jing Huang, Shangding Gu, Tuan Vinh, Xiatao Sun, Yanqiong Jiang, Yiran Hu, Zechen Xu, Ziao Yang, Chenguang Wang, Costas Spanos, Dongrong Yang, Haocheng Wang, Jiayi Cheng, Jicheng Wang, Junbo Peng, Kangqi Zhang, Keying Kuang, Lichao Sun, Ming Yin, Orson Xu, Vincent Siu, Wanzhe Liao, Wenqi Shi, Wenxi Deng, Xiaopan Zhang, Yangjunfeng Zhu, Yifan Shen, Yi Jiang, Yinsheng Li, Yongshuo Zong, Youbang Pan, Yuan Li, Yuchen Zhu, Yunshi Zhang, Zhenning Yang, Ziyan Rao, Alireza Rafiei, Andrew Glover, Bixuan Ren, Boqin Yuan, Chonghao Cai, Deren Lei, Hanwen Xing, Haoran Jin, Hector Gomez, Ishaan Singh Chandok, Jiaming Hu, Jianxin Sun, Justin Purnomo, Kunyang Sun, Lei Ding, Longtai Liao, Lukas Kim, Ming Liu, Qingpo Wuwu, Ran Jin, Ruiqi Zhang, Ting Sun, Wenbo Huang, Xuandong Zhao, Xuhuan Shen, Yifan Zhang, Yi Li, Yingze Wang, Yixin Wang, Yushi Yang, Zhen Dong, Zhiling Yan, Zhongxing Xu, Zibo Zhao, Zixin Jiang, Ali Emami, Andy Zeyi Liu, Angela Wu, Aylin Caliskan, Baihe Huang, Baixiang Huang, Benjamin Yang, Benran Zhang, Brian Zhao, Carlo Bosio, Charlie Guo, Che Liu, Chenchen Du, Chen Dai, Chenlong Zhang, Chenyu Li, Chenyu Zhu, Chi Zhang, Core Francisco Park, Daofeng Li, Dehao Wu, Dowling Wong, Erin Kasson, Fangyu Wu, Fan Xu, Fiona Shen, Haimin Hu, Haitao Li, Hang Zhou, Hannah Szlyk, Hanning Zhang, Hanwen Zhang, Hao Fan, Haoran Ye, HaoYang Shang, Haoyue Bai, Hengyu Fu, Hongwei Li, Huiqi Wang, Jiachen Li, Jialu Wang, Jiaming Liu, Jiawei Shen, Jierui Zuo, Jingxuan Fan, Jingyan Li, Jirong Yang, Jonah So, Jorin Overwiening, Joseph Cavanagh, Joseph Lee, Junge Zhang, Kaiyuan Liu, Kaustubh Dhole, Kevin Qinghong Lin, Kunpeng Zhang, Kyle Montgomery, Layna Paraboschi, Liana, Lihui Liu, Liwei Zhou, Lu Lu, Mengyuan "Millie" Wu, Mikko Tolonen, Mingxuan Yan, Minqi Shi, Murphy Niu, Pan Lu, Patricia Cavazos-Rehg, Qijian Zhao, Qiuyang Mang, Rimika Jaiswal, Ruiling Xu, Run-Ze Fan, Ruoxi Ning, Santanu Bosu Antu, Scott Rankin, Sen Yang, Shi Gu, Shi Qiu, Sichen Zhu, Sihan Xu, Siqi Zhu, Song Zhang, Sophia Sun, Stefano De Castro, Swati Rajwal, Tianneng Shi, Tianxing Chen, Tianyue Zhou, Ting Wang, Ventus Shaw, Vladislav Susoy, Weihang Xiao, Wei Wang, Wenbo Lv, Wenjin Zhang, Wenshuo Zhang, Wenyu Liang, Xiaohu Yu, Xiao Li, Xiaoli Yang, Xin Lan, Xinyu Wang, Xuan Liu, Xuewen Chen, Xu Pan, Yan Zhu, Yicheng Lyu, Yifan Gu, Yimin Zhao, Ying Zhang, Yin Ren, Yipu Chen, Yiren Rong, Yi Shao, Yishu Zhu, Yizhi Zhang, Yongyi Xiong, Yuangang Li, Yuchen Wu, Yuhaohua Zheng, Yuhao Wu, Yunfu Deng, Yushan Li, Yu Wu, Yuzheng Xu, Yuzhou Nie, Zehao Wang, Zhehang Du, Zhengming Yu, Zhihan Jiang, Zhi Li, Zhixu Li, Zhuoran Yi, Tao Sun, Zhun Wang, Justin Xu, Helena Casademunt, Wenjia Chen, Cheng Cheng, Nawen Deng, Rao Fu, Tianfu Fu, Yifan Han, Zhenyu He, Qiao Jin, Langlang Li, Yuetai Li, Sylvia Liu, Subhabrata Mukherjee, Yunqi Ouyang, He Ren, Dawei Shi, Haoran Wu, Zhiyue Wu, Hannah Yao, Jenny Yu, Rhea Zhan, Luqing Zhou, Blake Zhu, Junfan Zhu, Sanjiv Das, Yanjun Gao, Wenbo Guo, Erika J Schneider, Bing Liu, Zhiyong Lu, Yian Ma, Mark Mueller, Radha Poovendran, Somayeh Sojoudi, Huaxiu Yao, Yinglun Zhu, Yegor Denisov-Blanch, Changran Hu, Vincent Koc, Hezheng Yin, Naomi Xia, Xinyi Bai, Arsh Shah Dilbagi, Jing Gu, Yutong Liu, Lin Shi, Qizheng Zhang, Yuexing Hao, Xiaoming Li, Xiangyi Li, Bingran You, Wenjun Ke, and Dawn Song. Agents' Last Exam. In Advances in Neural Information Processing Systems, 2026. URL https://arxiv.org/abs/2606.05405. Accepted; to appear.

[139] Yang Tian, Sizhe Yang, Jia Zeng, Ping Wang, Dahua Lin, Hao Dong, and Jiangmiao Pang. Predictive Inverse Dynamics Models are Scalable Learners for Robotic Manipulation. In International Conference on Learning Representations, pages 92033–92052, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e5b5c402bb7bd5e60bede6961d6fe39e-Abstract-Conference.html.

[140] Vals AI. Robo-bench. Online benchmark report, 2026. URL https://www.vals.ai/benchmarks/robo_bench. Updated: 2026-10-13.

[141] Sai H. Vemprala, Rogerio Bonatti, Arthur Bucker, and Ashish Kapoor. ChatGPT for Robotics: Design Principles and Model

Abilities. IEEE Access, 12:55682–55696, 2024. doi: 10.1109/ACCESS.2024.3387941. URL https://doi.org/10.1109/ACCESS.2024.3387941.

[142] Andrew Wagenmaker, Zhiyuan Zhou, and Sergey Levine. Behavioral Exploration: Learning to Explore via In-Context Adaptation. In Proceedings of the 42nd International Conference on Machine Learning, volume 267 of Proceedings of Machine Learning Research, pages 61885–61912. PMLR, 2025. URL https://proceedings.mlr.press/v267/wagenmaker25a.html.

[143] Naoki Wake, Atsushi Kanehira, Kazuhiro Sasabuchi, Jun Takamatsu, and Katsushi Ikeuchi. GPT-4V(ision) for Robotics: Multimodal Task Planning From Human Demonstration. IEEE Robotics and Automation Letters, 9(11):10567–10574, 2024. doi: 10.1109/LRA.2024.3477090. URL https://github.com/microsoft/GPT4Vision-Robot-Manipulation-Prompts.

[144] Hanwen Wang, Weizhi Zhao, Xiangyu Wang, Siyuan Huang, He Lin, Boyuan Zheng, Rongtao Xu, Gang Wang, Yao Mu, He Wang, Lue Fan, Hongsheng Li, Zhaoxiang Zhang, and Tieniu Tan. DexJoCo: A benchmark and toolkit for task-oriented dexterous manipulation on MuJoCo. arXiv preprint arXiv:2605.16257, 2026. URL https://arxiv.org/abs/2605.16257.

[145] Qi Wang, Tianyi Wang, Chengyang Li, Shikun Ban, Yurun Chen, Yizhong Ge, Jason Qin, Chengtai Li, and Wentao Zhu. Towards the Harness of Embodied Agents. arXiv preprint arXiv:2608.11246, 2026. URL https://arxiv.org/abs/2608.11246.

[146] Shaoan Wang, Aocheng Luo, Fei Huang, Jingyi Xu, Xiaoyang Wang, Yueyu Wang, Qianli Ma, Fan Yang, Ran Mei, Jia Wei, Jiangpeng Hu, Xuhao Liu, Hongming Chen, Yuanbin Shao, Yiyang Lin, Ziliang Li, Liang Pan, Xinhang Liu, Yuntao Ma, and Tingxiang Fan. LightNav-0: Eliciting VLM Spatial Intelligence for Generalist Embodied Navigation. arXiv preprint arXiv:2608.30935, 2026. doi: 10.48550/arXiv.2608.30935. URL https://arxiv.org/abs/2608.30935.

[147] Shuo Wang, Yongcai Wang, Wanting Li, Xudong Cai, Yucheng Wang, Maiyue Chen, Kaihui Wang, Zhizhong Su, Deying Li, and Zhaoxin Fan. Aux-Think: Exploring Reasoning Strategies for Data-Efficient Vision-Language Navigation. In Advances in Neural Information Processing Systems, volume 38, pages 30892–30916. Curran Associates, Inc., 2025. doi: 10.52202/085713-1040. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/2c90bf5bffe6497730ecade3a3a37458-Abstract-Conference.html.

[148] Siyin Wang, Junhao Shi, Zhaoyang Fu, Xinzhe He, Feihong Liu, Chenchen Yang, Yikang Zhou, Zhaoye Fei, Jingjing Gong, Jinlan Fu, Mike Zheng Shou, Xuanjing Huang, Xipeng Qiu, and Yu-Gang Jiang. World Action Models: The Next Frontier in Embodied AI. arXiv preprint arXiv:2605.12090, 2026. URL https://arxiv.org/abs/2605.12090.

[149] Zihan Wang, Seungjun Lee, Yinghao Xu, and Gim Hee Lee. Image2Sim: Scaling Embodied Navigation via Generative Neural Simulator. arXiv preprint arXiv:2607.05765, 2026. doi: 10.48550/arXiv.2607.05765. URL https://arxiv.org/abs/2607.05765. Accepted at the Conference on Neural Information Processing Systems 2026.

[150] Meng Wei, Chenyang Wan, Jiaqi Peng, Xiqian Yu, Yuqiang Yang, Delin Feng, Wenzhe Cai, Chenming Zhu, Tai Wang, Jiangmiao Pang, and Xihui Liu. Ground Slow, Move Fast: A Dual-System Foundation Model for Generalizable Vision-Language Navigation. In International Conference on Learning Representations, pages 12380–12396, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/14da7aea05debb963b3d8d46449d51a0-Abstract-Conference.html.

[151] Songlin Wei, Hongyi Jing, Boqian Li, Zhenyu Zhao, Jiageng Mao, Zhenhao Ni, Sicheng He, Sheng Zang, Xiawei Liu, Kaidi Kang, Jie Liu, Weiduo Yuan, Marco Pavone, Di Huang, and Yue Wang. Ψ_0: An Open Foundation Model Towards Universal Humanoid Loco-Manipulation. In Proceedings of Robotics: Science and Systems, 2026. doi: 10.15607/RSS.2026.XXII.021. URL https://roboticsproceedings.org/rss22/p021.html.

[152] Songlin Wei, Zhenhao Ni, Jie Liu, Zhenyu Zhao, Junjie Ye, Hongyi Jing, Junkai Xia, Xiawei Liu, Michael Leong, Liang Heng, Di Huang, and Yue Wang. SIMPLE: Simulation-Based Policy Learning and Evaluation for Humanoid Loco-manipulation. arXiv preprint arXiv:2606.08278, 2026. URL https://arxiv.org/abs/2606.08278.

[153] Yule Wen, Yixin Ye, Yanzhe Zhang, Diyi Yang, and Hao Zhu. Real-Time Reasoning Agents in Evolving Environments. In International Conference on Learning Representations, 2026. URL https://arxiv.org/abs/2511.04898.

[154] Hongtao Wu, Ya Jing, Chilam Cheang, Guangzeng Chen, Jiafeng Xu, Xinghang Li, Minghuan Liu, Hang Li, and Tao Kong. Unleashing Large-Scale Video Generative Pre-training for Visual Robot Manipulation. In International Conference on Learning Representations, 2024. URL https://openreview.net/forum?id=NxoFmGgWC9.

[155] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. In Advances in Neural Information Processing Systems, volume 37, 2024. URL https://arxiv.org/abs/2404.07972.

[156] Zihao Xin, Wentong Li, Yixuan Jiang, Ziyuan Huang, Bin Wang, Piji Li, Jianke Zhu, Jie Qin, and Shengjun Huang. AgentVLN: Towards Agentic Vision-and-Language Navigation. In European Conference on Computer Vision, volume 17033 of Lecture Notes in Computer Science, pages 114–132. Springer Nature Switzerland, 2026. doi: 10.1007/978-3-032-37167-6_7. URL https://link.springer.com/chapter/10.1007/978-3-032-37167-6_7.

[157] Xinda Xue, Junjun Hu, Minghua Luo, Shichao Xie, Jintao Chen, Zixun Xie, Kuichen Quan, Wei Guo, Zedong Chu, Mu Xu, and Zhengzhou Zhu. OmniNav: A Unified Framework for Prospective Exploration and Visual-Language Navigation. In International Conference on Learning Representations, pages 106216–106231, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/ad6363efb7af02f7db13d087c7e649bd-Abstract-Conference.html.

[158] Jenai Xuning Yang, Ashna Khetan, and Shun Zhang. Astranomy: An Empirical Study of GPT-6 Astra as a Native Robot Policy. Technical report, NVIDIA, 2026. URL https://research.nvidia.com/labs/srl/projects/astranomy/.

[159] Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, Marziyeh Movahedi, Manling Li, Heng Ji, Huan Zhang, and Tong Zhang. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents. In Proceedings of the 42nd International Conference on Machine Learning, volume 267 of Proceedings of Machine Learning Research, pages 70576–70631. PMLR, 2025. URL https://proceedings.mlr.press/v267/yang25f.html.

[160] Ruihan Yang, Qinxi Yu, Yecheng Wu, Rui Yan, Borui Li, An-Chieh Cheng, Xueyan Zou, Yunhao Fang, Xuxin Cheng, Ri-Zhao Qiu, Hongxu Yin, Sifei Liu, Song Han, Yao Lu, and Xiaolong Wang. EgoVLA: Learning Vision-Language-Action Models from Egocentric Human Videos. arXiv preprint arXiv:2507.12440, 2025. doi: 10.48550/arXiv.2507.12440. URL https://arxiv.org/abs/2507.12440. Accepted to the IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026.

[161] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A High-Fidelity Simulation Benchmark for Analysis of Task Generalist Policies. In Proceedings of Robotics: Science and Systems, 2026. doi: 10.15607/RSS.2026.XXII.096. URL https://www.roboticsproceedings.org/rss22/p096.html.

[162] Yue Yang, Shuo Cheng, Yu Fang, Homanga Bharadhwaj, Mingyu Ding, Gedas Bertasius, and Daniel Szafir. LiLo-VLA: Compositional Long-Horizon Manipulation via Linked Object-Centric Policies. arXiv preprint arXiv:2602.21531, 2026. URL https://arxiv.org/abs/2602.21531.

[163] Zhenjie Yang, Yideng Zhang, Dongjie Zhang, Chenyu Jiang, Xianshuai Liu, Yufeng Li, Zuhao Ge, Xingyu Jiao, Zheng Zhang, Kaiyu He, He Wang, Yuwen Zhong, Yi Deng, Muyun Jiang, Xianliang Huang, Haisheng Su, Donghang Zhang, Jian Zhang, Xue Yang, Hongyang Li, Zuxuan Wu, Yu-Gang Jiang, Xiaosong Jia, and Junchi Yan. Bench2Dex: Benchmarking Visuo-Tactile Bimanual Dexterous Manipulation Across Dexterous Hands. arXiv preprint arXiv:2609.15726, 2026. URL https://arxiv.org/abs/2609.15726.

[164] Zhiqin Yang, Chenxin Li, Xiaomeng Hu, Yibin Liu, Weidong Huang, Jiankai Sun, Haitao Li, Zijian Wu, Yuzhi Huang, Fanding Huang, Hanwen Sun, Jiashun Liu, Jingqi Tong, Mingxin Huang, Shaoli Hu, Shijue Huang, Tianyi Bai, Xinyuan Wang, Yunlong Lin, Zhengyang Tang, Zhexin Zhang, Zhuo Chen, Xierui Song, Juntao Dai, Boyuan Chen, Jiaming Ji, Fangneng Zhan, Mengkang Hu, Wei Xue, Yonggang Zhang, Han Hu, Tsung-Yi Ho, and Yike Guo. RobotWorld: Benchmarking Multimodal Agents for Robot Use Across Diverse Tasks and Embodiments. arXiv preprint arXiv:2610.10409, 2026. doi: 10.48550/arXiv.2610.10409. URL https://arxiv.org/abs/2610.10409.

[165] Yifei Yao, Yude Zou, Guanbo Jiao, Ning Gao, Yukai Wang, Jiantong Chen, Zihou Zhu, Xinyu Li, Ningyu Yan, Shuai Wang, Jianming Ma, Haoyu Guo, Xing Gao, Weinan Zhang, and Chunhua Shen. How Far Does a Frontier Agent Get as an Embodied Policy? Technical report, Shanghai Artificial Intelligence Laboratory, 2026. URL https://internrobotics.github.io/EBench-Astra-Tech-Report/.

[166] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi "Jim" Fan, and Joel Jang. World Action Models are Zero-shot Policies. arXiv preprint arXiv:2602.15922, 2026. URL https://arxiv.org/abs/2602.15922.

[167] Chenghao Yin, Da Huang, Di Yang, Jichao Wang, Nanshu Zhao, Chen Xu, Wenjun Sun, Linjie Hou, Zhijun Li, Junhui Wu, Zhaobo Liu, Zhen Xiao, Sheng Zhang, Lei Bao, Rui Feng, Zhenquan Pang, Jiayu Li, Qian Wang, and Maoqing Yao. Genie Sim 3.0 : A High-Fidelity Comprehensive Simulation Platform for Humanoid Robot. arXiv preprint arXiv:2601.02078, 2026. URL https://arxiv.org/abs/2601.02078.

[168] Yue Yu, Bin Zhu, Junhui Li, Jiayu Wang, Shijie Zhou, and Jingjing Chen. ReTrace: Cross-Stage Failure Attribution for Self-Improving Robotic Manipulation. arXiv preprint arXiv:2610.15814, 2026. doi: 10.48550/arXiv.2610.15814. URL https://arxiv.org/abs/2610.15814.

[169] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do World Action Models Need Test-time Future Imagination? arXiv preprint arXiv:2603.16666, 2026. URL https://arxiv.org/abs/2603.16666.

[170] Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3D Diffusion Policy: Generalizable Visuomotor Policy Learning via Simple 3D Representations. In Proceedings of Robotics: Science and Systems, 2024. doi: 10.15607/RSS.2024.XX.067. URL https://roboticsproceedings.org/rss20/p067.html.

[171] Alex L. Zhang, Thomas L. Griffiths, Karthik R. Narasimhan, and Ofir Press. VideoGameBench: Can Vision-Language Models Complete Popular Video Games? arXiv preprint arXiv:2505.18134, 2025. URL https://arxiv.org/abs/2505.18134.

[172] Jiazhao Zhang, Gengze Zhou, Hale Yin, Yiyang Huang, Zixing Lei, et al. Qwen-RobotNav technical report: A scalable navigation model designed for an agentic navigation system. arXiv preprint arXiv:2606.18112, 2026.

[173] Kaifeng Zhang, Shuo Sha, Hanxiao Jiang, Matthew Loper, Hyunjong Song, Guangyan Cai, Zhuo Xu, Xiaochen Hu, Changxi Zheng, and Yunzhu Li. Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions. In Proceedings of the IEEE International Conference on Robotics and Automation, 2026. URL https://real2sim-eval.github.io/.

[174] Wenbo Zhang, Kaixuan Wang, Yutao Ouyang, Xiaoyu Huang, Liyang Li, Kailun Su, Weiyang Jin, Wenhao Chai, Haotian Liang, Zhiyang Dou, Yue Chen, and Tianxing Chen. An Unexpected Robot Policy: Early Evaluations of GPT-6 Astra on RoboDojo and Beyond. arXiv preprint arXiv:2609.24170, 2026. URL https://arxiv.org/abs/2609.24170.

[175] Yixian Zhang, Huanming Zhang, Feng Gao, Xiao Li, Zhihao Liu, Yi Nie, Chunyang Zhu, Jiaxing Qiu, Yuchen Yan, Jiyuan Liu, Wenhao Tang, Jiaji Rao, Zhengru Fang, Changxu Wei, Yu Wang, Wenbo Ding, and Chao Yu. Harness VLA: Steering Frozen VLAs into Reliable Manipulation Primitives via Memory-Guided Agents. arXiv preprint arXiv:2607.08448, 2026. URL https://arxiv.org/abs/2607.08448.

[176] Zhanguang Zhang, Zhiyuan Li, Behnam Rahmati, Rui Heng Yang, Yintao Ma, Amir Rasouli, Sajjad Pakdamansavoji, Yangzheng Wu, Lingfeng Zhang, Tongtong Cao, Feng Wen, Xinyu Wang, Xingyue Quan, and Yingxue Zhang. Do World Action Models Generalize Better than VLAs? A Robustness Study. arXiv preprint arXiv:2603.22078, 2026. URL https://arxiv.org/abs/2603.22078.

[177] Zihan Jack Zhang, Sravanthi Machcha, Sabrina Zou, Tzu Kit Chan, and Jay Chooi. GPT-6 Astra vs MolmoAct2 on Bimanual Robotic Manipulation. Online research report; StationeryBench, 2026. URL https://openai.robocurve.org/stationerybench/.

[178] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. In Proceedings of Robotics: Science and Systems, 2023. doi: 10.15607/RSS.2023.XIX.016. URL https://roboticsproceedings.org/rss19/p016.html.

[179] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, Hao Zhang, Joseph Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In Advances in Neural Information Processing Systems, volume 36, pages 46595–46623, 2023. doi: 10.52202/075280-2020. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html.

[180] Yangqing Zheng, Shunqi Mao, Dingxin Zhang, and Weidong Cai. LLM-Enhanced Rapid-Reflex Async-Reflect Embodied Agent for Real-Time Decision-Making in Dynamically Changing Environments. In IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshop on Embodied AI, 2025. URL https://embodied-ai.org/papers/2025/18_LLM_Enhanced_Rapid_Reflex_A.pdf.

[181] Gengze Zhou, Yicong Hong, and Qi Wu. NavGPT: Explicit Reasoning in Vision-and-Language Navigation with Large Language Models. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 38, pages 7641–7649, 2024. doi: 10.1609/aaai.v38i7.28597. URL https://ojs.aaai.org/index.php/AAAI/article/view/28597.

[182] Qinhong Zhou, Sunli Chen, Yisong Wang, Haozhe Xu, Weihua Du, Hongxin Zhang, Yilun Du, Joshua B. Tenenbaum, and Chuang Gan. HAZARD Challenge: Embodied Decision Making in Dynamically Changing Environments. In International Conference on Learning Representations, 2024. URL https://arxiv.org/abs/2401.12975.

[183] Xueyang Zhou, Yangming Xu, Guiyao Tie, Yongchao Chen, Guowen Zhang, Duanfeng Chu, Pan Zhou, and Lichao Sun. LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization. arXiv preprint arXiv:2510.03827, 2025. URL https://arxiv.org/abs/2510.03827.

[184] Chuning Zhu, Raymond Yu, Siyuan Feng, Benjamin Burchfiel, Paarth Shah, and Abhishek Gupta. Unified World Models: Coupling Video and Action Diffusion for Pretraining on Large Robotic Datasets. In Proceedings of Robotics: Science and Systems, 2025. doi: 10.15607/RSS.2025.XXI.015. URL https://www.roboticsproceedings.org/rss21/p015.html.

[185] Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In Proceedings of the 7th Conference on Robot Learning, volume 229 of Proceedings of Machine Learning Research, pages 2165–2183. PMLR, 2023. URL https://proceedings.mlr.press/v229/zitkovich23a.html.

[186] Jing Zuo, Lingzhou Mu, Fan Jiang, Chengcheng Ma, Mu Xu, and Yonggang Qi. FantasyVLN: Unified multimodal chain-of-thought reasoning for vision-language navigation. arXiv preprint arXiv:2601.13976, 2026.

Appendix

A Contributors

Project Leads: Zhongang Cai*, Zhonghua Wu*, Qingyi Tao*.

Simulation Benchmarks: Quanzhou Li*, Han Chong Chua*, Ruisi Wang*, Xiaoming Wu*, Tianqi Liu*, Bo Liu*, Pengfei Liu*, Jing Zuo*, Hongcan Xiao*, Qinyue Li.

Harness Framework: Weiye Xu*, Yuyao Shao*, Silei Wu*, Jiapeng Luo*, Hanming Deng*, Tailai Chen*, Linjun Dai*, Ziyi Shang*, Xiaowan Li, Donghao Wu, Tianbo Zhong, Chenjing Ding.

Real-World Validation: Tianqi Liu*, Ziang Cao*, Hanfei Zhao*, Jiaxu Wu*, Yuexiao Ma*, Xiaoming Wu*, Ziyi Wang*, Weichen Fan*, Zexin He*, Zhi Hou*, Beichen Wen, Zilong Lin, Long Zhuo, Zhichao Liao, Dawei Su.

Benchmark Prototype: Zerui Gong*, Zhe Yang*, Oscar Qian*, Baiqiao Yin*.

Analysis & Presentation: Wanqi Yin*, Chenyang Gu.

Collaborators: Liang Pan, Quan Wang, Haiwen Diao, Xiao Fu, Chunxiao Liu, Bo Li, Hokin Deng.

Advisory Board: Lei Yang*, Chen Change Loy*, Lewei Lu*, Ziwei Liu*, Dahua Lin*, Atsushi Yamashita, Xingang Pan, Dacheng Tao, Xiaogang Wang.

* Core Contributors.

B Related Work

B.1 Foundation Models as Embodied Agents

Large language and vision-language models have long been used to bring semantic knowledge into robot control, but mostly at the level of high-level reasoning. SayCan [68] grounds an LLM's proposals in the affordances of a pretrained skill library, and Inner Monologue [65] closes the loop by feeding success detection and scene descriptions back to the planner. Code as Policies [98] prompts the model to write programs that call perception and control APIs, while VoxPoser [64] and ReKep [66] have it compose 3D value maps or relational keypoint constraints that a motion planner then optimizes. Gemini Robotics-ER [48] extends this line with a model trained specifically for embodied reasoning. In all of these systems the foundation model specifies *what* to do, while a predefined skill set, planner or separately trained controller decides *how*. We instead ask whether a frontier model can act as the controller itself, emitting end-effector actions in closed loop from its own observations.

A second line of work, which has grown rapidly in 2026, borrows the harness paradigm of coding agents [74, 110] and holds that embodied performance depends as much on the infrastructure around the model as on the model itself. Some harnesses redesign the action interface: RoboDawn [54] and Show-Harness [28] expose discrete, semantically meaningful motion commands that a frozen VLM can issue directly, and Robo-Harness K1 [97] equips the model with calibrated perception tools such as depth, object tracking and grasp hypotheses. Others focus on context and verification: Thea [145] maintains a scene graph as persistent context and treats evaluation outcomes as exit codes, and Zetta [36] evolves code-based runtime critics and recovery skills around a frozen base policy. Guava [101] distills a frontier-model harness into a compact open model, while RHO [39], HARBOR [96] and EmbodiedSWE [130] use coding agents offline to build policy repositories, automate reinforcement-learning pipelines, or write programs against the full simulator state. Related studies examine how agents adapt from in-context experience [7, 142] and how reasoning latency interacts with a changing environment [153, 180]. Our ablation in Sec. 3 complements these system-level proposals by isolating the contribution of individual harness components, namely the action interface, subagent roles and memory, under a common protocol.

B.2 Robot Policy Learning

The dominant paradigm for robot control learns a policy from demonstrations. Early large-scale efforts such as BC-Z [71] and RT-1 [17] trained language-conditioned policies on large collections of teleoperated episodes, while Implicit BC [44], BeT [128], ACT [178] and Diffusion Policy [30] introduced action representations better suited to multimodal,

high-precision behavior, and 3D Diffusion Policy [170] further conditions on point clouds. These policies are typically trained and evaluated per task, and most established manipulation benchmarks, including several used in this report, were designed around this setting (Sec. 5).

Vision-language-action models (VLAs) inherit semantic knowledge from web-scale pretraining by fine-tuning a vision-language model to output actions. RT-2 [185] first demonstrated this transfer, OpenVLA [82] provided an open-source counterpart, and Octo [52] and RDT [103] trained generalist policies on cross-embodiment data such as Open X-Embodiment [117] and DROID [79]. The π series [12, 13, 122] pairs a VLM backbone with flow-matching or tokenized action experts and reaches open-world generalization in unseen homes, while GR00T N1 [116] and Gemini Robotics [48] extend the recipe to humanoids and to frontier-scale backbones. Recent systems such as Xiaomi-Robotics-0 [18] and MolmoAct2 [41] target real-time execution and deployable action reasoning. Grounding language in physical action nonetheless remains data-hungry, which has motivated the alternatives below.

World action models (WAMs) start instead from video or world-model backbones and predict how the scene will evolve together with the actions that produce it [129, 148]. UniPi [37] generates videos and recovers actions with an inverse-dynamics model, GR-1 and GR-2 [22, 154] pretrain on large-scale video, and Seer [139], VPP [60], UVA [94] and UWM [184] couple future prediction and action generation more tightly through predictive representations or joint diffusion. In 2026 this family has scaled rapidly. Cosmos Policy [83] and LingBot-VA [93] turn large video foundation models into policies, Fast-WAM [169] finds that most of the benefit comes from video modeling during training rather than from imagining the future at test time, and DreamZero [166] reports that WAMs generalize zero-shot to new tasks and environments substantially better than VLAs. A robustness study on LIBERO-Plus and RoboTwin 2.0-Plus likewise finds WAMs more robust to visual and language perturbations [176].

The most recent systems blur the boundary between the two families. Motus [10] and BagelVLA [61] train unified models in which understanding, video-generation and action experts share attention, VLA-JEPA [137] augments a VLA with a latent world model, and $\pi_{0.7}$ [123] conditions its action expert on subgoal images produced by a world model. Despite their differences, all of these approaches learn a policy from robot data. We ask the complementary question of how far a general-purpose frontier model can go as a policy without benchmark-specific policy training, and our harness ablation (Sec. 3) also invokes a VLA ($\pi_{0.5}$) as a callable tool.

B.3 Embodied Benchmarks

LIBERO [99] remains the most widely used simulation suite for language-conditioned manipulation. LIBERO-Plus [42] and LIBERO-PRO [183] perturb its scenes and instructions to expose brittleness and memorization, and LIBERO-Long++ and LIBERO-Ultra-Long [162] stress visual clutter and chains of 9 to 16 skills. SIMPLER [95] evaluates real-world policies in matched simulation, RoboCasa365 [112] scales simulated household tasks for training and benchmarking generalist robots, RoboDojo [26] unifies simulated and real-robot evaluation of generalist policies, and RoboLab [161] adds high-fidelity scenes with controlled perturbations. For dexterous control, Bench2Dex [163] benchmarks visuo-tactile bimanual manipulation across hand designs, and DexJoCo [144] provides task-oriented dexterous tasks covering tool use, bimanual coordination and long-horizon execution in MuJoCo.

For navigation, Room-to-Room [3] established instruction following on a discrete navigation graph, VLN-CE [87] moved the task to continuous environments with low-level actions, and LHPR-VLN [134] extends it to long-horizon, multi-stage instructions.

Household benchmarks broaden the scope further. BEHAVIOR and BEHAVIOR-1K [8, 90, 135] derive everyday activities from human needs and evaluate full-length household tasks. Housekeep [76] and TIDEE [127] test commonsense tidying without explicit instructions, HAZARD [182] places agents in dynamically changing environments, PARTNR [21] evaluates multi-agent planning, and EmbodiedBench [159] and IS-Bench [106] evaluate multimodal LLM agents on vision-driven tasks and on interactive safety, respectively. Simulation platforms such as Genie Sim [167] continue to raise the fidelity of these evaluations.

Benchmarks lose discriminative power as leading systems saturate them [80, 120]. Evaluation of language agents has responded with harder tasks [4, 20, 138] and with outcome-based protocols in which an agent acts in an environment by any means and is scored on the result [74, 110, 155], a protocol that EmbodiedSWE [130] brings to robotics. As discussed in Sec. 5, most of the embodied benchmarks above were designed for policies trained by imitation, and we argue that general-purpose agents call for a different design.

B.4 Empirical Studies of Frontier Models in Embodied Settings

Each generation of frontier models has prompted empirical studies of its embodied abilities. Shortly after the release of ChatGPT and GPT-4V, Vemprala et al. [141] and Wake et al. [143] examined how these models could be prompted to plan and program robot tasks, and NavGPT [181], MapGPT [24] and Open-Nav [124] studied zero-shot vision-and-language navigation with LLMs reasoning over textual or map-based descriptions of the scene. GPT-4 also generated dense end-effector trajectories and replanned with external visual feedback [89]. CaP-X [46] compared action abstractions, interaction modes, and perceptual grounding, exposing the dependence of model performance on supplied tools and interfaces. Our previous study [19] evaluated GPT-5 on eight spatial-intelligence benchmarks under a unified taxonomy and found unprecedented strength that nonetheless fell short of human performance across a broad range of tasks. The present report extends that effort from spatial to embodied intelligence, and from perception and reasoning to closed-loop control.

The release of GPT-6-Astra [118] set off a wave of such studies within weeks. In navigation, two evaluations on R2R-CE [33, 34] find that GPT-6-Astra rivals or surpasses prior zero-shot and supervised methods from monocular RGB input, yet still struggles to turn correct local judgments into reliable route execution and stopping decisions. In manipulation, Zhang et al. [174] report that GPT-6-Astra ranks first among public entries on the RoboDojo leaderboard while other frontier LLMs succeed in fewer than 2% of trials, but also that one-shot demonstrations lower rather than raise its success rate and that real-robot trials had to be halted because of unsafe actions. Astranomy [158] compares direct, agentic, and policy-assisted control, while Ditto-Bench [104] examines action interfaces, memory, and reasoning effort in contact-rich manipulation. Other studies examine how demonstration formats and accumulated interaction memory affect in-context adaptation [29, 100, 165, 168], the reuse of body knowledge and prior experience together with sim-to-real transfer [57], direct manipulation on LIBERO [35], token-efficient agents that compose classical primitives with VLA calls [132], and GPT-6-Astra's capabilities as an embodied policy more broadly [47, 67, 164]. RoboMME studies examine full-history direct control [72] and hierarchies combining Astra planning, learned execution, and lightweight monitoring [23]. Real-robot studies demonstrate coding-agent control and correction [73], while broader evaluations expose reliability gaps and substantial execution time and inference costs [5, 109, 121, 177]. Vals Robo-bench [140] incorporates inference delays through continuously running physics and wall-clock budgets. The Embodied Agent Arena [62] compares seven VLMs on 1,000 cases drawn from 32 benchmarks and finds that GPT-6-Astra excels at precise estimation and contact localization but not at coordinating goal-directed sequences, Rasheed et al. [125] observe a similar split between semantic and metric-geometric tasks across computer vision, and RoboQuest [102] examines evidence acquisition and exploration stopping decisions. RoboHarm [136] reports that GPT-6-Astra completed 60 of 100 deliberately harmful robot-arm trials and refused only two.

These studies typically examine one benchmark family, one capability or one harness at a time, and their conclusions do not always agree: demonstrations lower GPT-6-Astra's success on RoboDojo in Zhang et al. [174], yet RoboICL [100] reports gains when demonstrations are combined with interaction memory, and RoboDawn [54] reports similar gains from one-shot demonstrations with a frozen VLM. Our study differs in evaluating manipulation and navigation benchmarks under a single protocol, ablating harness design, validating on real robots, and drawing on these results to propose principles for the next generation of embodied benchmarks.

C Full Simulation Results

C.1 LIBERO-PRO

Tasks and demonstrations. LIBERO-PRO [183] applies task-preserving perturbations to the four LIBERO suites and separates each suite into Task and Swap variants. We evaluate the complete official benchmark: eight suites, ten tasks per suite, and ten fixed initial states per task, for 800 episodes in total. Before evaluation, we freeze a package selected by the exact base task, combining matching Task/Swap task-family notes with global strategy, perception, verification, and recovery notes distilled from a separate public exploration phase. The package contains no reported evaluation trajectory or outcome, current initial state, or reusable pixels, coordinates, object suffixes, or action sequences. We refer to our configuration as **GPT-6-Astra + ICL**. It uses maximum reasoning effort and direct robot control, with both the VLA and SAM3 disabled. All 800 episodes are valid and uncensored.

Observation and control settings. GPT-6-Astra observes calibrated RGB images from a fixed third-person `agentview` camera and the Panda wrist camera (native 256×256 observations, with 1024×1024 inspection renders), together

Embodiment / benchmark	Published baselines				GPT-6-Astra (ours)		
	SOTA		Agentic SOTA		Up to 1×	Up to 10×	
	Method	SR	Method	SR		Cutoff at 10×	Cutoff at 1×
Tabletop robotic arms							
LIBERO-Long++ [162]	LiLo-VLA [162]	82.0	–	–	97.5	100.0	97.5
LIBERO-PRO [183]	π_{RLinf} [175]	50.0	RPent / GPT-6 Astra [175]	92.6	98.3	99.3	98.3
LIBERO-Ultra-Long [162]	LiLo-VLA [162]	44.0	–	–	82.2	98.9	82.2
RoboDojo [26]	Awomo-0.5 [6]	29.6	PhysicalRSI [58]	31.4	54.2*	81.9	27.6
RoboLab‡ [161]	FLUX 3 Action [15]	42.9	–	–	57.1	69.2	57.1
Humanoid							
SIMPLE [152]	Psi-0 [151]	92.5	–	–	96.3	98.8	97.5
Mobile platforms							
BEHAVIOR [8]	Rondo [8]	14.8	Agent [8]	5.3	37.0	–	–
Dexterous hands							
DexJoCo [144]	$\pi_{0.5}$ [12]	40.3	–	–	78.2	87.3	78.2
Bench2Dex [163]	$\pi_{0.5}$ [12]	25.5	–	–	25.6	61.5	25.6
Navigation							
VLN-CE RxR [87]	Qwen-RobotNav [172]	76.5	–	–	82.0	90.0	84.0
VLN-CE R2R [87]	Robostral Navigate [16]	77.4	GPT-6-Astra (ultra) [33]	81.3	80.0	88.0	85.0
LHPR-VLN [134]	FantasyVLN [186]	2.4	Claude Code Opus-5	5.3	21.1	28.9	25.4

Table 2 Benchmark-level success rates (%) underlying Fig. 1. For each benchmark, we list the best published result (SOTA) with its method, the best published agentic method (Agentic SOTA), and GPT-6-Astra + ICL under the standard (1×) and cumulative extended (up to 10×) budgets. Benchmarks are grouped by embodiment. Published baselines are reported under the standard budget. The 1× execution in this table is performed separately from 10×, with the respective budget included in the context during execution. Dashes mean no published agentic result or no extended-budget run. ∗ denotes the result with VLA model, while ‡ denotes the zero-shot results without access to the training data.

with the measured end-effector pose and gripper state. The agent interface also provides calibration-based pixel back-projection into the current scene; LIBERO-PRO does not expose object poses or evaluator diagnostics to the planner. The underlying action is the benchmark's seven-dimensional Panda OSC-pose command at 20 Hz: three translational components, three axis–angle rotational components, and one gripper command. The interface compiles model-selected Cartesian motion, wrist-rotation, and gripper primitives into these actions. A unit translational or rotational component corresponds to 5 cm or 0.5 rad, respectively.

Budgets and continuation. The standard 1× boundary is the first of 10,000 environment steps or 100 planner turns. An unsuccessful episode then continues from the same simulator state and planner conversation to a cumulative 10× environment-step budget; the continuation is not terminated by the 100-turn 1× cap. The frozen 1× outcome is never replaced by the continuation. Table 3 reports success rate (SR) for every paired Task/Swap base task, the four family aggregates, and all 800 episodes, while integrating prior agentic and policy-based methods at their reported suite-level granularity.

Results and comparison. At 1×, **GPT-6-Astra + ICL** succeeds in 786 of 800 episodes (98.25%), exceeding the published RPent/GPT-6-Astra baseline with low reasoning [175], the strongest prior result, by 5.62 percentage points. It obtains 99–100% SR on six suites; the remaining errors concentrate in Long Task (97%) and Long Swap (90%). Continuing to the cumulative 10× boundary raises the result to 794/800 (99.25%), including 99% on Long Task and 95% on Long Swap. Table 3 also reports the mean test-time token and wall-clock cost at both boundaries.

Task family / base task	Published baselines				GPT-6-Astra + ICL (ours)								
	Task / Swap SR (%)					Task / Swap SR (%)		Tokens (millions)			Time (minutes)		
	GPT-6-Astra [175]	ASPIRE [105]	π_{RLinf} [175]	$\pi_{0.5}$ [12]	N	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit
Spatial	100 / 98	60 / 51	42 / 59	1 / 20	200	100 / 100	100 / 100	1.61	0.36	0.36	18.8	13.4	13.4
Black bowl: between plate/ramekin → plate	–	–	–	–	20	100 / 100	100 / 100	1.67	0.32	0.32	19.0	10.1	10.1
Black bowl: next to ramekin → plate	–	–	–	–	20	100 / 100	100 / 100	1.76	0.34	0.34	23.0	11.9	11.9
Black bowl: table center → plate	–	–	–	–	20	100 / 100	100 / 100	1.71	0.30	0.30	18.7	10.3	10.3
Black bowl: on cookie box → plate	–	–	–	–	20	100 / 100	100 / 100	1.42	0.38	0.38	17.3	12.5	12.5
Black bowl: top cabinet drawer → plate	–	–	–	–	20	100 / 100	100 / 100	1.72	0.45	0.45	18.3	15.1	15.1
Black bowl: on ramekin → plate	–	–	–	–	20	100 / 100	100 / 100	1.40	0.27	0.27	18.1	10.5	10.5
Black bowl: next to cookie box → plate	–	–	–	–	20	100 / 100	100 / 100	1.44	0.47	0.47	18.7	20.8	20.8
Black bowl: on stove → plate	–	–	–	–	20	100 / 100	100 / 100	1.88	0.31	0.31	18.9	14.1	14.1
Black bowl: next to plate → plate	–	–	–	–	20	100 / 100	100 / 100	1.83	0.35	0.35	18.7	13.9	13.9
Black bowl: on cabinet → plate	–	–	–	–	20	100 / 100	100 / 100	1.30	0.37	0.37	17.0	14.6	14.6
Object	100 / 99	95 / 98	71 / 78	1 / 17	200	100 / 100	100 / 100	1.76	0.28	0.28	19.6	18.1	18.1
Alphabet soup → basket	–	–	–	–	20	100 / 100	100 / 100	1.56	0.26	0.26	21.6	12.6	12.6
Cream cheese → basket	–	–	–	–	20	100 / 100	100 / 100	1.60	0.31	0.31	19.1	15.7	15.7
Salad dressing → basket	–	–	–	–	20	100 / 100	100 / 100	2.42	0.26	0.26	21.4	17.8	17.8
BBQ sauce → basket	–	–	–	–	20	100 / 100	100 / 100	1.52	0.30	0.30	20.8	21.1	21.1
Ketchup → basket	–	–	–	–	20	100 / 100	100 / 100	1.83	0.31	0.31	18.9	25.6	25.6
Tomato sauce → basket	–	–	–	–	20	100 / 100	100 / 100	2.11	0.28	0.28	21.0	24.7	24.7
Butter → basket	–	–	–	–	20	100 / 100	100 / 100	1.45	0.29	0.29	16.6	22.1	22.1
Milk → basket	–	–	–	–	20	100 / 100	100 / 100	1.45	0.29	0.29	19.3	16.7	16.7
Chocolate pudding → basket	–	–	–	–	20	100 / 100	100 / 100	2.15	0.26	0.26	19.0	13.3	13.3
Orange juice → basket	–	–	–	–	20	100 / 100	100 / 100	1.55	0.23	0.23	18.7	11.7	11.7
Goal	88 / 99	45 / 81	45 / 42	2 / 38	200	99 / 100	100 / 100	2.04	0.43	0.43	20.1	17.1	17.2
Open middle cabinet drawer	–	–	–	–	20	100 / 100	100 / 100	1.54	0.94	0.94	16.8	29.7	29.7
Bowl → stove	–	–	–	–	20	90 / 100	100 / 100	1.46	0.48	0.49	18.0	16.3	16.8
Wine bottle → cabinet top	–	–	–	–	20	100 / 100	100 / 100	1.49	0.27	0.27	20.3	11.2	11.2
Open top drawer; bowl inside	–	–	–	–	20	100 / 100	100 / 100	2.78	0.66	0.66	24.1	21.7	21.7
Bowl → cabinet top	–	–	–	–	20	100 / 100	100 / 100	2.32	0.35	0.35	20.8	13.3	13.3
Push plate to stove front	–	–	–	–	20	100 / 100	100 / 100	2.55	0.54	0.54	20.3	24.9	24.9
Cream cheese → bowl	–	–	–	–	20	100 / 100	100 / 100	2.97	0.28	0.28	20.7	14.4	14.4
Turn on stove	–	–	–	–	20	100 / 100	100 / 100	1.66	0.15	0.15	23.8	9.9	9.9
Bowl → plate	–	–	–	–	20	100 / 100	100 / 100	2.11	0.26	0.26	18.3	13.5	13.5
Wine bottle → rack	–	–	–	–	20	100 / 100	100 / 100	1.50	0.40	0.40	17.8	16.1	16.1
Long-horizon	85 / 72	38.3 / 22.6	49 / 14	1 / 8	200	97 / 90	99 / 95	1.83	1.02	2.92	19.8	27.5	46.6
Soup and tomato sauce → basket	–	–	–	–	20	100 / 100	100 / 100	2.16	0.61	0.61	22.1	20.0	20.0
Cream cheese and butter → basket	–	–	–	–	20	100 / 90	100 / 90	1.74	0.57	4.41	17.8	16.7	52.8
Turn on stove; moka pot on stove	–	–	–	–	20	100 / 100	100 / 100	2.47	0.71	0.71	24.2	22.8	22.8
Black bowl in bottom drawer; close drawer	–	–	–	–	20	80 / 100	100 / 100	1.60	1.65	1.91	19.9	35.0	39.7
White mug left; yellow-white mug right	–	–	–	–	20	100 / 100	100 / 100	1.70	0.87	0.87	19.9	32.3	32.3
Book → back of caddy	–	–	–	–	20	90 / 90	90 / 90	1.55	0.50	3.78	19.1	21.7	56.1
White mug on plate; pudding to right	–	–	–	–	20	100 / 100	100 / 100	1.97	0.64	0.64	21.6	20.9	20.9
Soup and cream cheese → basket	–	–	–	–	20	100 / 100	100 / 100	1.46	0.64	0.64	17.4	23.4	23.4
Both moka pots → stove	–	–	–	–	20	100 / 90	100 / 100	1.57	1.04	1.09	17.3	32.7	33.9
Yellow-white mug in microwave; close	–	–	–	–	20	100 / 30	100 / 70	2.11	3.00	14.51	19.2	49.1	164.0
Overall / mean	**92.63**	**61.36**	**50.00**	**11.00**	**800**	**98.25**	**99.25**	1.81	0.52	1.00	19.6	19.0	23.8

Note. Group aggregate rows precede their ten base-task rows. Within a family, each performance entry is Task / Swap SR; the overall row is a single SR over all 800 episodes. The four baseline columns retain GPT-6-Astra and the three strongest representative policy baselines from the suite-level comparison. The GPT-6-Astra column denotes the published RPent configuration with low reasoning, which also uses the official RPent memory; our method uses maximum reasoning and no VLA or SAM3 calls. Dashes mean unavailable task-level baseline results, not zero. N is the number of our evaluation episodes. Official RPent memory was already publicly available; for the final evaluation, we reorganized this memory by task into ICL packages. Because the official exploration cost was not reported, the ICL column provides, for reference, the actual mean cost of our reproduction of 40 independent task-specific explorations. The packages produced by these reproduced explorations were not used in the final evaluation. Test costs exclude ICL; up-to-10× includes the original 1× execution and continuation in the same simulator state and conversation, and is not directly comparable with standard-budget baselines. Tokens are observed provider input-plus-output usage and are lower bounds because 117 episodes have partial usage accounting; wall time is fully observed.

Table 3 Task-level performance and computational cost on LIBERO-PRO [183]. Each task row pairs the official Task and Swap perturbations of one base task (ten fixed initial states per perturbation). Four representative published baselines are integrated into the family aggregate rows because their task-level results are not reported. Our GPT-6-Astra + ICL results use the standard (1×) and cumulative extended (up to 10×) budgets.

C.2 LIBERO-Long++ and LIBERO-Ultra-Long

Tasks and demonstrations. LIBERO-Long++ and LIBERO-Ultra-Long [162] evaluate compositional manipulation under visual clutter and reordered skill chains. Long++ contains six scenarios in their original and reordered forms, yielding 12 configurations; Ultra-Long contains three scenarios under the original ordering and two reorderings, yielding nine configurations. We follow the released protocol and run ten independent bare-random resets per configuration, for 120 Long++ and 90 Ultra-Long episodes. Before testing, we construct one frozen package per composite task solely from the released atomic-skill training split, which contains 4,050 trajectories (150 per atomic skill; 351,999 transitions). The package summarizes canonical, geometric-variation, and recovery examples as phase, contact, verification, and failure cues with labeled agent-view and wrist-view contact sheets. It contains no evaluation rollout, initial state, reward, success label, or test-specific coordinates. We refer to our configuration as **GPT-6-Astra + ICL**. It uses maximum reasoning effort and direct robot control, with both the VLA and SAM3 disabled.

Benchmark / task	Published baselines			GPT-6-Astra + ICL (ours)								
	SR / AP (%)				SR / AP (%)		Tokens (millions)			Time (minutes)		
	LiLo-VLA [162]	$\pi_{0.5}$ [12]	OpenVLA-OFT [81]	N	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit
Long++: Original	78.0 / 88.0	83.0 / 93.0	7.0 / 11.0	60	98.3 / 99.6	100.0 / 100.0	1.68	1.04	1.08	20.3	56.8	57.9
Stove, then moka pot	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.95	0.80	0.80	21.1	53.0	53.0
Soup, then cream cheese	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.58	1.26	1.26	19.9	58.5	58.5
Soup, then tomato sauce	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.56	1.16	1.16	17.7	44.5	44.5
Cream cheese, then butter	–	–	–	10	90.0 / 97.5	100.0 / 100.0	1.62	1.22	1.49	20.2	66.4	72.8
White mug, then yellow-white mug	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.97	1.00	1.00	22.4	64.0	64.0
White mug, then pudding	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.43	0.80	0.80	20.4	54.3	54.3
Long++: Reordered	85.0 / 89.0	0.0 / 0.0	0.0 / 0.0	60	96.7 / 98.9	100.0 / 100.0	1.76	1.04	1.06	20.3	49.4	50.0
Moka pot, then stove	–	–	–	10	80.0 / 93.3	100.0 / 100.0	3.28	1.12	1.22	24.4	57.9	61.5
Cream cheese, then soup	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.21	1.03	1.03	22.1	62.2	62.2
Tomato sauce, then soup	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.48	1.23	1.23	17.6	75.7	75.7
Butter, then cream cheese	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.81	1.06	1.06	21.2	48.1	48.1
Yellow-white mug, then white mug	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.21	0.94	0.94	17.1	31.3	31.3
Pudding, then white mug	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.59	0.86	0.86	19.2	21.3	21.3
LIBERO-Long++ average	**82.0 / 89.0**	**42.0 / 46.0**	**3.0 / 5.0**	**120**	**97.5 / 99.2**	**100.0 / 100.0**	1.72	1.04	1.07	20.3	53.1	53.9
Ultra-Long: Original	53.0 / 79.0	0.0 / 1.0	0.0 / 0.0	30	63.3 / 90.1	100.0 / 100.0	2.08	9.19	10.18	21.2	185.9	199.4
Complete Kitchen Organization	–	–	–	10	50.0 / 83.3	100.0 / 100.0	2.09	4.35	5.80	21.1	118.7	142.5
Organize Table	–	–	–	10	70.0 / 95.0	100.0 / 100.0	2.14	16.86	17.75	20.3	268.0	280.3
Cooking Preparation Setup	–	–	–	10	70.0 / 92.0	100.0 / 100.0	2.00	6.35	7.00	22.2	171.1	175.3
Ultra-Long: Variant 1	37.0 / 84.0	0.0 / 0.0	0.0 / 0.0	30	93.3 / 98.1	96.7 / 98.3	2.03	8.37	8.38	21.3	84.0	84.1
Complete Kitchen Organization	–	–	–	10	100.0 / 100.0	100.0 / 100.0	2.31	4.14	4.14	23.7	73.6	73.6
Organize Table	–	–	–	10	80.0 / 94.4	90.0 / 95.0	2.32	15.27	15.30	20.7	109.0	109.3
Cooking Preparation Setup	–	–	–	10	100.0 / 100.0	100.0 / 100.0	1.48	5.69	5.69	19.4	69.3	69.3
Ultra-Long: Variant 2	43.0 / 74.0	0.0 / 0.0	0.0 / 0.0	30	90.0 / 98.8	100.0 / 100.0	1.94	9.11	9.59	21.8	62.8	67.0
Complete Kitchen Organization	–	–	–	10	100.0 / 100.0	100.0 / 100.0	2.12	4.64	4.64	23.6	50.3	50.3
Organize Table	–	–	–	10	80.0 / 97.5	100.0 / 100.0	2.14	17.77	19.20	19.7	91.7	104.2
Cooking Preparation Setup	–	–	–	10	90.0 / 99.0	100.0 / 100.0	1.56	4.92	4.94	22.2	46.4	46.6
LIBERO-Ultra-Long average	**44.0 / 79.0**	**0.0 / 0.3**	**0.0 / 0.0**	**90**	**82.2 / 95.7**	**98.9 / 99.4**	2.02	8.89	9.38	21.4	110.9	116.8
Overall / mean	**69.0 / 86.0**	**28.0 / 31.0**	**2.0 / 4.0**	**210**	**92.4 / 98.1**	**99.6 / 99.8**	1.85	4.40	4.63	20.8	77.9	80.9

Note. Ordering-level aggregate rows precede their task rows; highlighted rows report the two benchmark averages and the cross-benchmark overall mean. Dashes indicate that the published baselines report only ordering-level aggregates, not task-level results. N is the number of our evaluation episodes; published baselines use their released evaluation protocols. The overall performance row macro-averages the nine base scenarios (six Long++ and three Ultra-Long), whereas its resource columns average the 21 task preparations and 210 scored episodes. ICL is a one-time task-specific preparation cost; evaluation costs exclude ICL. The up-to-10× columns include the original 1× execution and continuation in the same simulator state and conversation, and are not directly comparable with standard-budget baselines. Tokens are observed provider input-plus-output usage and are lower bounds because usage accounting is partial for 8/120 Long++ and 27/90 Ultra-Long episodes; wall time is fully observed.

Table 4 Task-level performance and computational cost on LIBERO-Long++ and LIBERO-Ultra-Long [162]. Published baselines are integrated into the aggregate rows because their task-level results are not reported. Our GPT-6-Astra + ICL experiment evaluates ten episodes per task at the standard (1×) and cumulative extended (up to 10×) budgets. Performance entries are success rate (SR) / average progress (AP), both in percent.

Observation and control settings. GPT-6-Astra observes calibrated RGB images from a fixed third-person `agentview` camera and the Panda wrist camera (native 256×256 observations, with 1024×1024 inspection renders), the measured

end-effector pose and gripper state, and the released ordered-skill and instance-pose feedback used by our LiLo adapter. Calibration-based pixel back-projection supplies current-scene 3D geometry. The underlying action is the benchmark's seven-dimensional Panda OSC-pose command at 20 Hz: three translational components, three axis–angle rotational components, and one gripper command. The agent interface compiles model-selected Cartesian motion, wrist-rotation, and gripper primitives into these actions. A unit translational or rotational component corresponds to 5 cm or 0.5 rad, respectively.

Budgets and continuation. The standard $1\times$ step budgets are 600 or 800 steps on Long++ and 1,800, 2,000, or 3,200 steps on Ultra-Long. Success rate (SR) requires completion of the entire ordered chain within this boundary. Average progress (AP) is the fraction of the uninterrupted correct skill prefix completed at the same boundary. An unsuccessful episode continues from the same simulator state and planner conversation to a cumulative $10\times$ step budget; the $1\times$ outcome and prefix snapshot remain frozen. There is no separate planner-turn cap; a 12-hour planner wall-clock guard handles infrastructure censoring but does not change the environment-step budget. Prior work does not report an agentic method under this protocol. Table 4 reports every task and compares **GPT-6-Astra + ICL** with LiLo-VLA and two learned-policy baselines in the corresponding ordering-level aggregate rows.

Results and comparison. At $1\times$, **GPT-6-Astra + ICL** achieves 97.5% SR and 99.2% AP on LIBERO-Long++, and 82.2% SR and 95.7% AP on LIBERO-Ultra-Long, exceeding LiLo-VLA by 15.5 and 38.2 percentage points in SR, respectively. The cumulative $10\times$ results reach 100.0% SR/AP on Long++ and 98.9% SR with 99.4% AP on Ultra-Long. Macro-averaged over the nine scenarios, the corresponding results are 92.4%/98.1% at $1\times$ and 99.6%/99.8% at $10\times$. Table 4 also reports one-time preparation cost per composite task and mean test-time cost per episode.

C.3 RoboDojo

Tasks and demonstrations. We refer to our experiment as **GPT-6-Astra + ICL**. We evaluate all 42 RoboDojo tasks across 5 groups [26]. Generalization, Precision, Long-Horizon and Memory contain 34 tasks in total; Open contains 8 tasks. Each task uses `layout_id=0-4`, giving 210 evaluation episodes. We fix `seed=0`; each layout is initialized with `env.reset(seed=[layout_id])`. For each task in Generalization, Precision, Long-Horizon and Memory, GPT-6-Astra studies 10 demonstrations of that task. The 8 Open tasks have no task-specific demonstrations and share one study of 340 demonstrations from the other 4 groups, 10 per task.

Observation and control settings. Our RoboDojo experiments use the dual-arm ARX X5 robot, and GPT-6-Astra receives 640×480 RGB images from head, left-wrist and right-wrist cameras, measured joint positions, gripper states and end-effector poses. Privileged information is withheld: object poses, depth images, segmentation masks, contact information, hidden simulator state, task source code and evaluator diagnostics. For each arm, GPT-6-Astra outputs a target end-effector position (x, y, z) in meters in the environment coordinate frame, a target orientation as a unit quaternion (w, x, y, z), and an open or close gripper command. Each target differs from the current measured pose by at most 5 cm in position and 0.35 rad in orientation. Inverse kinematics converts these end-effector targets into joint targets, with each control step limiting end-effector translation to 2 cm, rotation to 0.1 rad and each joint's rotation to 0.05 rad. Control runs at 25 Hz; an action request can repeat a target for 1–10 control steps. After each action request, GPT-6-Astra uses the returned images and measured robot state to choose the next action.

Budgets and continuation. We evaluate GPT-6-Astra + ICL with the $1\times$ and $10\times$ step limits. Each task has a standard control-step limit B ranging from 200 to 1,900 steps, with evaluation limits of B and $10B$, respectively. We first run each episode for up to B control steps, or until earlier termination, and measure the $1\times$ result from this initial execution. If the episode has not terminated, we continue the same episode up to a total of $10B$ steps and record the $10\times$ result. Execution ends when the simulator terminates the episode or GPT-6-Astra stops.

Task group / task	Published baselines			GPT-6-Astra + ICL (ours)								
		Mean score / success rate			Mean score / success rate		Token usage (millions)			Time (minutes)		
	N	GPT-6-Astra [174]	Awomo-0.5 [6]	N	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit	ICL	1× step limit	up to 10× step limit
Generalization	600	35.32 / 32.67	37.70 / 30.89	60	34.92 / 28.33	81.42 / 75.00	3.03	15.58	52.47	28.9	82.2	329.2
Stack bowls	50	68.80 / 64.00	74.07 / 70.67	5	66.00 / 60.00	100.00 / 100.00	2.55	14.99	18.98	28.2	99.2	131.0
Push T	50	68.00 / 68.00	5.33 / 5.33	5	20.00 / 20.00	100.00 / 100.00	3.31	12.62	21.54	30.8	79.0	144.4
Pack objects into box	50	11.00 / 0.00	31.33 / 10.67	5	24.00 / 0.00	80.00 / 60.00	3.42	21.32	98.96	26.9	100.7	569.5
Fold clothes	50	76.80 / 72.00	74.93 / 70.67	5	60.00 / 60.00	84.00 / 80.00	3.37	9.49	26.50	30.1	40.6	146.0
Hang mugs	50	0.60 / 0.00	36.33 / 18.67	5	19.00 / 0.00	88.00 / 80.00	3.25	17.91	59.06	31.3	91.9	338.8
Sweep blocks	50	8.00 / 8.00	1.33 / 1.33	5	0.00 / 0.00	80.00 / 80.00	4.07	21.08	69.73	31.4	123.0	353.8
Pour liquid into cup	50	8.00 / 8.00	64.00 / 64.00	5	60.00 / 60.00	60.00 / 60.00	2.40	8.26	18.24	25.7	50.8	117.3
Make toast	50	0.00 / 0.00	19.33 / 12.00	5	5.00 / 0.00	45.00 / 40.00	2.46	28.44	128.47	27.4	167.0	786.0
Arrange largest number	50	76.00 / 68.00	26.07 / 14.67	5	54.00 / 40.00	100.00 / 100.00	2.66	11.65	18.96	28.7	53.1	103.0
Sort nesting dolls by size	50	16.00 / 16.00	22.67 / 22.67	5	40.00 / 40.00	100.00 / 100.00	2.69	17.64	21.24	29.5	65.0	91.6
Store laptop and headphones	50	0.80 / 0.00	18.93 / 5.33	5	8.00 / 0.00	40.00 / 0.00	3.27	15.48	137.86	27.6	82.1	1125.3
Stack blocks	50	89.80 / 88.00	78.07 / 74.67	5	63.00 / 60.00	100.00 / 100.00	2.91	8.11	10.08	28.6	34.0	43.1
Precision	400	12.65 / 4.00	39.66 / 33.50	40	28.62 / 15.00	91.00 / 87.50	3.40	9.17	37.86	32.5	35.4	198.0
Fasten screws	50	36.40 / 16.00	69.00 / 48.00	5	12.00 / 0.00	80.00 / 60.00	3.14	22.38	178.63	32.7	65.9	922.5
Insert tubes	50	6.00 / 0.00	79.47 / 71.33	5	40.00 / 0.00	100.00 / 100.00	2.79	8.15	11.55	33.8	25.3	43.4
Plug in charger	50	0.00 / 0.00	7.33 / 7.33	5	40.00 / 40.00	100.00 / 100.00	3.18	5.54	7.22	33.0	27.1	39.9
Pour balls into vase	50	4.00 / 4.00	39.33 / 39.33	5	80.00 / 80.00	100.00 / 100.00	3.65	8.00	18.09	36.8	45.8	93.8
Play xylophone	50	8.00 / 8.00	30.67 / 30.67	5	0.00 / 0.00	100.00 / 100.00	3.69	9.66	15.57	30.3	36.9	79.2
Deposit coin	50	16.00 / 2.00	24.27 / 19.33	5	20.00 / 0.00	100.00 / 100.00	3.55	5.32	12.02	29.5	26.7	71.3
Insert key	50	14.40 / 0.00	11.50 / 0.00	5	15.00 / 0.00	66.00 / 60.00	4.03	3.75	25.04	33.5	19.5	173.1
Build tower	50	16.40 / 2.00	55.67 / 52.00	5	22.00 / 0.00	82.00 / 80.00	3.18	10.57	34.78	30.2	36.1	160.8
Long-Horizon	400	21.45 / 8.25	40.36 / 26.75	40	41.37 / 20.00	95.00 / 90.00	3.24	16.71	38.42	27.3	66.9	195.8
Put bottles into dustbin	50	35.30 / 10.00	94.33 / 91.33	5	61.00 / 40.00	100.00 / 100.00	2.83	7.85	10.13	28.1	36.3	50.8
Play tic tac toe	50	10.70 / 0.00	58.70 / 36.00	5	65.00 / 20.00	100.00 / 100.00	3.73	10.79	13.54	30.1	34.4	45.7
Classify objects	50	69.00 / 56.00	31.20 / 20.67	5	51.00 / 40.00	100.00 / 100.00	3.14	17.84	33.42	29.8	80.3	151.0
Fill pen holder	50	5.00 / 0.00	48.27 / 26.00	5	28.00 / 0.00	100.00 / 100.00	2.89	21.18	55.06	26.1	99.2	263.2
Fill egg holder	50	5.50 / 0.00	7.20 / 0.67	5	27.00 / 20.00	70.00 / 60.00	4.58	14.59	84.66	37.0	66.2	658.9
Organize table	50	39.50 / 0.00	49.50 / 9.33	5	45.00 / 0.00	90.00 / 60.00	2.92	30.53	65.50	10.2	121.2	237.4
Play stacking toy	50	6.60 / 0.00	3.67 / 0.00	5	14.00 / 0.00	100.00 / 100.00	2.73	20.91	34.19	28.2	65.4	122.5
Make kong	50	0.00 / 0.00	30.00 / 30.00	5	40.00 / 40.00	100.00 / 100.00	3.09	9.95	10.84	29.1	32.4	36.9
Memory	300	43.03 / 38.67	42.27 / 41.22	30	73.67 / 70.00	96.67 / 96.67	1.54	10.11	10.55	6.7	27.1	29.8
Cover blocks	50	49.30 / 46.00	100.00 / 100.00	5	86.00 / 80.00	100.00 / 100.00	1.20	6.66	6.78	6.1	16.6	17.2
Match and pick from conveyor	50	62.00 / 62.00	66.00 / 66.00	5	100.00 / 100.00	100.00 / 100.00	1.57	4.50	4.50	5.6	11.4	11.4
Swap T	50	18.00 / 18.00	0.00 / 0.00	5	0.00 / 0.00	100.00 / 100.00	1.51	5.59	7.32	7.2	10.4	22.6
Press by number	50	70.00 / 70.00	67.33 / 67.33	5	100.00 / 100.00	100.00 / 100.00	1.41	5.92	5.92	7.6	21.5	21.5
Imitate sorting sequence	50	58.90 / 36.00	15.60 / 9.33	5	56.00 / 40.00	80.00 / 80.00	1.48	27.97	28.77	7.6	80.1	83.9
Swap blocks	50	0.00 / 0.00	4.67 / 4.67	5	100.00 / 100.00	100.00 / 100.00	2.07	9.99	9.99	6.3	22.3	22.3
Open	400	34.36 / 31.00	24.09 / 22.92	40	17.63 / 15.00	73.00 / 67.50	8.13	8.64	39.91	105.3	42.1	227.2
Align blocks	50	50.00 / 50.00	58.00 / 58.00	5	0.00 / 0.00	100.00 / 100.00	8.13	4.48	5.83	105.3	20.0	26.9
Solve equation	50	40.00 / 40.00	50.00 / 50.00	5	20.00 / 20.00	100.00 / 100.00	8.13	4.07	6.21	105.3	17.7	26.5
Stack blocks by language	50	48.00 / 40.00	14.67 / 11.33	5	8.00 / 0.00	100.00 / 100.00	8.13	5.20	9.58	105.3	20.2	35.5
General pickup	50	84.00 / 84.00	60.67 / 60.67	5	100.00 / 100.00	100.00 / 100.00	8.13	3.46	3.46	105.3	16.8	16.8
Classify objects by language	50	46.00 / 30.00	8.73 / 3.33	5	4.00 / 0.00	100.00 / 100.00	8.13	14.63	43.29	105.3	62.7	183.1
Pick from conveyor by image	50	4.00 / 4.00	0.00 / 0.00	5	0.00 / 0.00	20.00 / 20.00	8.13	14.44	49.26	105.3	87.1	355.1
Store tools in toolbox	50	2.50 / 0.00	0.67 / 0.00	5	5.00 / 0.00	40.00 / 0.00	8.13	10.36	150.49	105.3	47.5	909.0
Pour by language	50	0.40 / 0.00	0.00 / 0.00	5	4.00 / 0.00	24.00 / 20.00	8.13	12.51	51.16	105.3	65.0	265.1
Overall / mean	**2100**	**29.28 / 23.10**	**36.64 / 30.56**	**210**	**37.19 / 27.62**	**86.40 / 81.90**	3.05	12.47	38.63	27.7	54.9	216.6

Note. N denotes the number of evaluation episodes. Performance entries report mean score (0–100) / success rate (%). Group means average tasks within each group; the overall mean averages all 42 tasks. ICL costs average the study costs associated with each task's 5 evaluated layouts. Evaluation costs average all episodes, including successes, failures and timeouts, and exclude ICL costs; the 10× costs include both initial execution and continuation. Tokens count input (including cached input) and output. Time excludes simulator action calls and trajectory replay. The overall ICL cost averages 34 task-specific estimates and 1 shared estimate for the Open tasks, while the overall evaluation cost averages 210 episodes.

Table 5 Performance and computational cost by task group and task on RoboDojo [26]. The published GPT-6-Astra baseline is reported by the RoboDojo team [174]. Our ICL experiment using GPT-6-Astra evaluates 5 episodes per task (210 in total). Group aggregate rows precede their task results. Results are reported at the standard control-step limit (1×) and with an extended budget of up to 10× that limit. Token usage and time refer to our ICL experiment using GPT-6-Astra.

C.4 RoboLab

Benchmark and tasks. RoboLab [161] evaluates task-generalist manipulation policies in high-fidelity simulation. Its RoboLab-120 task library comprises 120 language-conditioned tasks organized along three competency axes: visual recognition, relational understanding, and procedural reasoning. Tasks span three difficulty levels and include object selection, spatial rearrangement, stacking, and reorientation. The benchmark supports evaluation of policies trained on real-world data without requiring co-training on simulation data.

Policies and evaluation metric. Table 6 compares GPT-6-Astra with nine baselines spanning vision-language-action models (VLAs) and world action models (WAMs). We report success rate (SR), expressed as a percentage, together with policy type and model size. Baseline results are taken from the official RoboLab leaderboard for the default language instructions; each listed baseline is evaluated over 1,200 episodes. Model sizes follow the reported parameter counts. These published results provide reference points for the agent evaluation; the table does not establish a comparison at matched inference cost.

Policy	Type	SR (%)	Size
GPT-6-Astra 10x	Agent	**69.2**	–
GPT-6-Astra 1x	Agent	57.1	–
FLUX 3 Action [15]	WAM	42.9	7B
Cosmos3-Nano-Policy [115]	WAM	36.8	16B
$\pi_{0.5}$ [12]	VLA	28.0	3.3B
DreamZero [166]	WAM	25.7	14B
Cosmos3-Edge-Policy [115]	WAM	22.9	4B
π_0-FAST [122]	VLA	15.5	3B
GR00T N1.6 [116]	VLA	7.2	3B
π_0 [13]	VLA	5.0	3.3B
paligemma-binning [9]	VLA	3.4	3B

Note. Baseline scores and model sizes follow the default-instruction results on the official RoboLab leaderboard [161] (accessed 15 October 2026). Each listed baseline reports 1,200 evaluation episodes (120 tasks with 10 episodes per task). VLA: vision-language-action model; WAM: world action model. B denotes billions of parameters; – indicates that model size is not reported here.

Table 6 Overall performance on RoboLab. Success rate (SR) is reported in percent; higher is better. Policies are ordered by success rate, and the best result is shown in bold.

Results. Under the standard ($1\times$) budget, GPT-6-Astra achieves a success rate of 57.1%, exceeding all baselines listed in Table 6. It exceeds FLUX 3 Action, the strongest listed baseline at 42.9%, by 14.2 percentage points. Among the listed VLAs, $\pi_{0.5}$ performs best at 28.0%, trailing GPT-6-Astra by 29.1 points. The aggregate results show an advantage for GPT-6-Astra on this comparison, while task-level outcomes would be needed to identify which competency axes account for the difference.

C.5 Bench2Dex

Tasks and demonstrations. We evaluate all 26 Bench2Dex [163] tasks across 12 arm–hand embodiments under Full randomization (`inv_cov`, unseen split). The paired cohort contains three fixed episodes per task (indices 5–7), totaling 78 episodes. Preparation uses all 100 released seen-split replay recordings per task, including rendering variants of shared motion trajectories. Demonstrations expose four-camera RGB, active-joint positions, and demonstrated joint targets. Discrepancies between recorded and current task instructions are flagged, with the current instruction remaining authoritative.

Observation and control settings. GPT-6-Astra receives stereo left/right and wrist left/right RGB images at 640×480, measured active-joint positions, robot kinematics, and camera calibration. Camera intrinsics and camera-to-robot-base transforms are obtained from the simulator after randomization. We supply kinematics and calibration as setup information readily accessible in a calibrated robotic deployment, consistent with our objective of assessing GPT-6-Astra's capabilities given available test-time information. Object states, depth, segmentation, contacts, and evaluator feedback are excluded. Actions are absolute active-joint targets executed at 20 Hz, with three 60 Hz physics steps per target. GPT-6-Astra chooses the action-chunk length, and our wrapper adds no interpolation or clipping.

Task	Embodiment	N	Success rate (%)			Tokens (millions)			Time (minutes)		
			$\pi_{0.5}$ [12]	GPT-6-Astra 1×	GPT-6-Astra 10×	ICL	Test 1×	Test 10×	ICL	Test 1×	Test 10×
Baking Tray Prep (32)	IIWA7 + Sharpa	3	20.0	0.0	100.0	4.40	8.30	54.82	27.8	30.2	186.4
Canned Food Tray Arrangement (26)		3	6.0	0.0	33.3	2.61	9.01	52.91	22.0	23.5	144.0
Jigsaw Puzzle Assembly (73)		3	0.0	0.0	0.0	3.82	13.73	72.62	28.6	47.2	236.7
Gaming Desk Setup (80)	JAKA ZU7 + DexHand021	3	40.0	0.0	0.0	3.55	14.16	59.65	34.9	40.4	189.0
Medicine Shoebox Pack (61)	Panda + Orca	3	24.0	66.7	100.0	3.39	13.05	18.70	24.6	32.8	57.9
Tool Box Loading (22)		3	68.0	66.7	100.0	4.65	10.50	19.09	37.7	24.5	34.8
Sports Ball Cup Sort (64)	Panda + Allegro	3	22.0	66.7	100.0	4.41	10.54	12.36	32.8	32.7	40.2
Faucet Cup Water Fill (67)	RM65 + Revo2	3	18.0	33.3	100.0	3.65	13.49	41.29	36.4	32.7	108.5
Stationery Category Sorting (24)		3	0.0	0.0	66.7	3.25	14.69	99.48	26.7	37.6	300.0
Bimanual Piano Melody (79)	xArm7 + Ability	3	4.0	66.7	100.0	3.84	8.60	11.28	26.5	18.4	28.4
Wine Glass Plate Balance (03)		3	38.0	0.0	0.0	4.32	10.63	11.17	30.2	24.0	25.1
Cleaner Box Loading (09)	xArm7 + LEAP	3	48.0	0.0	100.0	2.99	8.66	25.29	23.2	26.0	55.3
Shoebox Accessory Pack (62)		3	48.0	0.0	0.0	2.70	9.88	23.59	26.6	24.7	72.8
Toilet Lid Cleaner Pour (51)		3	46.0	33.3	66.7	3.52	6.46	6.77	24.0	18.3	19.6
Breadbasket Fast-Food Loading (60)	UR5 + RH5DG2	3	6.0	0.0	66.7	3.32	15.81	39.18	30.8	41.0	98.8
Citrus Plate Loading (07)		3	64.0	0.0	33.3	3.56	8.21	11.25	28.3	24.0	29.7
Fridge Wine Interhand Pour (34)		3	10.0	0.0	66.7	3.06	14.95	74.84	29.6	32.3	182.8
Fruit Bowl Loading (06)	UR5 + RH56DFX	3	38.0	0.0	33.3	2.27	9.23	18.62	21.5	25.2	56.0
Screwdriver Box & Hammer (12)		3	32.0	66.7	100.0	4.14	7.72	7.93	27.8	21.5	21.8
Trash Disposal (42)		3	36.0	66.7	100.0	5.32	5.86	12.06	36.8	16.0	22.9
Fridge Fruit Shelf Sorting (43)	UR5 + Shadow	3	4.0	33.3	66.7	3.06	16.60	32.93	27.3	37.2	89.3
Soup Serving (76)		3	0.0	0.0	0.0	3.88	20.24	79.85	31.5	60.7	268.3
Frypan Stand & Pour (08)	UR5 + Schunk	3	12.0	0.0	0.0	3.02	12.20	28.57	25.0	34.7	71.4
Microwave Bowl Loading (44)		3	62.0	33.3	100.0	4.63	12.14	27.94	29.7	26.7	63.7
Ball Box Loading (27)	UR5 + Wuji	3	8.0	100.0	100.0	4.08	8.92	8.92	31.5	18.2	18.2
Condiment Box Loading (21)		3	8.0	33.3	66.7	3.50	8.75	24.49	26.7	21.8	57.2
Overall / mean		**78**	**25.5**	**25.6**	**61.5**	3.65	11.24	33.68	28.8	29.7	95.3

Note. The $\pi_{0.5}$ is the current state-of-the-art baseline under Full randomization reported on the official Bench2Dex project website [163] (accessed 6 October 2026), achieving 25.5% SR (331/1300 successes). It is evaluated on 50 episodes per task under the standard budget, separately from the GPT-6-Astra episodes. ICL costs are incurred once per task; test costs are means over all three episodes, including failures. The 10× costs include the original 1× execution and its extension, excluding preparation and discarded/interrupted attempts. Tokens include cached input; reported test token counts are lower bounds because usage accounting is partial. Times measure tool-inclusive inference wall time, excluding simulator stepping and replay. The final row averages ICL costs over 26 tasks and test costs over 78 episodes.

Table 7 Task-level performance and computational cost on Bench2Dex [163] under Full randomization. Tasks are grouped by embodiment following the benchmark paper. GPT-6-Astra uses three episodes per task (78 total) at the standard (1×) and extended (10×) interaction budgets. Token and time costs refer to GPT-6-Astra.

Budgets and continuation. The standard budget follows the benchmark, $B_t = \lceil 1.5E_t/3 \rceil$, where E_t is the expert reference duration in physics steps. B_t ranges from 593 to 1,542 control steps. SR uses official stable-success early stopping, with a 0.5 s dwell except for Faucet Cup Water Fill, which requires 3 s. Only standard-budget timeouts receive additional interaction up to a total of $10B_t$, while successes and other terminal failures retain their outcomes. Because the original 1× runs were already complete, we reconstruct timeout endpoints using the original reset configuration and exact recorded-action replay, then resume the original episode context. Replay must match archived numerical states and boundary stage histories within fixed tolerances. The continuation additionally permits an observation-grounded explicit give-up, counted as failure.

Results and comparison. GPT-6-Astra achieves 25.6% SR at 1×, comparable to the published $\pi_{0.5}$ baseline of 25.5%, and reaches 61.5% at 10×, an increase of 35.9 percentage points. The baseline uses 50 episodes per task, evaluated separately from our three episodes per task. Reported test-token counts in Table 7 are lower bounds because usage accounting is partial; tokens-per-success values derived from these counts are therefore also lower bounds and do not establish a cross-benchmark token-cost ranking.

C.6 DexJoCo

Tasks and demonstrations. We evaluate all 11 DexJoCo [144] tasks (six single-arm and five bimanual) under `rand_full`, using five fixed test seeds per task (100000–100004; 55 episodes). This regime randomizes object placement and table height, external camera pose, lighting, and table texture, while dynamics randomization is disabled. For each task, preparation has access to 100 released `rand_full` replay recordings through a restricted reader that exposes robot proprioception, demonstrated actions, timestamps, and selected RGB videos. GPT-6-Astra also receives the pinned robot model. Each task's resulting notes, reference examples, and control utilities are frozen as one package before testing.

Task	N	Success rate (%)			Tokens (millions)			Time (minutes)		
		$\pi_{0.5}$ [12]	GPT-6-Astra 1×	GPT-6-Astra 10×	ICL	Test 1×	Test 10×	ICL	Test 1×	Test 10×
Hammer Nail	5	17.3	100.0	100.0	4.87	4.71	4.71	35.3	13.5	13.5
Click Mouse	5	54.7	100.0	100.0	4.98	2.84	2.84	27.8	6.1	6.1
Pick Bucket	5	78.7	100.0	100.0	3.36	9.21	9.21	22.1	16.7	16.7
Pinch Tongs	5	18.7	80.0	80.0	3.95	3.00	3.00	29.6	6.7	6.7
Fold Glasses	5	39.3	60.0	100.0	4.00	15.96	19.46	24.4	32.6	41.5
Water Plant	5	75.3	100.0	100.0	5.33	1.64	1.64	32.3	4.9	4.9
Unlock iPad	5	32.7	20.0	20.0	4.38	3.66	3.66	32.6	10.7	10.7
Hanoi	5	33.3	100.0	100.0	4.09	5.87	5.87	28.1	12.9	12.9
Assembly	5	0.7	60.0	80.0	6.29	16.76	31.95	37.2	48.2	96.7
Microwave	5	69.3	100.0	100.0	9.80	6.24	6.24	38.0	13.4	13.4
Photograph	5	23.3	40.0	80.0	7.26	12.46	27.46	43.0	30.3	85.6
Overall / mean	**55**	**40.3**	**78.2**	**87.3**	5.30	7.49	10.55	31.9	17.8	28.0

Note. The $\pi_{0.5}$ success rates are the rand-full results in DexJoCo Table II [144], which reports a 40.3% mean over 11 tasks and evaluates three training seeds with 50 test episodes per task per seed. GPT-6-Astra uses five test seeds per task. ICL preparation is incurred once per task. Test costs average all five scored episodes, including failures and the scored step-zero Pinch Tongs action-envelope abort. 10× test costs include the original 1× session and any continuation, excluding preparation and discarded invalid attempts. Tokens include cached input. Time is Codex session wall time minus simulator stepping, including other model and tool work. The final row averages ICL costs over 11 tasks and test costs over 55 episodes.

Table 8 Task-level performance and computational cost on DexJoCo [144] under rand-full randomization. GPT-6-Astra uses five test episodes per task at the standard (1×) and extended (10×) interaction budgets. Token and time costs refer to GPT-6-Astra.

Observation and control settings. At test time, GPT-6-Astra receives 640×640 RGB from one external camera and one wrist camera for single-arm tasks, or two wrist cameras for bimanual tasks. Public robot state contains end-effector pose and 16 hand-joint values per arm (23 or 46 values). Object state, depth, contacts, rewards, and success flags are withheld. Actions specify absolute world-frame end-effector position and rotation vector plus 16 hand-joint targets per arm (22 or 44 values), executed at 30 Hz. GPT-6-Astra may submit 1–30 actions per decision. The interface converts rotation vectors to the quaternion format expected by the simulator.

Budgets and continuation. DexJoCo's task-specific standard limits range from 1,000 to 1,500 action steps. Success is determined by the simulator's task-specific success flag. Only episodes that exhaust the standard limit without a terminal outcome continue toward $10B$, for which a fresh GPT-6-Astra session resumes the same live simulator instance with the episode's observation and action history. No reset or trajectory replay is used for these continuations. For `click_mouse`, we reproduce the benchmark client's 30-step initial alignment and count those steps against the task budget.

Results and comparison. GPT-6-Astra achieves 78.2% SR at 1×, exceeding the published $\pi_{0.5}$ baseline of 40.3% by 37.9 percentage points, and reaches 87.3% at 10×, a further increase of 9.1 percentage points. The baseline reports results over three training seeds with 50 test episodes per task per seed, while GPT-6-Astra uses five test seeds per task.

C.7 LHPR-VLN

Benchmarks and tasks. LHPR-VLN [134] evaluates long-horizon vision-language navigation. Given a global natural-language instruction, the policy must reach multiple navigation goals in a strictly specified order. Following these instructions requires visually grounding successive goals, tracking completed stages, and using earlier observations to guide subsequent movements.

Evaluation subset. The official unseen test split contains 205 episodes across 27 scenes. Many episodes contain subtask target objects that are not visible along the reference trajectories. We manually checked target visibility along each

episode's reference trajectory and retained an episode only when every subtask target object was visible. The resulting subset contains 114 episodes across 25 scenes, with two to four goals per episode. The complete retained episode list appears below.

Protocol and metrics. Each episode starts in a fresh session with the global instruction and a frozen ICL workspace containing exploration notes, code, and observations from training scenes that do not overlap with the test scenes. Test episodes do not share interaction history. During navigation, the policy receives three RGB views at a resolution of 366×366, facing left, forward, and right at $+60°$, $0°$, and $-60°$. Available actions are 0.25 m forward motion, $30°$ left or right turns, and `STOP`. Three consecutive blocked forward actions trigger a random turn and a forward action, both counted toward the action budget. Ground-truth trajectories, target coordinates, and scores are withheld. To submit a goal, the policy issues `STOP`, which advances the stage regardless of success. A submission succeeds within 1 m geodesic distance of the original target or a matching target object in the specified room. Environment feedback identifies the current stage without revealing submission success. The $1\times$ and $10\times$ settings allow 500 and 5,000 actions per sequence. Extended evaluation retains successful $1\times$ outcomes and evaluates only initially unsuccessful episodes with the larger budget, preserving earlier scores when they are higher. Metrics are averaged over all 114 episodes. SR measures full-sequence success. ISR, CSR, and CGT describe individual-goal and sequential success, with definitions in the table note.

Policy	Type	SR	ISR	CSR	CGT	Size	Tokens
GPT-6-Astra Max + ICL $10\times$	Agent	**28.95**	**62.87**	**52.97**	**52.58**	–	2.587
GPT-6-Astra Max + ICL $1\times$	Agent	21.05	54.68	44.55	43.91	–	1.317
Claude Code Opus-5	Agent	5.26	26.87	17.82	17.84	–	–
FantasyVLN [186]	VLN	2.44	11.01	9.64	8.99	7B	–
Aux-think [147]	VLN	0.65	3.16	2.04	1.47	8B	–
MGDM [134]	VLN	0.00	2.35	1.65	2.92	7B	–

Note. SR is the success rate for complete sequences. ISR is the independent success rate for individual goals. CSR is the conditional success rate, which gives greater weight to successes that follow a successful subtask. CGT further weights CSR by ground-truth path length. Tokens are mean test-time input plus output tokens per episode, including failures. Cached input and reasoning output are counted once. The $10\times$ cost includes $1\times$ consumption and additional evaluation for initially unsuccessful episodes. Exploration and technical retries are excluded. – indicates an unreported value.

Table 9 Overall performance on LHPR-VLN [134]. SR, ISR, CSR, and CGT are percentages, with higher values indicating better performance. Token consumption is reported in millions. Policies are ordered by SR. Bold values mark the highest navigation scores.

Results. GPT-6-Astra Max + ICL achieves 21.05% SR at $1\times$, compared with 5.26% for Claude Code Opus-5 and 2.44% for FantasyVLN in Table 9. Extended evaluation raises SR to 28.95%, increasing completed episodes from 24 to 33. ISR rises from 54.68% to 62.87%, while CSR and CGT improve by 8.42 and 8.67 percentage points. Average test-time token consumption increases from 1.317 M to 2.587 M, or $1.96\times$ the initial cost.

Discussion. Full-sequence success remains below 30% under extended evaluation despite individual-goal success above 60%. Reaching some goals improves ISR, whereas SR requires every goal to be completed in order. A failed submission advances the sequence irreversibly, making goal verification and progress tracking consequential. Aggregate scores alone do not establish how much either contributes to the remaining failures. Baseline comparisons do not control for inference cost, and the reported token consumption should be considered alongside success rates.

Episode IDs. Each batch/index pair below identifies an episode as `batch_b_episode_e`, where b is the batch number and e is the listed episode index.

Batch 1: `2, 10, 19, 20, 55, 56, 60, 61, 62, 65, 81, 82, 84, 94, 103, 119, 128, 130`

Batch 2: `1, 17, 20, 35, 52, 54, 69, 78, 85, 87, 88, 99, 103`

Batch 3: `10, 24, 26, 36, 44, 46, 63, 69, 83, 86, 91, 97, 101, 103, 106, 107, 112`

Batch 4: `31, 32, 36, 58, 66, 68, 110, 111, 112, 114`

Batch 5: `3, 4, 7, 16, 21, 28, 38, 40, 45, 52, 53, 64, 72, 74, 75, 86, 88, 100`

Batch 6: `34, 37`

Batch 7: `13, 47, 56, 66, 74, 82, 83, 85, 99, 120, 123, 146, 160`

Batch 8: `5, 30, 63, 66, 77, 82, 85, 100, 112, 115, 134, 141, 146, 154, 156, 161, 166, 180, 191, 205, 206, 211, 223`

C.8 VLN-CE

Benchmarks and tasks. VLN-CE [87] evaluates instruction-guided navigation in continuous indoor environments. Given a natural-language route instruction, the policy must navigate from a specified starting pose to a single goal and stop. This requires grounding landmarks, tracking progress along the route, and choosing movements from visual observations without a predefined navigation graph.

Evaluation subsets. R2R [3] uses the 100-episode val-unseen subset from Open-Nav [124], which covers 10 scenes and is also adopted by SmartWay [131] and AgentCanvas [1]. For RxR [88], we select 100 English guide episodes from the val-unseen split of 3,669 English episodes across 11 scenes, matching the R2R subset's episode counts in each of the same 10 scenes. With seed 0, we draw 20,000 stratified 100-episode subsets and retain the candidate with the smallest sum of two-sample Kolmogorov–Smirnov distances to the full English split for instruction word count and reference-path length. We then attempt 60,000 random within-scene episode replacements, accepting a change only if it reduces this sum. Instruction length is the whitespace-separated word count, and reference-path length is the sum of Euclidean distances between consecutive reference waypoints. The complete episode lists for both subsets appear below.

Protocol and metrics. Each episode starts in a fresh session with a frozen ICL workspace prepared using training scenes that do not overlap with the test scenes. Test episodes do not share interaction history. Observations contain a single forward-facing RGB image at 512×512 resolution. Forward actions move 0.25 m. R2R uses 15° left or right turns, while RxR uses 30° turns and permits 30° upward or downward camera tilts. Ground-truth routes, goal coordinates, and evaluation scores are withheld. `STOP` terminates the episode and succeeds within 3 m geodesic distance of the goal. The $1\times$ and $10\times$ limits are 500 and 5,000 actions. Extended evaluation is applied to initially unsuccessful episodes. Reported extended results retain prior successes and the better per-episode value of each metric. Metrics are averaged over each 100-episode subset. NE is the final geodesic distance to the goal. OS is the oracle success rate, measuring whether the trajectory ever enters the 3 m success region. SR measures successful completion, and SPL [2] weights success by path efficiency. nDTW [69] measures alignment with the reference trajectory using normalized dynamic time warping.

Policy	Size	Type	Observation	R2R val-unseen				RxR val-unseen			
				NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
GPT-6-Astra Max + ICL $10\times$	–	Agent	Mono. RGB	**2.34**	**91**	**88**	**76.98**	**1.36**	**90**	**73.52**	**82.98**
GPT-6-Astra Max + ICL $1\times$	–	Agent	Mono. RGB	3.45	84	80	68.47	2.08	82	65.61	77.85
GPT-6-Astra Ultra [33]	–	Agent	Mono. RGB	2.9	83.7	81.3	71.5	–	–	–	–
Robostral Navigate [16]	8B + 121M	VLN	Mono. RGB	3.2	81.3	77.4	74.2	3.47	75.1	68.7	–
Qwen-RobotNav-8B [172]	8B	VLN	Pano. RGB	3.53	78.5	72.1	66.6	3.58	76.5	65.7	72.5
ABot-N1 [53]	4B + 2B	VLN	3-view RGB	3.32	75.2	70.9	67.5	3.13	73.9	63.9	–
Image2Nav [149]	4B	VLN	Mono. RGB	3.71	76.1	70.3	65.6	3.74	70.7	59.1	71.8
GroundingVLN [92]	4B	VLN	3-view RGB-D	3.66	74.8	69.9	64.1	3.54	75.1	62	–
OmniNav [157]	3B	VLN	3-view RGB	3.74	74.6	69.5	66.1	3.77	73.6	62	–
LightNav-0 [146]	4B	VLN	Mono. RGB	3.91	73.7	68.5	62.8	3.66	73.6	64.5	67.4
AstraNav-World [25]	3B + 5B	WAM	3-view RGB	3.86	73.9	67.9	65.4	3.82	72.9	61.5	–
SEDualVLN [63]	7B + GPT-4o	VLN	Mono. RGB	3.75	73.7	67.3	62.5	4.12	63.9	52.4	72.8
AgentVLN [156]	3B	VLN	Mono. RGB-D	3.88	73.5	67.2	64.7	3.92	69.5	61.3	–
TAMP-Nav [43]	7B	VLN	4-view RGB-D	3.85	74.5	66.2	58.8	4.32	65.7	56.9	–
AwareVLN [55]	8B	VLN	Mono. RGB	4.02	73.5	65.4	55.1	3.95	67.6	56.1	–
DualVLN [150]	7B	VLN	Mono. RGB	4.05	70.7	64.3	58.5	4.58	61.4	51.8	70.0

Note. Higher OS, SR, SPL, and nDTW and lower NE indicate better performance. Mono. and Pano. denote monocular and panoramic observations. RGB-D includes depth. Image2Nav uses a 180° horizontal field of view. Sizes denote reported backbone or component sizes. – indicates an unreported value.

Table 10 Overall performance on VLN-CE [87] R2R and RxR val-unseen. OS, SR, SPL, and nDTW are percentages, and NE is in meters. Bold values mark the best reported navigation scores.

Results. Extended evaluation raises SR from 80% to 88% on R2R and from 82% to 90% on RxR, adding eight successes per subset in Table 10. SPL improves by 8.51 and 7.91 percentage points, respectively. NE decreases from 3.45 m to 2.34 m on R2R and from 2.08 m to 1.36 m on RxR. R2R OS rises from 84% to 91%, while RxR nDTW increases from 77.85% to 82.98%.

Discussion. On R2R, OS exceeds SR by three percentage points under extended evaluation, identifying episodes that reach the goal region without stopping successfully. The RxR nDTW gain indicates closer alignment with reference routes. SPL remains below SR on both datasets, indicating that successful outcomes can still involve detours. Our results cover 100 episodes per dataset. Differences in episode coverage, observation access, and inference budgets limit direct comparisons with the listed baselines.

Episode IDs.

R2R: `7, 11, 13, 40, 42, 52, 70, 94, 116, 140, 150, 156, 166, 171, 176, 181, 187, 190, 191, 207, 218, 226, 232, 244, 247, 259, 265, 275, 308, 312, 321, 330, 338, 348, 362, 371, 377, 387, 403, 411, 423, 432, 439, 447, 454, 461, 469, 479, 513, 516, 526, 531, 546, 550, 559, 568, 576, 586, 602, 609, 620, 643, 655, 670, 677, 698, 705, 715, 721, 739, 743, 748, 755, 765, 781, 787, 804, 810, 821, 824, 842, 1051, 1056, 1061, 1071, 1077, 1084, 1085, 1087, 1092, 1106, 1117, 1133, 1139, 1142, 1148, 1284, 1301, 1307, 1406`

RxR: `198, 311, 377, 683, 886, 901, 912, 1108, 1109, 1196, 1575, 1719, 1866, 1914, 2175, 2220, 2258, 2264, 2274, 2463, 2485, 2561, 2638, 2694, 2800, 2823, 2917, 3468, 3514, 3550, 3564, 3591, 3644, 3673, 4027, 4061, 4101, 4150, 4252, 4312, 4326, 4449, 4502, 4517, 4635, 4753, 5247, 5348, 5431, 5499, 5503, 5843, 5922, 5950, 5968, 6004, 6134, 6180, 6237, 6289, 6319, 6362, 6597, 6623, 6632, 6703, 6968, 6995, 7008, 7121, 7234, 7284, 7433, 7471, 7488, 7569, 7732, 7733, 7741, 7819, 8127, 8246, 8259, 8340, 8686, 8928, 9142, 9156, 9242, 9457, 9514, 9632, 9803, 9848, 10025, 10188, 10387, 10789, 10917, 10971`

C.9 SIMPLE

Tasks and evaluation scope. SIMPLE [152] evaluates humanoid loco-manipulation on a Unitree G1 with two Dex3 hands. Our study uses the six-task evaluation set reported in Table 1 of the benchmark paper: XMovePick, BendPick, Handover, Tabletop, XMoveBendPick, and PickBetweenTables. Each task has three difficulty levels: L0 varies table texture and distractor type and placement; L1 adds extreme lighting changes; L2 further perturbs the target object's initial position. With 10 released evaluation instances per task–level, this set contains 180 candidate episodes. Outcome coverage differs across the four GPT-6-Astra settings (zero-shot or ICL, at $1\times$ or $10\times$), as specified below.

Coverage and comparison subsets. At the standard ($1\times$) budget, zero-shot (ZS) evaluation has complete outcomes for five tasks (XMovePick, BendPick, Handover, Tabletop, and XMoveBendPick), totaling 150 instances. Four of these tasks also have complete ICL outcomes, forming Common-4: XMovePick, BendPick, Handover, and Tabletop, with 120 instances. Cross-budget comparisons use Matched-8, the eight task–level combinations with complete outcomes in all four settings: XMovePick L0–L2, BendPick L0–L1, and Tabletop L0–L2, totaling 80 instances across three tasks. PickBetweenTables has only partial, outcome-selected coverage and is included solely in the reference aggregates, outside both comparison subsets.

Demonstration preparation. For task-level in-context learning (ICL), GPT-6-Astra explores the released training demonstrations through an information-bounded reader and compiles task notes, reference numerical data, image indices, and runnable helpers into a single hash-frozen package; the same package is shared across L0/L1/L2, every test episode starts from a fresh session, and no model weights are updated. Preparation accesses 99–100 released demonstrations per task; XMoveBendPick has no ICL package and is reported zero-shot only.

Observation and control settings. At test time GPT-6-Astra receives a head-mounted left RGB image (typically 640×360), a 32-column robot-interface state, the task text, and the current episode's history. It is additionally given nominal robot metadata (static URDF, joint order, and limits), robot-only forward/inverse kinematics, and NumPy/SciPy code tools; object pose, contacts, and online scoring feedback are withheld. Actions are continuous physical chunks of shape $(T, 36)$—finger and arm joint angles, torso roll/pitch/yaw, base height, base translation (v_x, v_y), a turn flag, and a target yaw—with the chunk length T chosen by the model. Each action row is executed at 50 Hz by the task-specific official System 0 controller: the decoupled whole-body controller for the Teleop tasks, and the Ψ_0/AMO

Method	XMovePick			BendPick			Handover			Tabletop			XMoveBendPick			PickBetweenTables			Six-task total (reference only)	Matched-8 total
	L0	L1	L2	L0	L1	L2	L0	L1	L2	L0	L1	L2	L0	L1	L2	L0	L1	L2		
GPT-6-Astra, ZS · 1×	9/10	9/10	10/10	7/10	7/10	5/10	7/10	4/10	4/10	9/10	9/10	5/10	9/10	9/10	7/10	0/6[a]	0/7[a]	0/4[a]	110/167 (65.9%)	65/80 (81.3%)
GPT-6-Astra, ICL · 1×	10/10	10/10	8/10	10/10	10/10	8/10	9/10	9/10	6/10	10/10	10/10	9/10	–	–	–	2/3[a]	0/2[a]	0/1[a]	111/126 (88.1%)[b]	77/80 (96.3%)
GPT-6-Astra, ZS · 10×	10/10	10/10	10/10	10/10	10/10	8/10	7/7[u]	4/4[u]	5/5[u]	10/10	10/10	9/10	10/10	10/10	10/10	6/6[a]	7/7[a]	4/4[a]	150/153 (98.0%)	**79/80 (98.8%)**
GPT-6-Astra, ICL · 10×	10/10	10/10	10/10	10/10	10/10	8/8[u]	9/9[u]	9/9[u]	6/6[u]	10/10	10/10	9/10	–	–	–	3/3[a]	2/2[a]	1/1[a]	117/118 (99.2%)[b]	**79/80 (98.8%)**
Psi-0 [151] · 1×	10/10	10/10	6/10	10/10	10/10	10/10	7/10	7/10	10/10	10/10	10/10	8/10	10/10	9/10	9/10	7/10	5/10	6/10	154/180 (85.6%)	74/80 (92.5%)
DreamZero [166] · 1×	10/10	10/10	10/10	9/10	9/10	8/10	7/10	8/10	9/10	9/10	10/10	7/10	0/10	0/10	1/10	5/10	3/10	3/10	118/180 (65.6%)	74/80 (92.5%)
ACT [178] · 1×	10/10	10/10	5/10	10/10	9/10	9/10	7/10	7/10	10/10	10/10	10/10	8/10	9/10	10/10	10/10	5/10	5/10	5/10	149/180 (82.8%)	72/80 (90.0%)
OpenPi π0.5 [12] · 1×	7/10	5/10	1/10	10/10	10/10	8/10	5/10	4/10	5/10	10/10	10/10	8/10	0/10	0/10	0/10	3/10	3/10	3/10	92/180 (51.1%)	61/80 (76.3%)
GR00T N1.6 [116] · 1×	10/10	10/10	7/10	7/10	7/10	6/10	1/10	3/10	3/10	9/10	9/10	7/10	4/10	4/10	1/10	0/10	0/10	0/10	88/180 (48.9%)	66/80 (82.5%)
Diff. Policy [30] · 1×	3/10	3/10	2/10	10/10	8/10	6/10	3/10	2/10	4/10	8/10	9/10	8/10	0/10	0/10	0/10	4/10	0/10	0/10	70/180 (38.9%)	51/80 (63.8%)
EgoVLA [160] · 1×	0/10	1/10	2/10	7/10	5/10	8/10	0/10	4/10	3/10	10/10	10/10	7/10	3/10	5/10	4/10	0/10	0/10	0/10	69/180 (38.3%)	42/80 (52.5%)
InternVLA-M1 [27] · 1×	0/10	0/10	0/10	5/10	5/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	3/10	5/10	7/10	0/10	0/10	0/10	25/180 (13.9%)	10/80 (12.5%)
H-RDT [11] · 1×	0/10	0/10	2/10	0/10	0/10	1/10	0/10	1/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	4/180 (2.2%)	2/80 (2.5%)

Note. For the Matched-8 total, **best** is in bold and second best underlined across all rows. Six-task totals are provided for reference only. [a] GPT-6-Astra PickBetweenTables results use an outcome-selected subset and a revised release metric; these episodes are included only in the reference Six-task total. [u] For 10×, denominators below 10 indicate incomplete outcome coverage; unknown episodes are excluded. The Matched-8 total uses the same eight complete task–level cells for every row (80 episodes): XMovePick L0–L2, BendPick L0–L1, and Tabletop L0–L2. Cells are selected by completeness, not success rate. This is a comparison on the shared complete subset, not an estimate for all SIMPLE tasks. [b] GPT-6-Astra ICL Six-task total covers 5 tasks (XMoveBendPick has no ICL). Six-task totals use differing denominators across methods (baselines /180; GPT-6-Astra includes XMoveBendPick and the PickBetweenTables subset), so this column is a descriptive reference, not a matched comparison.

Table 11 Per-level success on the six-task evaluation set from Table 1 of SIMPLE [152]: GPT-6-Astra vs. published baselines, under the standard (1×) and extended (10×) interaction budgets. Each cell reports successes / judged episodes at levels L0/L1/L2 under the indicated budget; published baselines report 1× only. The Matched-8 total compares the same eight complete task–level cells (80 episodes) across all rows. Six-task totals are provided for reference only.

motion-planning path for BendPick and Tabletop [91, 151]. The simulator is paused during inference, so reported action steps exclude thinking time.

Budgets and scoring. The standard (1×) per-episode action budget is 800 steps for XMovePick, Handover, Tabletop, and XMoveBendPick, 400 for BendPick, and 1,200 for PickBetweenTables; the extended budget is 10×. The Matched-8 comparison includes only complete task–level cells; partial results elsewhere use judged denominators, excluding unknown outcomes. Tabletop 1× outcomes are thresholded from the 10× trajectories, and one operator-stopped ICL episode (L2/E01) is counted as a failure by an explicit reporting rule. PickBetweenTables is an outcome-selected successful subset scored with a revised release metric. It is included only in the Six-task totals, which are provided for reference, and excluded from both the Matched-8 comparison in Table 11 and the Common-4 comparison in Table 12. Compute, sensing, tool access, inference latency, and rendering hardware are not matched across systems, so the comparison is descriptive rather than cost-controlled.

Results on the shared complete subset. Table 11 reports per-level success and two aggregate columns. The Matched-8 total compares all four GPT-6-Astra settings and the published baselines on the same 80 instances defined above. The subset is selected by completeness, not by success rate. On this subset, zero-shot success rises from 65/80 (81.3%) at 1× to 79/80 (98.8%) at 10×, while ICL rises from 77/80 (96.3%) to 79/80 (98.8%); both Ψ_0 [151] and DreamZero achieve 74/80 (92.5%) at 1×. Thus, ICL improves standard-budget performance by 15.0 percentage points on this subset, while both GPT-6-Astra settings reach the same extended-budget success rate. These results describe the shared complete subset and do not estimate performance on excluded task–level cells. The Six-task total retains all judged outcomes for reference only: task coverage and denominators differ across rows, and the ICL aggregate covers five tasks because XMoveBendPick has no ICL package.

Standard-budget ICL and efficiency. Table 12 separately compares zero-shot and ICL at 1× on the four common tasks (XMovePick, BendPick, Handover, and Tabletop), where all 120 episodes are judged in both settings. This is a broader subset than Matched-8 and is used only for the standard-budget comparison. GPT-6-Astra reaches 85/120 (70.8%) zero-shot and 109/120 (90.8%) with task-level ICL, a 20.0-percentage-point gain; the latter is comparable to the published Ψ_0 result of 108/120 (90.0%), and the one-case difference does not establish statistical superiority. Spatial generalization (L2) remains the main weakness at 1×: ICL is 39/40 at both L0 and L1 but 31/40 at L2. On the 81 episodes solved in both settings, ICL reduces mean action steps from 376.9 to 210.4 (44.2%), while mean test

Metric	GPT-6-Astra Zero-shot	GPT-6-Astra ICL	Psi-0 [151] (ref.)
Success rate			
Common-4	85/120 (70.8%)	**109/120 (90.8%)**	108/120 (90.0%)
Six-task (reference only)[c]	110/167 (65.9%)	111/126 (88.1%)	154/180 (85.6%)
Per-episode efficiency on matched successes[a]			
Mean action steps	376.9	210.4	–
Mean test tokens (millions)	2.64	6.40	–
Mean test wall-time (min)	47.0	31.4	–
ICL preparation (one-time, per task)			
Tokens (millions)	–	8.34	–
Wall-time (minutes)	–	90.4	–

Note. For Common-4 success at 1×, the **best** is in bold and the second best underlined across methods. [a] Per-episode means over the 81 cases solved in both settings. [c] Six-task success rates use differing denominators (GPT-6-Astra ZS /167, incl. XMoveBendPick and the PickBetweenTables subset; GPT-6-Astra ICL = 5 tasks, /126) and are a descriptive reference. ICL preparation is a one-time per-task cost; XMoveBendPick (no ICL) and PickBetweenTables (no comparable cost summary) are excluded from these efficiency figures.

Table 12 GPT-6-Astra standard-budget (1×) success, efficiency and cost on SIMPLE: zero-shot vs. ICL. Common-4 covers all 120 episodes from XMovePick, BendPick, Handover and Tabletop; it differs from the 80-episode Matched-8 subset in Table 11. Efficiency is measured on the 81 episodes solved in both settings. Six-task success is provided for reference only. Psi-0 is the published baseline.

token consumption rises from 2.64M to 6.40M. Mean test wall-clock time falls from 47.0 to 31.4 minutes; preparation additionally costs 8.34M tokens and 90.4 minutes per task, paid once. These cost figures use the matched-success set and should not be interpreted as measurements on the Matched-8 subset.

C.10 BEHAVIOR

Benchmarks and tasks. BEHAVIOR-1K [90] is a household embodied-AI benchmark covering 1,000 everyday activities grounded in human needs. Built on the OmniGibson simulator, it supports realistic interactions with rigid objects, deformable objects, and liquids. Its long-horizon activities require navigation and manipulation to achieve task-specific goal conditions.

We evaluate on all 100 tasks from the BEHAVIOR-1K 2026 Challenge, using public-test episode 0 with rollout seed 0 for each task and performing a single evaluation rollout per task instance. We retain the benchmark's default R1Pro dual-arm mobile manipulator, with the arm-controller modification described below. Each rollout starts from the official task-instance initialization and proceeds through closed-loop interaction conditioned on head-mounted and bilateral wrist RGB-D observations, robot proprioception, and the task instruction. The simulator operates at 30 native steps per second, and each model-issued action can be repeated for 1–100 simulation steps. A rollout terminates upon environment termination, exhaustion of the official task-specific step budget, or an explicit agent finish call. Following the official evaluator, the step budget is set to 1.5 times the task's mean human-demonstration length.

Protocol and Metrics. We use GPT-6 Astra with maximum reasoning effort for both demonstration-based in-context learning and evaluation. During the ICL stage, all official training data are provided including RGB videos, actions, proprioceptive states, and annotations. We adopt a 21-dimensional action interface, consisting of three base-control values, four torso joint targets, two six-dimensional absolute end-effector pose targets, and two gripper commands. Each end-effector target specifies a position and axis-angle orientation in the robot's current base frame, which the official inverse-kinematics controller translates into arm motion. For consistency with this interface, demonstration arm commands are converted from joint-space targets to end-effector poses using the official forward-kinematics implementation.

We assess performance using the official evaluator, reporting binary task success, final predicate-based Q-score, simulator step count and simulated time, and base and end-effector travel distances together with their official human-normalized counterparts. Model token usage and wall-clock durations for individual stages are recorded separately to characterize computational cost.

Results. Across the 100 evaluation rollouts on the BEHAVIOR-1K 2026 Challenge, our method successfully completes 37 tasks, achieving an overall task success rate of 37%.

Other Discussion. The aggregate task success rate increases from 1% with the 23-dimensional joint-space interface to 37% with the 21-dimensional end-effector-pose interface. This substantial improvement is consistent with the hypothesis that task-space action representations better align language-model planning with robotic execution. By specifying the desired position and orientation of each end effector, the model can express manipulation intent directly while delegating joint-level kinematic computation to the official controller.

Qualitative inspection of the trajectories suggests that the model can extract procedural knowledge from demonstrations, identify task stages, and adapt its plans based on visual feedback. However, failures due to imprecise grasp alignment, arm collisions, navigation obstructions, and ineffective recovery strategies highlight persistent challenges in contact-rich manipulation. Taken together, these findings suggest that action representation plays a critical role in embodied task performance, but do not establish that high-level reasoning alone is sufficient for reliable robotic manipulation.

D Harness Ablation Study

Configuration	**Average**	Generalization	Precision	Long-Horizon	Memory	Open	Tokens (M) ↓
Baseline	22.78 / 15.56	27.08 / 22.22	9.79 / 0.00	23.13 / 4.17	22.22 / 22.22	31.67 / 29.17	2.56
A. Agent Components							
w/ planning subagent	26.03 / 19.44	37.08 / 33.33	11.88 / 4.17	21.46 / 4.17	40.56 / 38.89	19.17 / 16.67	2.96
w/ execution subagent	21.56 / 15.28	36.39 / 33.33	14.79 / 4.17	10.63 / 0.00	25.56 / 22.22	20.42 / 16.67	2.55
w/ feedback subagent	**30.00 / 24.72**	30.83 / 27.78	16.88 / 12.50	24.38 / 8.33	33.33 / 33.33	44.58 / 41.67	2.74
w/ history compression	28.03 / 21.11	35.42 / 33.33	7.29 / 0.00	24.38 / 4.17	38.89 / 38.89	34.17 / 29.17	0.86
B. Action Representation and Chunking							
w/ directional motion commands	26.61 / 18.06	24.58 / 22.22	16.88 / 8.33	29.38 / 4.17	23.89 / 22.22	38.33 / 33.33	1.91
w/ joint-angle	16.08 / 9.72	9.86 / 5.56	5.21 / 0.00	19.79 / 0.00	22.22 / 22.22	23.33 / 20.83	1.12
w/ fixed action chunk ($L = 4$)	17.97 / 11.39	27.50 / 25.00	8.96 / 0.00	18.13 / 0.00	11.11 / 11.11	24.17 / 20.83	0.43
w/ adaptive action chunk	**26.36 / 20.00**	30.83 / 27.78	18.13 / 8.33	30.21 / 12.50	22.22 / 22.22	30.42 / 29.17	1.15
C. System 1–System 2 Collaboration							
VLA ($\pi_{0.5}$) [12]	11.41 / 6.91	13.37 / 8.17	12.40 / 5.50	23.54 / 14.67	5.78 / 4.56	1.98 / 1.67	–
w/ VLA open-loop ($\pi_{0.5}$) [12]	11.21 / 5.83	21.67 / 16.67	1.88 / 0.00	32.50 / 12.50	0.00 / 0.00	0.00 / 0.00	0.39
w/ VLA closed-loop	31.15 / 23.61	35.97 / 30.56	18.54 / 8.33	36.25 / 16.67	35.00 / 33.33	30.00 / 29.17	7.66
w/ VLA closed-loop ($L = 4$)	34.64 / 25.83	38.61 / 33.33	20.63 / 8.33	44.79 / 25.00	33.33 / 33.33	35.83 / 29.17	2.80
w/ ICL	39.24 / 29.67	34.92 / 28.33	28.63 / 15.00	41.38 / 20.00	73.67 / 70.00	17.63 / 15.00	9.57
w/ ICL + VLA closed-loop	**60.08 / 54.17**	54.58 / 50.00	49.38 / 45.83	60.42 / 41.67	66.67 / 66.67	69.38 / 66.67	19.33

Table 13 Harness Design Ablations. We evaluate the impact of harness design on GPT-6 Astra across 42 RoboDojo tasks, covering (A) agent components, (B) action representation and chunking, and (C) VLA collaboration and ICL. VLA denotes RoboDojo-fine-tuned $\pi_{0.5}$. L denotes actions per GPT decision in (B) and VLA-proposed actions reviewed per GPT call in (C). Performance entries report score / success rate (%), with "Average" computed as the unweighted mean across the five categories. Token consumption includes both input and output tokens and is reported in millions (M). ICL uses knowledge packages from the corresponding main tasks and a general-purpose knowledge package for Open tasks, with ICL-only results taken from Table 5.

All experiments are evaluated on 42 tasks of RoboDojo [26] under a consistent evaluation protocol, with three evaluation runs per task using different seeds, except for ICL, which follows Table 5. We report both task scores, which capture partial task completion, and success rates (SR), which measure full task completion. Performance is further evaluated across five dimensions: Generalization, Precision, Long-Horizon, Memory, and Open. We additionally report the average token consumption per episode, including both input and output tokens from GPT-6-Astra and auxiliary agents, in millions (M), as a proxy for inference cost. Unless otherwise specified, each ablation is compared against baseline followed by our RoboRSI [119] to isolate the impact of individual design choices.

D.1 Agent Components

We evaluate four components individually: (i) a planning subagent that decomposes tasks into subgoals and completion criteria, with its outputs incorporated into the main agent's context; (ii) an execution subagent that generates EEF actions from the main agent's subgoals; (iii) a feedback subagent that evaluates execution outcomes and suggests continuation, adjustment, or retry; and (iv) history compression that retains the latest four turns and consolidates earlier interactions into a rolling summary capped at 2,000 tokens.

This study investigates how different agent components affect the embodied performance of GPT-6-Astra and whether increasing agent complexity consistently improves task execution. As shown in Table 13, introducing a feedback subagent yields the largest improvement, increasing the average score by 7.22 points and the success rate by 9.16 percentage points, with only a 7.03% increase in token consumption. History compression improves the score by 5.25 points and the success rate by 5.55 percentage points while reducing token consumption by 66.41%, demonstrating benefits in both performance and efficiency. The planning subagent provides a smaller overall score improvement of 3.25 points at a 15.63% increase in token consumption, although its substantial gain in the Memory category indicates that its benefits vary across task categories. In contrast, the execution subagent reduces the score by 1.22 points at approximately the same token cost as the baseline. These results suggest that effective agent design depends on how additional reasoning supports task execution, rather than on the number of components alone.

Feedback and history compression offer complementary benefits: explicit outcome evaluation improves task performance with a modest increase in token consumption, while compressing earlier interactions improves performance and substantially reduces context overhead. These findings motivate further investigation of structured execution memory and targeted verification mechanisms for tracking task progress and failures. Future harness design should assess additional components by their contributions to both task performance and total token consumption.

E Full Real Robot Results

This appendix documents the real-world experiments of Sec. 4. App. E.1 summarizes all platforms, and App. E.2–E.9 document each platform, from the simplest to the most complex embodiment, in the same five parts: setting, context provided, tasks and results, findings, and visualization.

E.1 Platforms and Results

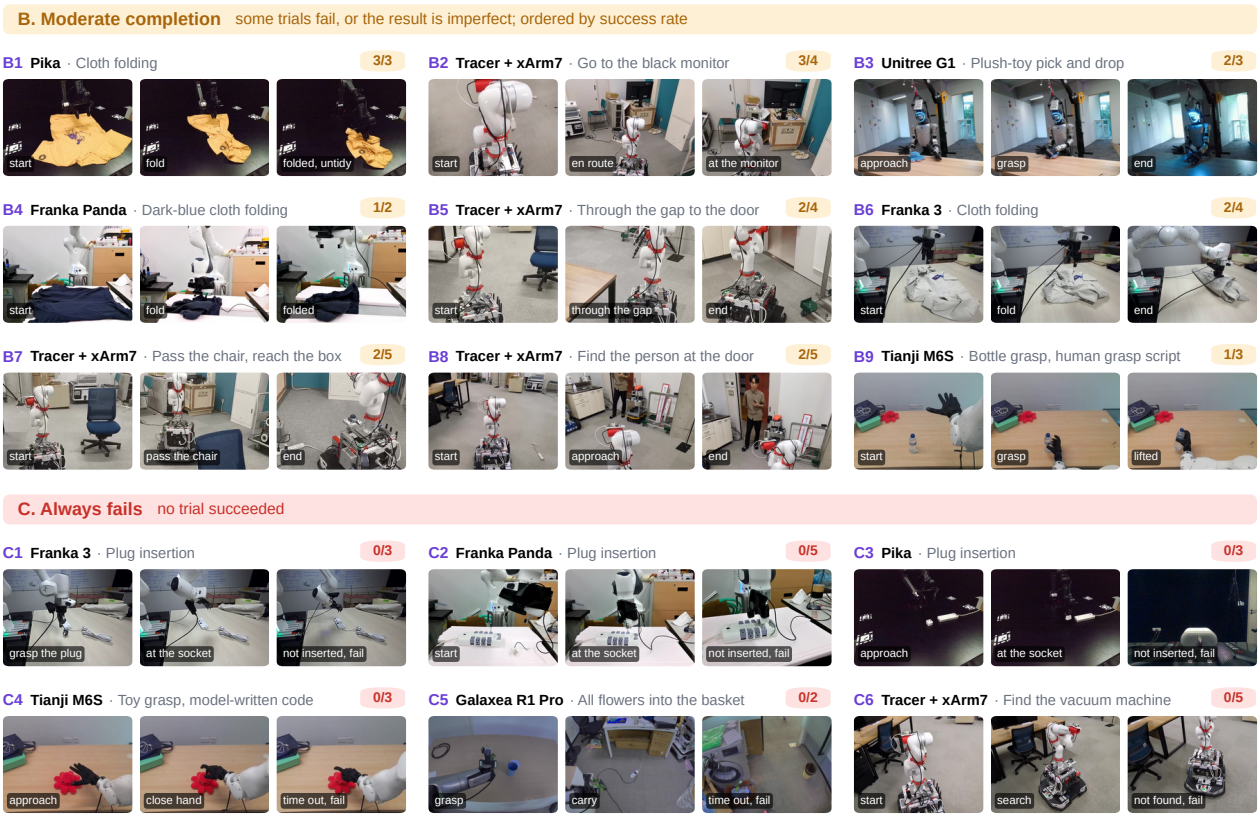
Table 14 summarizes the cameras, action interfaces, supplied context, and task outcomes across all eight platforms. Figure 13 provides task visualizations grouped by outcome, while the following subsections detail each platform's protocols and results.

Table 14 Real-world platforms, the context supplied to GPT-6-Astra, and results (successes / trials), ordered from simple to complex embodiments. No demonstrations are used on any platform. Cameras list only the streams given to GPT-6-Astra.

Platform	Cameras	Action interface	Context beyond the task instruction	Tasks and results
Single arm with gripper				
Franka 3, Robotiq 2F-85	Wrist ZED Mini + third-person ZED 2	End-effector pose + grasp label	Robot settings	Block stacking 6/6 Cloth folding 2/4 Plug insertion 0/3
Franka Panda, Franka Hand	Calibrated wrist webcam	Absolute end-effector pose + gripper open/close	Tool schemas, joint limits, safety constraints	Tape-roll stacking 3/3 Towel folding 2/2; cloth folding 1/2 Plug insertion 0/5
AgileX PiPER, Pika gripper	Wrist + side view	End-effector pose and gripper commands; IK by local tools	Step-level task prompt, admissible grasp points	Container nesting 3/3 Cloth folding 3/3 Plug insertion 0/3
Dual arm with grippers				
AgileX Piper, two arms	Front + two wrist views	Controls both arms directly without predefined protocols	Robot specs and official manuals	12 tasks, 14/14: batteries to plates, batteries to box, blocks to basket, items to grids, cup stacking, basket lift, bimanual milk-carton hold, pens to holder, towel-roll pyramid, towel folding, T-shirt folding, flower insertion
Dual arm with dexterous hands				
Tianji M6S CCS-809, Wuji Gen-1 hands	Head Azure Kinect	End-effector pose + grasp label	Tool schemas, joint limits, hand presets; human-written grasp scripts	Toy grasp 3/3 Bottle grasp 1/3[†] Model-written grasp code 0/3
Wheeled mobile manipulator				
Galaxea R1 Pro	Head stereo, five chassis, two wrist D405	GPT-6-Astra controls exposed robot parts directly	One-time exploration on the robot with GPT-6-Astra	Find washer and press button 2/2[‡] Flowers to basket 0/2[§] 6 tabletop manipulation tasks, 9/9: basket lift, towel rolls to tray, sponges to tank, clothes to basket, spill wiping and tidying, bottles and towels to tray
AgileX Tracer + xArm7	Arm-mounted RealSense D455	Three navigation waypoints	None (no robot model, SLAM or collision detection)	Navigation and search 9/23 (five tasks)
Humanoid				
Unitree G1 (29 DoF), two Wuji Gen-1 hands	Head Azure Kinect	14 arm-joint targets, binary hand open/close; IMU turn, timed walk at 0.15 m/s, stop	Tool schemas, joint limits, binary hand presets, locomotion and safety constraints	Toy pick-and-drop 2/3 Turn, approach and stop 3/3

[†] Succeeded after human feedback. [‡] Judged on the button press. [§] Both runs ended by a torso over-temperature fault.

Figure 13 Full gallery of zero-shot real-world tasks, grouped by outcome. Three frames per task in temporal order; chips give successes over all counted trials. (A) Tasks in which every counted trial succeeded; for the dual-arm Piper and the Galaxea R1 Pro tabletop tasks, counts are those of the reported rollouts (App. E.5 and E.7). A1 and A2, and A12 and A13, show two rollouts of the same task. (B) Tasks with moderate completion, ordered by success rate: some trials fail, or, as in B1, every counted trial succeeds but the result is imperfect (the folded cloth remains untidy). (C) Tasks in which no trial succeeded.



E.2 Single Arm with Gripper — Franka 3: Block Stacking, Cloth Folding and Plug Insertion

Setting

Robot	Franka Research 3 (7-DoF) with a Robotiq 2F-85 parallel gripper
Cameras given to GPT-6-Astra	Third-person Stereolabs ZED 2 and wrist ZED Mini; RGB only
Action interface	End-effector pose and a grasp label, issued as short trajectory segments at 15 Hz with gripper open/close commands
Harness	DROID-style setup on ROS 2: each segment is executed by the joint-trajectory controller, and the measured joint, end-effector and gripper states are returned; exterior and wrist videos are recorded at 15 fps; Model: GPT-6-Astra, Reasoning Effort: max
Budget and success rule	One hour per trial; an operator marks each run as success or failure when it ends. The recorded runs last 7–16 min

Context provided

- Task instruction, robot settings, two RGB views and robot-state feedback.
- No demonstrations and no semantic manipulation primitives.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Place the yellow block into the box, then put the green block into the box as well, stacking it on top of the yellow block	6	6	7–11 min*
First, lay the collar area flat; then, fold the left sleeve inward, followed by the right sleeve. Finally, fold the bottom hem of the garment up toward the collar	4	2	10–15 min*
Grasp the plug and insert it into the socket, aligning it with the slots	3	0	9–16 min*

*Trial counts and times are those of the recorded runs. Two folding trials ended in execution errors and are counted as failures; the three plug-insertion trials were marked failed by the operator.

Findings

- **Rigid pick-and-place transfers from simulation.** Block stacking succeeds in all 6 trials across different initial configurations, matching simulation.
- **Folding is judged without a shared criterion.** The operator marks a folding run as a success from the final state of the cloth; how flat or well aligned the fold must be is not specified, and the judgment was not unified with the other platforms (App. E.4), so folding results are not directly comparable across them.
- **Contact-rich tasks end in protective stops.** With thin cloth the gripper easily touches the table and the robot halts; a first failed insertion typically collides with the socket and halts the robot. A simulator would let execution continue and leave room for further attempts.

Visualization

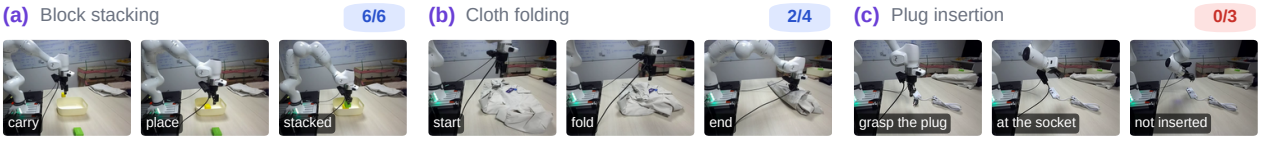


Figure 14 Franka 3: (a) block stacking, (b) cloth folding and (c) plug insertion, exterior view; in (c) the plug reaches the socket but is not inserted.

E.3 Single Arm with Gripper — Franka Panda: Tape-Roll Stacking, Towel Folding and Plug Insertion

Setting

Robot	Franka Panda with the Franka Hand parallel gripper
Cameras given to GPT-6-Astra	One calibrated wrist webcam (intrinsics and hand–eye), 640×480 RGB at 25 fps
Action interface	Tool calls: camera image, robot and gripper state, end-effector pose, absolute end-effector moves, gripper open and force-controlled close
Harness	ROS 2 backend; collision-scene setup and gripper timeouts are handled by the backend or operator
Budget and success rule	No common time limit: runs last 7–37 min, and the successful dark-blue-cloth run took 196 min. A person scores each trial afterwards: it is a success only when the instructed physical goal and its spatial constraints are achieved and the model reports completion; all other trials count as failures. Each physical motion request is approved by the operator before it is executed

Context provided

- Task instruction, tool schemas, joint limits and safety constraints.
- No demonstrations; no pick, place, stack, fold or insert tools, no custom IK, joint trajectories or shell access.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Pick one tape roll and place it on another	3	3	10 min*
Fold the white towel	2	2	20 min*
Fold the dark-blue cloth	2	1	9 min; 196 min
Insert the white plug into the socket	5	0	22, 37, 7, 19, 27 min

*Only one trial of this task has a model log; the time is that trial's.

Findings

- **Sub-tasks are solved in isolation.** When inserting the plug, GPT-6-Astra grasps it without checking whether the resulting pose admits insertion.

- **Cautious termination under uncertainty.** In trial 1 (117 tool calls, 10.6 M tokens), GPT-6-Astra stops pushing when small moves produce diminishing motion, withdraws 2.2 mm and reports the insertion as unverified.
- **No recovery from hardware faults.** When the driver reports a collision, GPT-6-Astra declares the task failed.

Visualization

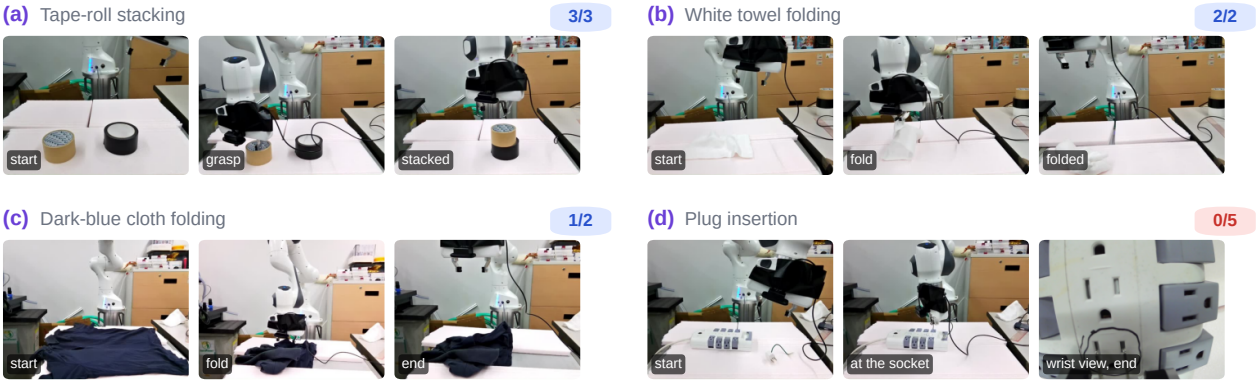


Figure 15 Franka Panda: (a) tape-roll stacking, (b) white-towel folding, (c) dark-blue-cloth folding (the successful run) and (d) plug insertion, ending with the final wrist view. Frames (a)–(c) come from time-compressed recordings and are labelled by phase.

E.4 Single Arm with Gripper — PiPER with Pika Gripper: Container Nesting, Cloth Folding and Plug Insertion

Setting

Robot	One AgileX PiPER arm (6 joints) with a Pika parallel gripper
Cameras given to GPT-6-Astra	Wrist RealSense D405 (640×480 RGB) and a third-person Azure Kinect DK from the front right (1280×720 RGB). Depth is not given as an image, but a tool returns the depth and 3D point of a queried wrist-image pixel
Action interface	GPT-6-Astra outputs target end-effector poses, motion paths and gripper commands; local tools solve IK, check and execute the trajectory, and return the measured state with new images of both views
Harness	`gpt-6-astra`, with reasoning effort `high` for container nesting and plug insertion, and `max` for folding; every trial starts from a fresh context, without the trajectories or interaction records of other trials
Budget and success rule	60 minutes from the first model request (plus 60 s grace); a person judges success from the final state: containers must be stably nested in the intended order, the plug must be correctly seated in the socket, and folding is assessed qualitatively from the overall folded result

Context provided

- **Task prompt:** The task goal, step guidance and manipulation advice (e.g. cup wall or rim grasps).
- **System prompt:** Tool and coordinate conventions, joint-coordination principles, and instructions to verify action outcomes and replan when needed.
- **Observations and feedback:** Current images, measured robot state, calibration, joint limits and within-trial tool feedback. The model chooses grasp points, poses and routes; no demonstrations or reference trajectories are provided.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Nest the containers into the bowl	3	3	15–24 min
Fold the clothes	3	3	32–60 min
Insert the plug into the socket	3	0	7–14 min*

Two earlier folding runs with an incomplete task prompt are not counted. *Two of the three insertion runs ended in a J6 collision fault of the joint driver.

Findings

- **Re-planning from feedback.** When the receiving bowl is displaced by contact, GPT-6-Astra detects the change, raises the container and realigns before placing it (run 3, 07:06–10:40); after an empty grasp it re-grasps the cloth (folding run 2, 12:46–17:21).
- **The prompt shapes the strategy.** Unless alternative grasp points are named, GPT-6-Astra keeps trying to grasp a cup by its handle even when the pose is awkward or unreachable.
- **Insertion is limited by observation and reachability.** The pins and the socket are poorly visible, and poses that would reveal them are limited by joint reachability. Reorienting the initially upward-facing plug for insertion can also bring the arm close to its joint limits.
- **Folding is judged without a shared criterion.** An operator judges success from the final cloth state: a smaller, layered cloth packet remaining on the table after release, without requiring perfectly flat or aligned folds. All three reported trials met this practical criterion, although wrinkles and uneven edges remain (Fig. 13, B1). One run reached the 60-minute limit and was subsequently accepted based on its final state. Since the success criterion was not standardized across platforms (App. E.2), the reported folding success rates are not directly comparable.

Visualization

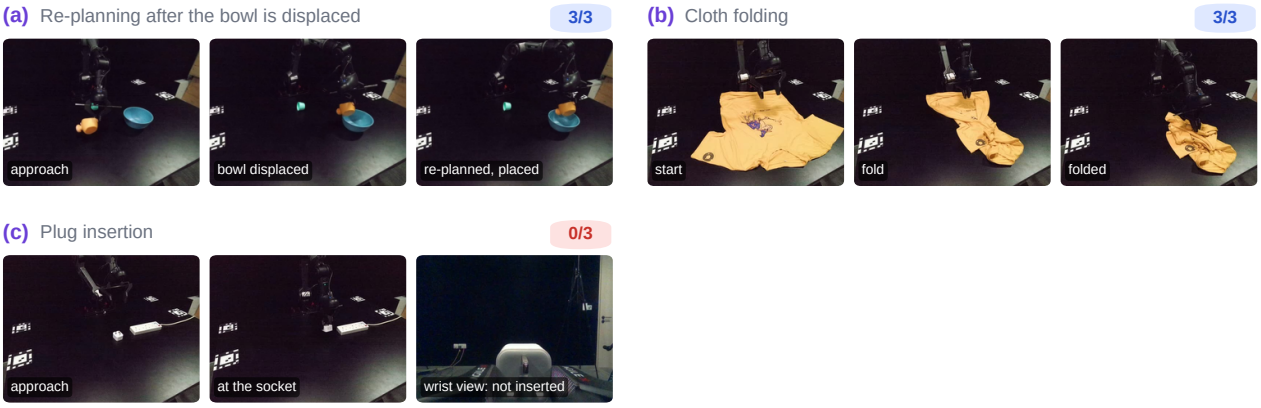


Figure 16 PiPER with Pika gripper: (a) re-planning after the receiving bowl is displaced, (b) cloth folding and (c) plug insertion, whose last frame is the wrist view with the plug covering the socket.

E.5 Dual Arm with Grippers — AgileX Piper Series: Zero-Shot Bimanual Tasks

Setting

Robot	Two AgileX Piper/PiperX arms with parallel grippers
Cameras given to GPT-6-Astra	One front view and one wrist view per arm
Action interface	GPT-6-Astra controls both arms directly, without predefined control protocols
Harness	GPT-6-Astra is called through the Codex SDK within the Agentic Orchestration Policy framework[1]
Budget and success rule	No predefined time limit per rollout. GPT-6-Astra decides when a task is complete; a person then judges success from the physical outcome

Context provided

- The task instruction, verbatim as in the table below.
- A one-time exploration of each robot before the rollouts, which exposes detailed robot specifications such as the URDF model, camera specifications and links to the official documentation.
- No demonstrations; every task is run zero-shot.

[1] https://github.com/ZexinHe/Agentic-Orchestration-Policy

Tasks and results

These are demonstration rollouts that illustrate what GPT-6-Astra can do; each reported rollout succeeded. Times are elapsed wall-clock times.

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Put the batteries into their nearest plates.	2	2	16–20 min
Put the toy blocks on the table into the basket.	1	1	18 min
Put the items in respective grids.	1	1	11 min
Stack the paper cups to the one marked with X.	1	1	18 min
Lift the fruit basket in the air, with a clear margin above the table, and hold still.	1	1	6 min
Hold the milk brick in the air with both hands, showing a clear margin above the table.	1	1	20 min
Put the pens into the pen holder.	2	2	11–14 min
Put the batteries into their nearest box.	1	1	21 min
Stack the towel rolls into a pyramid on the tray, three at the bottom, two in the middle, and one on top.	1	1	15 min
Fold the towel into a quarter square, from bottom to top, then from left to right.	1	1	14 min
Fold the T-shirt into a square, from bottom to top.	1	1	8 min
Insert the plastic flowers into the vase.	1	1	21 min

Findings

- **Instructions with relations and order.** GPT-6-Astra follows spatial relations (“nearest plate”, “the cup marked X”) and prescribed orders (fold bottom-to-top, then left-to-right; a 3-2-1 pyramid).
- **Bimanual coordination.** Both arms are used together to hold a carton or lift a basket clear of the table, showing that GPT-6-Astra can coordinate two arms without a bimanual controller.
- **Time varies across rollouts.** For tasks with more than one rollout, the elapsed time varies, partly because of network latency and Codex retries in some runs.

Visualization

Figure 17 AgileX Piper/PiperX: all tabletop manipulation tasks, from simple placement to ordered folding and insertion.

E.6 Dual Arm with Dexterous Hands — Tianji M6S: Toy and Bottle Grasping

Setting

Robot	Tianji M6S CCS-809 with two Wuji Gen-1 dexterous hands
Cameras given to GPT-6-Astra	Head-mounted Azure Kinect; RGB only
Action interface	End-effector pose and a grasp label
Harness	Tool calls through the OpenAI Responses API; every model call, tool call and camera frame is archived. Collision-scene setup and gripper timeouts are handled by operator
Budget and success rule	60 minutes per attempt; a person approves the final state

Context provided

- Task instruction, tool schemas, joint limits, predefined hand open/close profiles, RGB observations and robot-state feedback.
- In the human-coding condition, grasp scripts written by a person; no demonstrations in either condition.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Grasp the toy, grasp code written by GPT-6-Astra	3	0	60 min
Grasp the toy, human-written grasp script	3	3	9, 22, 26 min
Grasp the bottle, human-written grasp script	3	1[†]	29, 56, 38 min

[†] After a timed-out attempt, success followed human feedback in the final 10 minutes. A fourth attempt in the model-written condition, with a human hint, also failed.

Findings

- **Direct dexterous control is hard.** Code written entirely by GPT-6-Astra failed, including after it summarized its failures and after a human hint.
- **Human priors make the task feasible.** Following human-written grasp scripts, the toy grasp succeeds in every trial.
- **Real-world perturbations matter.** Small disturbances and object deformation affect the grasp; together with overheating and safety, these are the main differences from simulation.

Visualization

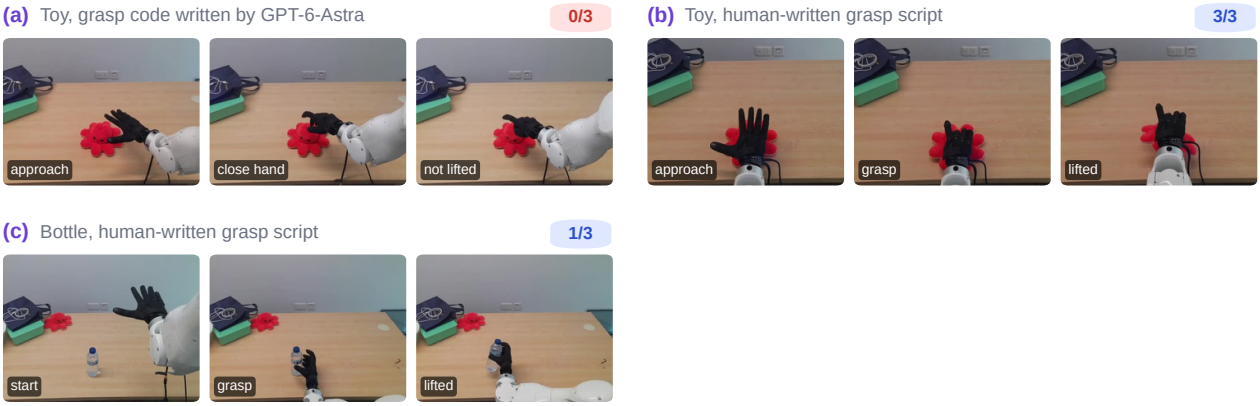


Figure 18 Tianji M6S with Wuji hands: (a) toy grasp with grasp code written by GPT-6-Astra (round 1; the toy is never lifted), (b) toy grasp and (c) bottle grasp with human-written grasp scripts. Frames from the head-camera recordings.

E.7 Wheeled Mobile Manipulator — Galaxea R1 Pro: Appliance Use, Flower Collection and Tabletop Tasks

Setting

Robot	Galaxea R1 Pro (2025): wheeled base, 4-axis torso, two 7-DoF arms with parallel grippers
Cameras given to GPT-6-Astra	Head stereo pair, five chassis cameras and two wrist RealSense D405 (RGBD); LiDAR
Action interface	GPT-6-Astra controls exposed robot parts directly, without predefined control protocols
Harness	Tabletop manipulation tasks use the Agentic Orchestration Policy framework (App. E.5); navigation tasks run in the Codex app directly
Budget and success rule	One hour per task; a task is failed unless the physical outcome is verified in fresh images before the deadline

Context provided

- For tabletop manipulation, only the two arms and grippers are exposed for control, and only the head and wrist cameras for observation.
- For navigation, the torso and chassis are additionally exposed for control, and the five chassis cameras and the LiDAR for observation.
- A one-time exploration of the robot before the rollouts, which exposes its specifications.
- No demonstrations; every task is run zero-shot.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Find washer and press button	2	2[‡]	41–52 min
Put all flowers into the basket near the washing machine	2	0[§]	60 min
Lift the basket in the air, and hold still.	2	2	6–9 min
Put the towel rolls into the tray, side by side.	2	2	13–18 min
Put the sponge cubes into the tank.	1	1	13 min
Put the clothes into the basket.	1	1	25 min
Clean up the dirty water with the blue towel. Put the clean white towel rolls into the tray.	1	1	15 min
Put the two bottles of water into the tray, upright in the center, then put the rolled towels onto the tray, one on each side.	2	2	32–43 min

[‡]Judged on the Warm-button press. The wash cycle did not start in either run, which we attribute to the washer: it requires the Start button to be pressed shortly after Warm, and it showed an unexpected fault code in one run. [§]0 of 4 and 1 of 4 flowers placed; both runs ended by a torso over-temperature fault. The last six tasks are tabletop manipulation runs.

Findings

- **Self-built context.** GPT-6-Astra discovers the robot's interfaces itself and records discrepancies with the documentation, such as a chassis message type that differs from the vendor guide.
- **Contact without force sensing.** GPT-6-Astra uses joint-effort change as a contact signal and approaches in small steps, and when reflective button surfaces corrupt the wrist depth, it cross-checks several depth patches against RGB.
- **Long runs meet hardware limits.** As a task proceeds, some runs hit hardware over-temperature faults that significantly affect completion; we regard this as a limitation of hardware reliability rather than of GPT-6-Astra.

Visualization

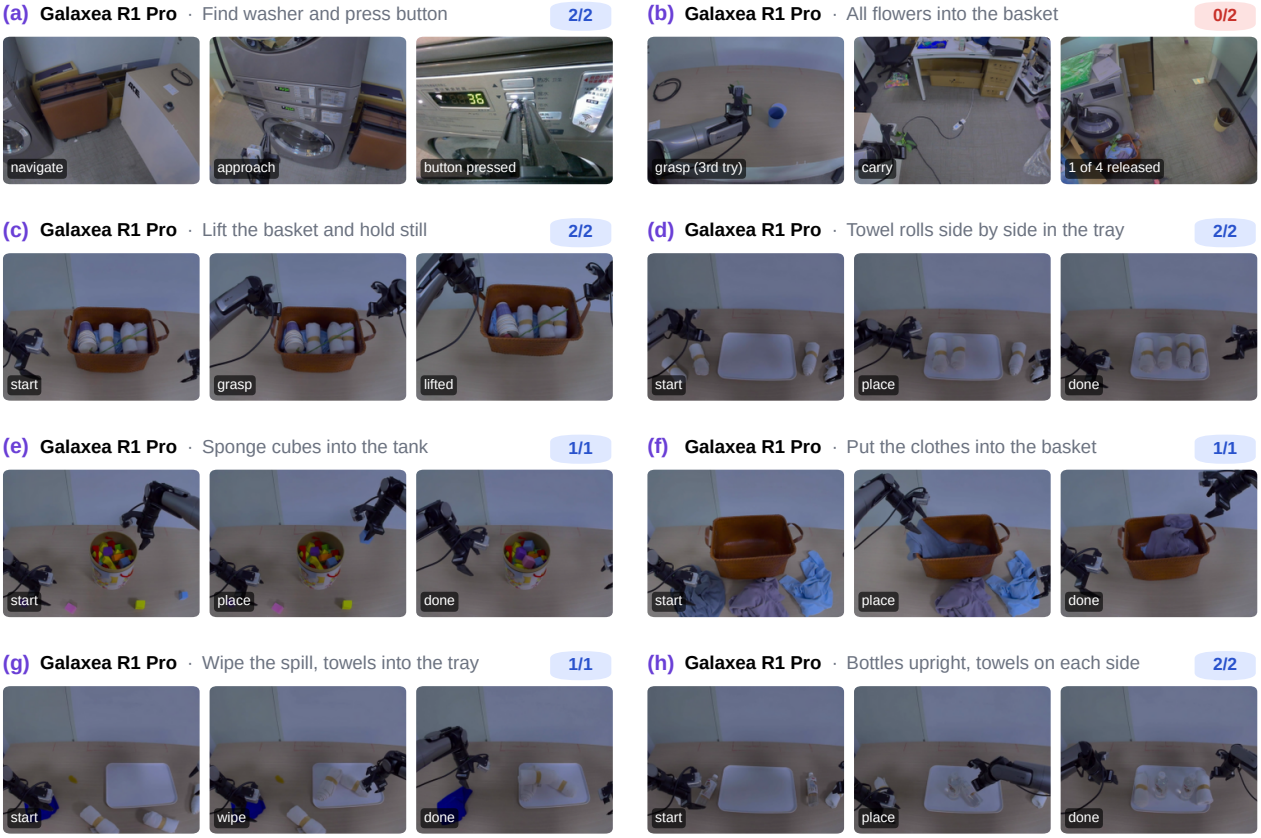


Figure 19 Galaxea R1 Pro: (a) washer task and (b) flower collection, from the task records; (c–h) tabletop manipulation tasks.

E.8 Wheeled Mobile Manipulator — AgileX Tracer + xArm7: Navigation and Search

Setting

Robot	AgileX Tracer 1.0 mobile base with a UFactory xArm7
Cameras given to GPT-6-Astra	Forward-facing RealSense D455 on the arm, uncalibrated, 1280×720 RGB
Action interface	Three navigation waypoints per decision, followed by a waypoint controller (cruise 0.7 m/s); the pose is estimated by integrating the commanded velocity, with no odometry, SLAM or collision detection
Harness	`gpt-6-astra` through the OpenAI API; each decision receives the latest RGB frame (downscaled to 640 px width) and returns the next waypoints
Budget and success rule	No fixed time limit; runs last 0.2–8.3 min and end when the model reports completion or being blocked, on a timeout, or by operator interruption. A person scores each trial afterwards: it is a success only when the instructed physical goal and its spatial constraints are achieved and the model reports completion; all other trials count as failures

Context provided

- Task instruction and RGB input only: no robot model, SLAM or collision detection, and no demonstrations.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Pass the blue chair on the left, then approach the wooden box	5	2	0.5–3.6 min
Pass through the tripod–table gap, then approach the door	4	2	2.5–8.3 min
Approach the black monitor on the left	4	3	0.7–2.8 min
Find the person at the door on the right	5	2	1.2–3.0 min
Find the vacuum machine	5	0	0.2–1.1 min

Findings

- **Losing track of progress.** The robot loses its heading and, in some trials, forgets that it has already reached the goal area.
- **Over-checking constraints.** When told to pass a chair on the left, GPT-6-Astra repeatedly confirms the side instead of progressing toward the goal, and loses its way in some trials.
- **No recovery after collisions.** After a collision GPT-6-Astra almost never plans a recovery trajectory.

Visualization

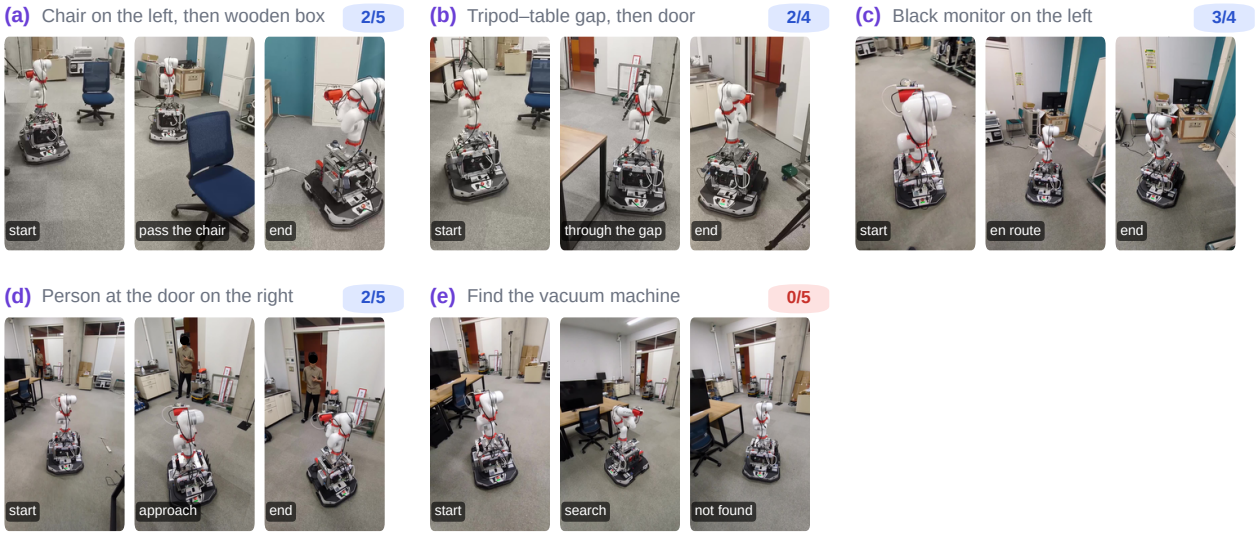


Figure 20 AgileX Tracer + xArm7: one recorded run per navigation task, third-person phone video. Chips give successes over all trials of the task.

E.9 Humanoid — Unitree G1: Tabletop Manipulation and Locomotion

Setting

Robot	Unitree G1 humanoid (29 body DoF) with two Wuji Gen-1 dexterous hands; the right thumb is mechanically frozen, and waist, legs and body height are locked during manipulation
Cameras given to GPT-6-Astra	Head-mounted Azure Kinect, RGB only; depth is recorded but neither given to the model nor used as a collision guard. Measured robot state is additionally available through tools
Action interface	Manipulation: `move_arms` (14 absolute arm-joint targets in radians, clamped to joint limits and interpolated by the offboard arm controller) and `set_hands` (binary open/close profile per hand, no individual-finger, grip-width or force control; hands driven by the onboard Jetson). Locomotion: `turn_to_heading` (IMU yaw feedback), `walk_forward` (fixed 0.15 m/s for a commanded duration, at most 1.5 s per step) and `stop`, executed by the vendor high-level controller; GPT-6-Astra never commands leg joints, and no translational odometry is available to the policy
Harness	Tool calls through the OpenAI Responses API (`gpt-6-astra`, reasoning effort `max`, mode `pro`, `store=false`); model calls, tool calls and camera frames are archived
Budget and success rule	Six retained runs, three per task; manipulation and locomotion use different tool sets and are evaluated separately. No common 60-minute cutoff applies: both successful manipulation runs exceeded one hour. A person judges the binary outcome from third-person video after the run, separately from the model's completion statement; the requested ∼10 cm stopping distance is approximate and the final gap was not independently measured. Operator emergency stop throughout; the locomotion runs additionally use a slack safety-gantry strap

Context provided

- Task instruction, embodiment description, tool schemas, joint limits, camera placement, binary hand open/close profiles, locomotion and safety constraints, RGB observations, robot-state feedback, and the current run's interaction history.
- No demonstrations, no demonstration-derived ICL package, and no semantic pick, place or navigation primitives; every motion is chosen by the model, one tool call at a time.

Tasks and results

Task (instruction given to GPT-6-Astra)	Trials	Success	Wall-clock time
Pick up the blue plush toy and then drop it back onto the table	3	2	74:31; ∼87:47
Turn, walk to directly in front of the table and stop about 10 cm from it without touching it[†]	3	3	8–13 min

[†] The locomotion instruction was given in Chinese; the wording above is a translation. Wall-clock times are listed for the two successful manipulation runs; per-run details below.

Run	Outcome	Tool calls	Image calls	Elapsed time
pickRun1	Failure	29	14	—[*]
pickRun2	Success	74	36	∼87 min 47 s
pickRun3	Success	65	32	74 min 31 s
walkRun1	Success	18	9	12 min 49 s
walkRun2	Success	12	6	8 min 13 s
walkRun3	Success	13	7	10 min 12 s

[*] Elapsed time unavailable from the archived records.

Manipulation succeeds in 2/3 trials and locomotion in 3/3, giving 5/6 overall. The failed manipulation run drops the toy off the table, and recovery is not completed within the fixed-body workspace. With three trials per task, these results establish feasibility rather than reliability.

Findings

- **Observation-conditioned approach under a safety constraint.** In walk run 1, GPT-6-Astra shortens successive forward commands from 0.8 s to 0.5, 0.3, 0.2 and 0.15 s after fresh images, and all three locomotion runs stop at a self-estimated 0.15–0.20 m rather than 0.10 m, citing the no-contact requirement. This shows conservative progress assessment, not metric stopping accuracy.

- **Valid joint targets do not guarantee faithful execution.** In pick run 3, all 26 arm motions return `stalled_short_of_target` (18) or `timeout` (8), with median final tracking error 0.07065 rad and maximum 0.1904 rad, yet the human-labelled outcome is a success: useful motion occurs without meeting the joint-target criterion. GPT-6-Astra notices the discrepancies and adjusts, but reliable error compensation is not established.
- **Slow supervision remains costly on hardware.** Across 108 archived API calls, median request latency is 24.84 s, and API request wall time accounts for 88.8–91.7% of elapsed run time; image capture takes 4.5–9.0%. The remainder includes tool execution and harness overhead and is not a direct measure of physical motion time. Both successful manipulation runs last over an hour.

Visualization

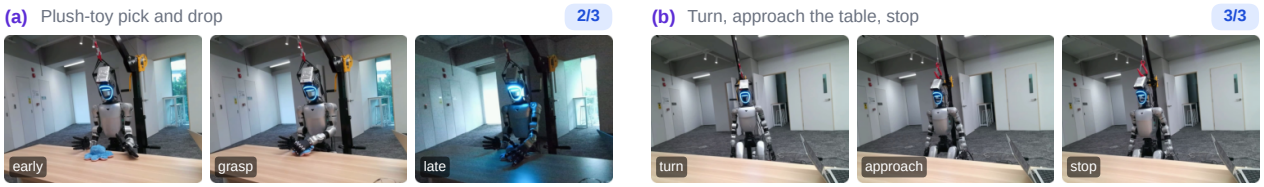


Figure 21 Unitree G1 with Wuji hands: fixed-body plush-toy pick-and-drop (2/3 successes) and table approach through the vendor velocity interface (3/3), evaluated separately. Frames are from third-person recordings; GPT-6-Astra received the head-camera RGB view and tool-returned robot state. Binary hand profiles were used.